

MCP Server Tool Reference

flowise-questionnaire-system

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Every tool below is documented with its real Python signature, its matching MCP prompt (slash-command shortcut), and a complete, real request/response example -- generated by actually calling each tool function against this project's own live production data (real modules, real routing edges, real matched Colectica/Forsta+ and Colectica/Colectica pairs), not hand-written, fabricated, or built from a synthetic fixture. Two of the largest real responses are truncated to a few representative entries for readability, each clearly marked with the true total count -- everything else is shown in full. Because this is real production data, the exact module names, UUIDs, discrepancy IDs, and counts below will drift as the underlying questionnaires change; call `list_modules` first to get current identifiers for your own environment rather than assuming these exact values still resolve. Server is read-only: it never writes to the database and never triggers expensive pipeline steps (extraction, graph building, matching, AI review) itself -- every tool reads data some other part of the app already computed and persisted.

Running the server

python manage.py runmcp -- its own standalone process, own port (8765 by default via \$MCP_SERVER_HOST/\$MCP_SERVER_PORT), own streamable-HTTP transport. Does not touch config/asgi.py, wsgi.py, or manage.py runserver. HTTP-only by design (no stdio transport). Serves HTTPS via a local self-signed cert in mcp_server/certs/ if present when run locally; in production, nginx terminates TLS and runmcp runs behind it as plain HTTP (--no-tls).

Authentication

Per-person token embedded in the URL path: https://<host>/t/<token>/mcp (not a header -- MCP client connector UIs generally only take a URL, e.g. Claude Desktop's "Add custom connector"). Admins generate/revoke tokens at /questionnaires/mcp-tokens/ (staff-only page). A bare /mcp (no token) always 404s; an unknown/revoked token 403s; revocation takes effect immediately (a live DB check, not a config reload).

Colectica / Forsta+ scope, at a glance

Most tools are Colectica-only and reject Forsta+ modules with a clear ToolError. Two tools (get_module_graph, evaluate_edge_condition) support either format on a single module. Two pairs of tools compare two modules against each other: the Colectica-vs-Forsta+ pair (get_routing_diff_report / get_routing_discrepancy_detail) requires one module of each format; the Colectica-vs-Colectica "version diff" pair (get_version_diff_report / get_version_diff_discrepancy_detail) requires both modules to be Colectica JSON -- e.g. two waves/versions of the same questionnaire. These are two independent comparators with different matching priorities and discrepancy types, not the same code reused.

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- `get_module_summary`
- `list_questions`
- `get_question`
- `get_routing_edges`
- `trace_variable`

Cross-format tools (Colectica + Forsta+)

- `get_module_graph`
- `evaluate_edge_condition`

Colectica vs Forsta+ routing diff

- `get_routing_diff_report`
- `get_routing_discrepancy_detail`
- `get_routing_simulation`

Colectica vs Colectica routing diff (version diff)

- `get_version_diff_report`
- `get_version_diff_discrepancy_detail`

CORE COLECTICA TOOLS

list_modules

Scope: Colectica-only

MCP prompt: /listModules

Description

List every Colectica JSON QuestionnaireModule: id, name, status, question_count, edge_count.

Python signature

```
list_modules(  
  
) -> dict
```

Example prompt shortcut

```
/listModules
```

"Using the list_modules tool, list every Colectica questionnaire module available, with its status, question count, and routing-edge count."

Example request

```
list_modules()
```

Example response (real output)

```
{  
  "count": 6,  
  "modules": [  
    {  
      "id": "082097ea-506f-42bc-b93e-df24afe88190",  
      "name": "W16",  
      "version": "v02",  
      "status": "extracted",  
      "question_count": 1981,  
      "edge_count": 4465  
    },  
    {  
      "id": "234f2404-7ed7-476a-afce-7e2dbe5d12e3",  
      "name": "colectica_test_2",  
      "version": "02",  
      "status": "extracted",  
      "question_count": 1803,  
      "edge_count": 4152  
    },  
    {  
      "id": "2ab9692f-5f5f-4d88-a714-002cac97c977",  
      "name": "W18",  
      "version": "v01",  
      "status": "uploaded",  
      "question_count": 0,  
      "edge_count": 0  
    },  
    {  
      "id": "3a2de4bf-432b-42f2-b74c-2ceffd868e8a",  
      "name": "W18",  
      "version": "v01",  
      "status": "extracted",  
      "question_count": 1803,  
      "edge_count": 4152  
    }  
  ]  
}
```

```
    "question_count": 1803,  
    "edge_count": 4152  
  },  
  {  
    "id": "c4fe3c04-1cda-40d7-9028-a43988c4c9dc",  
    "name": "Full Wave 18 Questionnaire",  
    "version": "v1",  
    "status": "uploaded",  
    "question_count": 0,  
    "edge_count": 0  
  },  
  {  
    "id": "c8171847-7442-41ef-b534-04682184a14f",  
    "name": "Benefits W18",  
    "version": "v02",  
    "status": "extracted",  
    "question_count": 20,  
    "edge_count": 35  
  }  
]  
}
```

get_module_summary

Scope: Colectica-only

MCP prompt: /showModuleSummary

Description

Full detail on one Colectica module: status, counts, timestamps.

Raises a clear ToolError (not an empty result) if the module exists but is a Forsta+ XML module.

Python signature

```
get_module_summary(  
    module_id_or_name: str  
) -> dict
```

Example prompt shortcut

```
/showModuleSummary
```

```
"Using the get_module_summary tool, give me the full status summary for module 'Benefits W18'  
(question/edge counts by type, whether a graph has been built, and any error message)."
```

Example request

```
get_module_summary(module_id_or_name="Benefits W18")
```

Example response (real output)

```
{  
  "id": "c8171847-7442-41ef-b534-04682184a14f",  
  "name": "Benefits W18",  
  "version": "v02",  
  "status": "extracted",  
  "error_message": "",  
  "question_count": 20,  
  "routing_edge_count": 35,  
  "conditional_edge_count": 8,  
  "loop_edge_count": 8,  
  "sequential_edge_count": 19,  
  "has_graph": true,  
  "created_at": "2026-07-17T15:28:17.626907+00:00",  
  "updated_at": "2026-07-17T15:28:41.482825+00:00"  
}
```

list_questions

Scope: Colectica-only

MCP prompt: /listQuestions

Description

All NormalizedQuestion rows for a Colectica module (name, text, type, options).

Python signature

```
list_questions(
    module_id_or_name: str
) -> dict
```

Example prompt shortcut

```
/listQuestions
```

"Using the list_questions tool, list every question in module 'Benefits W18' with its type and options."

Example request

```
list_questions(module_id_or_name="Benefits W18")
```

Example response (real output)

```
{
  "module_id": "c8171847-7442-41ef-b534-04682184a14f",
  "module_name": "Benefits W18",
  "count": 20,
  "questions": [
    {
      "name": "INTBENEFIT",
      "label": "Benefit module intro",
      "text": "One of the most important parts of our research is how people are getting by financially these days. We have found that we need to ask about a number of different types of income, otherwise our results could be misleading. Please think about ALL of the extra sources of income you receive, as well as any benefits or tax credits. `If you are at all unsure about whether something would count, you can check with me. {if GRIDVARIABLES.ModeType = 1 | 2}` We'd like to remind you that anything you tell us is completely confidential.",
      "question_type": "single_choice",
      "options": [
        {
          "code": "1",
          "label": "Continue",
          "exclusive": false,
          "is_missing": false,
          "category_identifier": "f0ffa159-b6c1-40b2-8e8b-0d25de6d91c1"
        },
        {
          "code": "-1",
          "label": "Don't know",
          "exclusive": false,
          "is_missing": true,
          "category_identifier": "22faf625-0f73-4b7b-9762-6000887ba8e6"
        },
        {
          "code": "-2",
          "label": "Refused",
          "exclusive": false,
          "is_missing": true,
          "category_identifier": "158ab848-b837-406d-9588-130060f4e3df"
        }
      ]
    }
  ]
}
```

```

    },
    {
      "code": "-8",
      "label": "Inapplicable",
      "exclusive": false,
      "is_missing": true,
      "category_idenfifier": "c70a3ae7-70eb-4685-905e-44c7644563c4"
    },
    {
      "code": "-9",
      "label": "Missing",
      "exclusive": false,
      "is_missing": true,
      "category_idenfifier": "1b429daf-d444-409c-a721-ec0334832fe3"
    }
  ],
  "help_text": "",
  "interviewer_instruction": "",
  "sequence_index": 1
},
{
  "name": "BENBASE",
  "label": "Income: Receives core benefits",
  "text": "First, are you currently receiving any of these payments `, either just yourself or jointly {if
hhsz > 1 & (livesp = 1 or livewith = 1)}`?",
  "question_type": "multi_choice",
  "options": [
    {
      "code": "1",
      "label": "Income Support",
      "exclusive": false,
      "is_missing": false,
      "category_idenfifier": "dd751356-0b0e-414e-afd8-b2a8554d12fa"
    },
    {
      "code": "2",
      "label": "Job Seeker's Allowance",
      "exclusive": false,
      "is_missing": false,
      "category_idenfifier": "333a7022-101b-46ab-861f-bdbd698187d3"
    },
    {
      "code": "3",
      "label": "Child Benefit",
      "exclusive": false,
      "is_missing": false,
      "category_idenfifier": "2abae12b-8b1b-47b0-b925-81004d0d61f3"
    },
    {
      "code": "4",
      "label": "Universal Credit",
      "exclusive": false,
      "is_missing": false,
      "category_idenfifier": "7428141f-0c04-461a-b1ba-b8b6fdfe08e6"
    },
    {
      "code": "96",
      "label": "None of these",
      "exclusive": true,

```



```

    "is_missing": false,
    "category_idenfifier": "81139977-d8a9-4121-b829-ac78ecee9d0"
  },
  {
    "code": "-1",
    "label": "Don't know",
    "exclusive": false,
    "is_missing": true,
    "category_idenfifier": "df93b9f7-0f81-44b1-bb42-a68ae122e3c1"
  },
  {
    "code": "-2",
    "label": "Refused",
    "exclusive": false,
    "is_missing": true,
    "category_idenfifier": "158ab848-b837-406d-9588-130060f4e3df"
  },
  {
    "code": "-8",
    "label": "Inapplicable",
    "exclusive": false,
    "is_missing": true,
    "category_idenfifier": "b04e36eb-7569-44d7-bf3f-e4ff7cc54012"
  },
  {
    "code": "-9",
    "label": "Missing",
    "exclusive": false,
    "is_missing": true,
    "category_idenfifier": "96fae899-a4cc-4e46-bf80-dff40b244afb"
  }
],
  "help_text": "\u2022 Child Benefit** is a payment for those who are responsible for a child aged under 16 or under 20 and in relevant education or training. It is only paid to one person, normally the mother (unless there is none in the household). Only record getting Child Benefit if it is paid to you (whether into your own account or a joint bank or building society account). \r\n\r\n**High Income Child Benefit Tax Charge**: If you receive Child Benefit and you or your partner have an individual income over \u00a3360,000 you may have to pay the High Income Child Benefit Charge. Please record that you receive Child Benefit even if the High Income Child Benefit Charge applies in your circumstances.\r\n\r\n\r\n\u2022 Income Support (IS)** is paid if you have no income or a low income but are working less than 16 hours a week. IS replaced Supplementary Benefit (sometimes called Social Security Benefit) and is often paid to top up other benefits or earnings from part time work. Income Support has been replaced by Universal Credit (UC) but some people still get IS. \r\n\r\nDo not include as Income Support any maintenance payments from a former or separated spouse or for child support which are paid through the Department for Work and Pensions or other government agencies.\r\n\r\n\r\n\u2022 Job Seeker's Allowance (JSA)** replaced Unemployment Benefit in October 1996. To receive JSA you must be capable of work and available for work and actively seeking work or are working less than 16 hours a week. There are two types of JSA - Contribution-based JSA (which is now known as \u2018New Style\u2019 JSA) and Income-based JSA (which is being replaced by Universal Credit). The Contribution-based New Style JSA is paid in the first 26 weeks of unemployment if enough National Insurance contributions have been paid. You will also make an agreement to look for work with the Department for Work and Pensions known as a Claimant Commitment (previously known as a Job Seeker\u2019s agreement).\r\n\r\n\r\n\u2022 Universal Credit** can be received if you\u2019re on a low income, out of work or you cannot work. If you are over state pension age you might receive Universal Credit if your partner is still of working age. Universal Credit was set up in October 2013 and is now the main working age benefit, replacing Income Support, income-based Job Seeker's Allowance, income-related Employment Support Allowance, Housing Benefit, Child Tax Credit and Working Tax Credit. Some people might still receive the benefits Universal Credit has replaced but new claims are made for Universal Credit. \r\n\r\n",
  "interviewer_instruction": "CODE ALL THAT APPLY",
  "sequence_index": 2
},

```

```

{
  "name": "BENCTC",
  "label": "Income:  Receives Child Tax Credit",
  "text": "Are you receiving Child Tax Credit ` , either just yourself or jointly {if hysize > 1 & (livesp = 1 or livewith = 1)}`?",
  "question_type": "single_choice",
  "options": [
    {
      "code": "1",
      "label": "Yes",
      "exclusive": false,
      "is_missing": false,
      "category_idenfifier": "13eebd1b-2ba6-4f29-a559-e95d9dec7fbd"
    },
    {
      "code": "2",
      "label": "No",
      "exclusive": false,
      "is_missing": false,
      "category_idenfifier": "1a5d0476-4269-467b-bc14-8c0dfed78e76"
    },
    {
      "code": "-1",
      "label": "Don't know",
      "exclusive": false,
      "is_missing": true,
      "category_idenfifier": "94a95a70-47e8-44d3-b6d6-761a8563dec9"
    },
    {
      "code": "-2",
      "label": "Refused",
      "exclusive": false,
      "is_missing": true,
      "category_idenfifier": "fb95b220-d9f4-4407-92e1-3277a4d04058"
    },
    {
      "code": "-8",
      "label": "Inapplicable",
      "exclusive": false,
      "is_missing": true,
      "category_idenfifier": "2f0c6df3-107e-40a9-84cd-a7e37c5e0830"
    },
    {
      "code": "-9",
      "label": "Missing",
      "exclusive": false,
      "is_missing": true,
      "category_idenfifier": "1d43b6f5-edaa-44e7-93e6-1a36aade019"
    }
  ],
  "help_text": "Child Tax Credit is paid to the person responsible for the care of the child(ren) so can be received by one person only in the household.  It is being replaced by Universal Credit.",
  "interviewer_instruction": "",
  "sequence_index": 3
},
{
  "name": "BENWCA",
  "label": "Work Capability Assessment",
  "text": "Universal Credit can include support for those with health conditions or disabilities that limit the

```

```

work they can do. Have you had a Work Capability Assessment for Universal Credit?",
  "question_type": "single_choice",
  "options": [
    {
      "code": "1",
      "label": "Yes, and I have received a decision outcome",
      "exclusive": false,
      "is_missing": false,
      "category_identifier": "94438e97-0011-4c20-8424-4d442d2f400e"
    },
    {
      "code": "2",
      "label": "Yes, and I am awaiting a decision outcome",
      "exclusive": false,
      "is_missing": false,
      "category_identifier": "67a28cf1-8afb-4329-a190-35e6f998f759"
    },
    {
      "code": "3",
      "label": "No",
      "exclusive": false,
      "is_missing": false,
      "category_identifier": "e450e928-e5b1-4551-9f30-c8c1e4695c49"
    },
    {
      "code": "-1",
      "label": "Don't know",
      "exclusive": false,
      "is_missing": true,
      "category_identifier": "df93b9f7-0f81-44b1-bb42-a68ae122e3c1"
    },
    {
      "code": "-2",
      "label": "Refused",
      "exclusive": false,
      "is_missing": true,
      "category_identifier": "158ab848-b837-406d-9588-130060f4e3df"
    },
    {
      "code": "-8",
      "label": "Inapplicable",
      "exclusive": false,
      "is_missing": true,
      "category_identifier": "b04e36eb-7569-44d7-bf3f-e4ff7cc54012"
    },
    {
      "code": "-9",
      "label": "Missing",
      "exclusive": false,
      "is_missing": true,
      "category_identifier": "96fae899-a4cc-4e46-bf80-dff40b244afb"
    }
  ],
  "help_text": "",
  "interviewer_instruction": "",
  "sequence_index": 4
},
{
  "name": "BENWCAOTC",

```

```

"label": "Work Capability Assessment outcome",
"text": "What was the outcome?",
"question_type": "single_choice",
"options": [
  {
    "code": "1",
    "label": "Fit for work - you will be expected to look for work or to increase your earnings",
    "exclusive": false,
    "is_missing": false,
    "category_idenfifier": "47f0f64c-b493-46ca-9346-de23a17deaf5"
  },
  {
    "code": "2",
    "label": "Limited capability for work - you may not be able to look for work now but you can prepare for
work with the aim of working at some time in the future",
    "exclusive": false,
    "is_missing": false,
    "category_idenfifier": "a222f8b6-8ed1-4407-aa11-d3de58ef9d64"
  },
  {
    "code": "3",
    "label": "Limited capability for work and work-related activity - you will not be asked to look for work,
or to prepare for work",
    "exclusive": false,
    "is_missing": false,
    "category_idenfifier": "21fed2b4-b5ca-4391-a4c4-9ec745bf5133"
  },
  {
    "code": "-1",
    "label": "Don't know",
    "exclusive": false,
    "is_missing": true,
    "category_idenfifier": "df93b9f7-0f81-44b1-bb42-a68ae122e3c1"
  },
  {
    "code": "-2",
    "label": "Refused",
    "exclusive": false,
    "is_missing": true,
    "category_idenfifier": "158ab848-b837-406d-9588-130060f4e3df"
  },
  {
    "code": "-8",
    "label": "Inapplicable",
    "exclusive": false,
    "is_missing": true,
    "category_idenfifier": "b04e36eb-7569-44d7-bf3f-e4ff7cc54012"
  },
  {
    "code": "-9",
    "label": "Missing",
    "exclusive": false,
    "is_missing": true,
    "category_idenfifier": "96fae899-a4cc-4e46-bf80-dff40b244afb"
  }
],
"help_text": "",
"interviewer_instruction": "",
"sequence_index": 5

```

```

},
{
  "name": "BENPEN",
  "label": "Income: Pensions",
  "text": "Are you receiving any of these types of payments ` , either just yourself or jointly {if hhsize > 1 & (livesp = 1 or livewith = 1)}`?",
  "question_type": "multi_choice",
  "options": [
    {
      "code": "1",
      "label": "State Retirement (Old Age) Pension",
      "exclusive": false,
      "is_missing": false,
      "category_idenfifier": "e47c05b4-b5a9-4b41-9f4b-78323ca05442"
    },
    {
      "code": "2",
      "label": "A pension from a previous employer",
      "exclusive": false,
      "is_missing": false,
      "category_idenfifier": "02578ce1-3776-4652-b34d-8fdc23ea65bf"
    },
    {
      "code": "3",
      "label": "A pension from a spouse's previous employer",
      "exclusive": false,
      "is_missing": false,
      "category_idenfifier": "46ffce60-86d5-46fe-b753-ab31d22197fd"
    },
    {
      "code": "4",
      "label": "Pension Credit including Guarantee Credit & Savings Credit",
      "exclusive": false,
      "is_missing": false,
      "category_idenfifier": "6cec64db-a140-4723-9c35-26eae0bcaa66"
    },
    {
      "code": "5",
      "label": "Private Pension or Annuity",
      "exclusive": false,
      "is_missing": false,
      "category_idenfifier": "abeb21c0-9d0f-4bf4-b245-07f3db393b49"
    },
    {
      "code": "6",
      "label": "Widow's or War Widow(er)'s Pension",
      "exclusive": false,
      "is_missing": false,
      "category_idenfifier": "73f52f5d-00d0-4019-aba2-2d79a2c3afb8"
    },
    {
      "code": "7",
      "label": "Widowed Mother's Allowance, Widowed Parent's Allowance, Bereavement Allowance or Bereavement Support Payment",
      "exclusive": false,
      "is_missing": false,
      "category_idenfifier": "b1b0000a-4eae-478d-a599-8ba02b87dea7"
    }
  ],
}

```

```

    "code": "8",
    "label": "War Pension Scheme, War Disablement Pension, Armed Forces Compensation Scheme",
    "exclusive": false,
    "is_missing": false,
    "category_idenfier": "2592d45c-1051-4a5c-9373-bb69fcd7332a"
  },
  {
    "code": "96",
    "label": "None of these",
    "exclusive": true,
    "is_missing": false,
    "category_idenfier": "9584075c-60ab-44c0-b5ac-fab085d80def"
  },
  {
    "code": "-1",
    "label": "Don't know",
    "exclusive": false,
    "is_missing": true,
    "category_idenfier": "df93b9f7-0f81-44b1-bb42-a68ae122e3c1"
  },
  {
    "code": "-2",
    "label": "Refused",
    "exclusive": false,
    "is_missing": true,
    "category_idenfier": "158ab848-b837-406d-9588-130060f4e3df"
  },
  {
    "code": "-8",
    "label": "Inapplicable",
    "exclusive": false,
    "is_missing": true,
    "category_idenfier": "b04e36eb-7569-44d7-bf3f-e4ff7cc54012"
  },
  {
    "code": "-9",
    "label": "Missing",
    "exclusive": false,
    "is_missing": true,
    "category_idenfier": "96fae899-a4cc-4e46-bf80-dff40b244afb"
  }
],
  "help_text": "
  A **pension from a previous employer** should include any pension income currently received
  from any employer's scheme, regardless of whether you are currently working or not.\r\n\r\n
  A **Pension from a spouse's previous employer** may be received if your spouse has died. \r\n\r\n
  **Pension Credit** is a means tested benefit paid if you are of State Pension age you will not receive it unless you have applied for it.
  \r\n\r\n
  A **Widow's/War Widow(er)'s Pension**\r\n
  **Widow\u2019s Pension** was replaced by Bereavement Allowance for new claimants on or after 9 April 2001. If you received Widow\u2019s Pension before 9 April 2001 you would
  continue to receive it. **Bereavement Support Payment** replaced Bereavement Allowance, for deaths **on or after 6 April 2017**.\r\n
  **War Widow(er)\u2019s Pension** is payable to widow/ers and children of someone killed in the Armed Forces or who died as a result of injury sustained in the Armed Forces
  **before 6 April 2005**. The Armed Forces Compensation Scheme compensates for any injury, illness or death which was caused by service **on or after 6 April 2005**.
  Do **not** include Widow's Benefit (which is a single lump sum payment) as a Widow\u2019s Pension.\r\n\r\n
  **Bereavement Allowance** replaced Widow\u2019s Pension **from 9 April 2001**, if you received Widow\u2019s Pension before that date you would continue to receive it.
  Bereavement Support Payment replaced Bereavement Allowance, Bereavement Payment and Widowed Parent\u2019s Allowance for deaths **on or after 6 April 2017**.\r\n\r\n
  **Bereavement Support Payment** replaced Bereavement Allowance, Bereavement Payment and Widowed Parent\u2019s Allowance for deaths **on or after 6 April 2017**.\r\n\r\n
  A **War Disablement Pension** is payable if you were injured or disabled while serving in the Merchant Navy, the Coastguard Service, the Nursing Service and

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the Home Guard, or were serving in the Polish forces under British command during the Second World war, or serving
as a civil defence volunteer, or you were a prisoner of war. Civilians injured as a result of enemy action in the
Second World War are also eligible. \r\nThe **War Pension Scheme** (WPS) compensates for any injury, illness or
death which was caused by service **before 6 April 2005**. \r\nThe **Armed Forces Compensation Scheme**
compensates for any injury, illness or death which was caused by service **on or after 6 April 2005**.\r\n\r\n",
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Disability Living Allowance [if REGION=3]`",
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In Scotland, Pension Age Disability Payment will replace Attendance Allowance by 22 April 2025 (having been piloted in 5 areas from 21 October 2024 and extended to more local authority areas by 24th March 2025).\r\n\r\n\u2022 Carer's Allowance is for people aged 16 or over who give regular and substantial care (for 35+ hrs per week) to a severely disabled person who gets one of: Disability Living Allowance care component at the middle or highest rate, Personal Independence Payment daily living component at either rate, Attendance Allowance at either rate, Constant Attendance Allowance at or above the normal maximum rate with an Industrial Injuries Disablement Benefit, Constant Attendance Allowance at the basic rate with a War Disablement Pension or Armed Forces Independence Payment. \r\n\r\nIf you are in paid work, a limit is set on the amount you can earn a week after tax, National Insurance and expenses and receive Carer's Allowance. If you are in full-time education or studying for 21 hours a week or more you cannot get Carer's Allowance. \r\n\r\nIf you get State Pension and/or Pension Credit you may not be eligible to receive Carer's Allowance it will depend on the amount of State Pension you receive each week.\r\n\r\nIf you live in Scotland and get Carer's Allowance, you may also get Carer's Allowance Supplement as a top-up payment. It is paid twice a year and is paid for by Social Security Scotland. By Spring 2025 it is expected that Carer Support Payment will replace Carer's Allowance in Scotland.\r\n\r\n\r\n\u2022 Carer Support Payment (CSP) replaces Carer's Allowance in Scotland. In a phased approach, from February 2024, in Scotland claimants of Carer's Allowance transfer to Carer Support Payment, with the move to CSP across Scotland completed in Spring 2025. You can receive CSP if you are aged 16 or over and usually live in Scotland and provide care for someone (for 35 hours or more a week). If you are in paid work, a limit is set on the amount you can earn a week after tax, National Insurance and expenses and receive CSP.\r\n\r\n\r\n\u2022 Disability Living Allowance (DLA) might be received by some people aged under State Pension age if help is needed with personal care and/or with getting around. There are two components: Care component or Mobility Component. Since 2013, DLA has been replaced by other benefits for adults. \r\n\r\nFrom age 16 but under State Pension age: Disability Living Allowance has been replaced by Personal Independence Payment (PIP) for people living in England or Wales aged from 16 but under State Pension age. If you live in Scotland and have not reached State Pension age you can get Adult Disability Payment instead of PIP. State Pension age: If you are of State Pension age and do not get DLA you can claim Attendance Allowance.\r\n\r\nChildren aged under 16: New claims for Disability Living Allowance are only accepted for children aged under 16 who live in England or Wales. Children aged under 16 who live in Scotland can apply for Child Disability Payment instead of DLA.\r\n\r\n\r\n\u2022 Employment and Support Allowance is for people with a disability or health condition that affects how much you can work. You can apply for New Style Employment and Support Allowance if you're under State Pension age and you have a disability or health condition that affects how much you can work. You also need to have worked either as an employee or have been self-employed and paid enough National Insurance contributions to be eligible to receive it. You can apply whether you're in or out of work. You cannot get New Style ESA if you claim Jobseeker's Allowance or get Statutory Sick Pay. \r\n\r\n\r\n\u2022 Incapacity Benefit is paid if you could not work because of an illness or disability before 31 January 2011 \u2013 it is being replaced with Employment and Support Allowance (ESA). Incapacity Benefit replaced NI Sickness Benefit and Invalidity Benefit in April 1995. \r\n\r\n\r\n\u2022 Personal Independence Payment (PIP) started to replace Disability Living Allowance (DLA) from 8 April 2013 for people aged 16 or over but under State Pension age with a long-term physical or mental health condition or disability who have difficulty doing certain everyday tasks or getting around because of your condition. There are two parts to PIP: a daily living part \u2013 if you need help with everyday tasks and a mobility part \u2013 if you need help with getting around (including not just a physical disability but difficulty getting around because of a cognitive or mental health condition).\r\n\r\nIf you are over State Pension age and have received PIP, you can make a new claim for PIP if you were eligible for it in the year before you reached State Pension age. Otherwise, if you are over State Pension age you can get Attendance Allowance instead of PIP.\r\n\r\nIf you live in Scotland, you can receive Adult Disability Payment instead of PIP.\r\n\r\n\r\n\u2022 Scottish Adult Disability Living Allowance: Adults who live in Scotland and currently receive Disability Living Allowance, after Spring 2025, might be moved onto Scottish Adult Disability Living Allowance. The payments of Scottish Adult Disability Living Allowance will be paid by Social Security Scotland rather than the Department for Work and Pensions. You might then change to receive Adult Disability Payment which replaces DLA and Personal Independence Payment in Scotland. No new claims for Scottish Adult Disability Living Allowance can be made.\r\n\r\n\r\n\u2022 Severe Disablement Allowance is being replaced with Employment and Support Allowance (ESA). Severe Disablement Allowance is paid to people of working age who were too unwell to work but no new claims for

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this benefit can be made. \r\n\r\n\u2022 **Sickness and Accident Insurance** includes any payments to compensate
for lost wages during time off work through sickness. \r\n",
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"help_text": "\u2022 Council Tax Reduction\u2022, sometimes called Council Tax Support and formerly Council Tax Benefit is paid by the local authority to the people who are liable for Council Tax (usually the householder(s). In multi-occupied accommodation (e.g. bedsits) the landlord is usually liable for Council Tax and so the council tax might be included in the rent. In these circumstances, the tenant will not be able to get the Council Tax Reduction. Local authorities are able to set their own eligibility criteria for Council Tax reduction. \r\n\r\nCouncil Tax reduction should not be confused with CT exemptions (e.g. for student households), or discounts (e.g. for one-adult households). \r\n\r\nRent rebates to help pay rates might apply if you live in Northern Ireland and are getting Universal Credit.\r\n\r\n\r\n\u2022 Housing Benefit\u2022 is benefit paid to help with housing costs, either by the DWP or the Local Authority if you\u2019re on a low income or claiming benefits. For working age households, it is being replaced by Universal Credit (for new claims). People of State Pension age can still make new claims for Housing Benefit. In most cases Housing Benefit is paid directly to you but sometimes the benefit might be paid directly to your landlord. Only tick that you get Housing Benefit at this question if it is paid directly to you.\r\n\r\n\u2022If you are living in Northern Ireland\u2022 and you or your partner are of State Pension age you can apply

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for Housing Benefit and Rate Relief (as a tenant or home owner) from the Housing Executive.\r\n\r\n\u2022**
Maternity Allowance** applies only to women not eligible for Statutory Maternity Pay. Usually women receiving
Maternity Allowance will be either self-employed or will have recently changed jobs. \r\n\r\n\u2022 **Working Tax
Credit** is a payment from HM Revenue and Customs (HMRC) which tops up the earnings of working people on low
incomes (it replaced Working Families Tax Credit and Disabled Person's Tax Credit in April 2003). The amount you
can receive is based both on the number of hours you work \u2013 which must be over 16 per week \u2013 and your
income (joint income for couples). Working Tax Credit has been replaced by Universal Credit for most people. New
claims for Working Tax Credit can only be made if you already get Child Tax Credit.\r\n\r\n\r\n\r\n\r\n\r\n\r\n\r\n\r\n\r\n",
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these types of payments ` , either just yourself or jointly {if hhsize > 1 & (livesp = 1 or livewith = 1)}`?",
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    },
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financial institutions. They cover tuition fees and living costs whilst at University, and are repaid once the
borrower starts earning over a certain amount.\r\n\r\n**Educational Grants and Scholarships** may be provided by
Local Authorities, the Department for Education, Research Councils, charities, prospective employers (e.g.
companies or the Armed Forces), educational institutions, family trusts and a number of other bodies such as Trade
Unions.\r\n\r\nFor **Trade Union/ Friendly Society Payments** include all payments from such bodies here with the
exception of Educational Grants and Sickness or Accident Insurance. Please include Strike Pay.\r\n\r\n**Maintenance

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or Alimony** can be received by men or women. In most cases such payments come direct but they may also come via a solicitor, a court or the DWP. If it is paid by the DWP, please do not include any Income Support already mentioned.\r\n\r\n**Payments from a family member** not living with you should not include payments for 'keep' from those living with you.",

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        "category_identifier": "9198c9d6-ac6e-4500-be16-60db38d954f5"
      }
    ],
    "help_text": "",
    "interviewer_instruction": "",
    "sequence_index": 11
  }

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    },
    {
      "name": "NFH",
      "label": "Further income sources",
      "text": "Can I just check, according to our records, you have in the past received `[ff_bentype01] - [ff_bentype47]`. Are you currently receiving `[ff_bentype01] - [ff_bentype47]` `either just yourself or jointly {if HHGRID.HHSIZE > 1 & (HHGRID.LIVESP = 1 or HHGRID.LIVewith = 1)}`?",
      "question_type": "single_choice",
      "options": [
        {
          "code": "1",
          "label": "Yes",
          "exclusive": false,
          "is_missing": false,
          "category_idenfier": "902ac320-5dcf-42b4-aa2b-56154b317aa6"
        },
        {
          "code": "2",
          "label": "No",
          "exclusive": false,
          "is_missing": false,
          "category_idenfier": "0691cab3-12db-4189-a531-19f6c3748c0f"
        },
        {
          "code": "-1",
          "label": "Don't know",
          "exclusive": false,
          "is_missing": true,
          "category_idenfier": "a20ecc9a-0168-402f-8623-24bee6498584"
        },
        {
          "code": "-2",
          "label": "Refused",
          "exclusive": false,
          "is_missing": true,
          "category_idenfier": "c9adb251-5595-436a-b601-fe74cd1e58b6"
        },
        {
          "code": "-8",
          "label": "Inapplicable",
          "exclusive": false,
          "is_missing": true,
          "category_idenfier": "fccc7c70-05ab-4aaa-b6d7-70a0c2f80c6e"
        },
        {
          "code": "-9",
          "label": "Missing",
          "exclusive": false,
          "is_missing": true,
          "category_idenfier": "6d6d95f3-f287-4bf1-8580-8190d162164b"
        }
      ],
      "help_text": "",
      "interviewer_instruction": "",
      "sequence_index": 12
    },
    {
      "name": "FICODE",
      "label": "Income receipt code",

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```

"text": "",
"question_type": "single_choice",
"options": [
  {
    "code": "1",
    "label": "NI Retirement/State Retirement (Old Age) Pension",
    "exclusive": false,
    "is_missing": false,
    "category_idenfifier": "4ade0afa-0280-49f7-9055-4b3032566c2f"
  },
  {
    "code": "2",
    "label": "A Pension from a previous employer",
    "exclusive": false,
    "is_missing": false,
    "category_idenfifier": "7cbf0468-250e-4e9a-b723-0a00197f7f7b"
  },
  {
    "code": "3",
    "label": "A Pension from a spouse's previous employer",
    "exclusive": false,
    "is_missing": false,
    "category_idenfifier": "41452729-a535-4d3f-86f1-8be3485c9845"
  },
  {
    "code": "4",
    "label": "A Private Pension/Annuity",
    "exclusive": false,
    "is_missing": false,
    "category_idenfifier": "88ff7317-37ac-40b6-af2a-fc81d6984070"
  },
  {
    "code": "5",
    "label": "A Widow's or War Widow's Pension",
    "exclusive": false,
    "is_missing": false,
    "category_idenfifier": "99ef6cf1-786b-4d6f-9ea8-a686517b0301"
  },
  {
    "code": "6",
    "label": "A Widowed Mother's Allowance / Widowed Parent's Allowance / Bereavement Allowance",
    "exclusive": false,
    "is_missing": false,
    "category_idenfifier": "93a20cc1-b047-4209-8700-6e610b38ec61"
  },
  {
    "code": "7",
    "label": "Pension Credit (includes Guarantee Credit & Saving Credit)",
    "exclusive": false,
    "is_missing": false,
    "category_idenfifier": "ca837629-7b22-4e31-8812-274a1c58bcb3"
  },
  {
    "code": "8",
    "label": "Severe Disablement Allowance",
    "exclusive": false,
    "is_missing": false,
    "category_idenfifier": "92d4cb92-e608-4b43-9f1a-d9f7dedbb132"
  },

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{
  "code": "9",
  "label": "Industrial Injury Disablement Allowance",
  "exclusive": false,
  "is_missing": false,
  "category_idenfifier": "339c46aa-7566-4357-90ef-e2ecb4ef99e7"
},
{
  "code": "10",
  "label": "`Disability Living Allowance [if REGION =1|2|4]`      `Disability Living Allowance / Scottish
Adult Disability Living Allowance [if REGION=3]`",
  "exclusive": false,
  "is_missing": false,
  "category_idenfifier": "8b3cab5d-3afd-45a8-92a3-81f22f86642c"
},
{
  "code": "11",
  "label": "Attendance Allowance",
  "exclusive": false,
  "is_missing": false,
  "category_idenfifier": "0325f523-5f8c-4e48-8ae0-aa4cd0a954b0"
},
{
  "code": "12",
  "label": "Carer's Allowance (formerly Invalid Care Allowance)",
  "exclusive": false,
  "is_missing": false,
  "category_idenfifier": "4ab029e4-c56c-4d83-9497-69e54825b766"
},
{
  "code": "13",
  "label": "War Disablement Pension",
  "exclusive": false,
  "is_missing": false,
  "category_idenfifier": "a11b2480-de9b-46fe-afde-ace8b88830f9"
},
{
  "code": "14",
  "label": "Incapacity Benefit",
  "exclusive": false,
  "is_missing": false,
  "category_idenfifier": "1a4a4dd4-779d-41e3-877f-cfc452489981"
},
{
  "code": "15",
  "label": "Income Support",
  "exclusive": false,
  "is_missing": false,
  "category_idenfifier": "e6c63f8d-5629-465a-9c98-f9b425a89133"
},
{
  "code": "16",
  "label": "Job Seeker's Allowance",
  "exclusive": false,
  "is_missing": false,
  "category_idenfifier": "d92cb98c-7ded-4112-9a75-971bca199254"
},
{
  "code": "18",

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    "label": "Child Benefit (including Lone-Parent Child Benefit payments)",
    "exclusive": false,
    "is_missing": false,
    "category_idenfifier": "2f5c9d9e-5c3b-43c7-b9ea-2e427d10ff75"
  },
  {
    "code": "19",
    "label": "Child Tax Credit",
    "exclusive": false,
    "is_missing": false,
    "category_idenfifier": "3c5c6363-9f85-46f9-ae74-5cf02c3b3f11"
  },
  {
    "code": "20",
    "label": "Working Tax Credit (includes Disabled Person's Tax Credit)",
    "exclusive": false,
    "is_missing": false,
    "category_idenfifier": "55026e61-6564-49a1-b69b-6b740d9a6037"
  },
  {
    "code": "21",
    "label": "Maternity Allowance",
    "exclusive": false,
    "is_missing": false,
    "category_idenfifier": "9d216045-5d57-4f1b-8596-c2ab4d2e7b16"
  },
  {
    "code": "22",
    "label": "Housing Benefit",
    "exclusive": false,
    "is_missing": false,
    "category_idenfifier": "e4879540-160d-4768-820d-2ce9588d968c"
  },
  {
    "code": "23",
    "label": "Council Tax Benefit",
    "exclusive": false,
    "is_missing": false,
    "category_idenfifier": "280c3b3b-f48d-488b-8f43-471e2c308fe1"
  },
  {
    "code": "24",
    "label": "Educational Grant (not Student Loan or Tuition Fee Loan)",
    "exclusive": false,
    "is_missing": false,
    "category_idenfifier": "b1eb92dc-f4ab-4972-a7c1-0f171acc0464"
  },
  {
    "code": "25",
    "label": "Trade Union / Friendly Society Payment",
    "exclusive": false,
    "is_missing": false,
    "category_idenfifier": "688b8b7d-ab2f-485f-b054-a50d2d7ee245"
  },
  {
    "code": "26",
    "label": "Maintenance or Alimony",
    "exclusive": false,
    "is_missing": false,

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    "category_identifier": "a4497d1a-3563-497a-b17c-91a1b8d122b8"
  },
  {
    "code": "27",
    "label": "Payments from a family member not living here",
    "exclusive": false,
    "is_missing": false,
    "category_identifier": "769a99c0-7292-4470-9777-335e50bc39fe"
  },
  {
    "code": "28",
    "label": "Rent from Boarders or Lodgers (not family members) living here with you",
    "exclusive": false,
    "is_missing": false,
    "category_identifier": "ba8e4805-c83d-40d8-9315-797f7106770d"
  },
  {
    "code": "29",
    "label": "Rent from any other property",
    "exclusive": false,
    "is_missing": false,
    "category_identifier": "b45101a6-8c0c-47f3-8709-3b9341c5c293"
  },
  {
    "code": "30",
    "label": "Foster Allowance / Guardian Allowance",
    "exclusive": false,
    "is_missing": false,
    "category_identifier": "05cc54b8-41f2-4330-96d4-12fef59d7c95"
  },
  {
    "code": "31",
    "label": "Rent Rebate",
    "exclusive": false,
    "is_missing": false,
    "category_identifier": "4ab82894-b59f-4bf7-a0cf-d4273ec052d1"
  },
  {
    "code": "32",
    "label": "Rate Rebate",
    "exclusive": false,
    "is_missing": false,
    "category_identifier": "f0383eaa-d86a-4162-bed1-0d5ddd40aa90"
  },
  {
    "code": "33",
    "label": "Employment and Support Allowance",
    "exclusive": false,
    "is_missing": false,
    "category_identifier": "36a2c51c-7625-4c44-ba5e-dfdd9b608aaf"
  },
  {
    "code": "35",
    "label": "Sickness and Accident Insurance",
    "exclusive": false,
    "is_missing": false,
    "category_identifier": "cd345453-4ad5-45ff-9077-95bae9bc0e61"
  },
  {

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    "code": "37",
    "label": "Other Disability Related Benefit or Payment",
    "exclusive": false,
    "is_missing": false,
    "category_idenfifier": "fabf5893-cd26-40cc-b555-1a5bc7755d8e"
  },
  {
    "code": "38",
    "label": "Any other regular payment",
    "exclusive": false,
    "is_missing": false,
    "category_idenfifier": "7c14fe52-3280-4a39-9d87-49b4f98fcf60"
  },
  {
    "code": "39",
    "label": "Any other state benefit",
    "exclusive": false,
    "is_missing": false,
    "category_idenfifier": "fb1ee672-d6f6-490a-974d-a40aca19769e"
  },
  {
    "code": "40",
    "label": "Universal Credit",
    "exclusive": false,
    "is_missing": false,
    "category_idenfifier": "3cab78d-a099-4df8-9208-f53f3a8271ef"
  },
  {
    "code": "41",
    "label": "Personal Independence Payments",
    "exclusive": false,
    "is_missing": false,
    "category_idenfifier": "2bccf9d7-d426-4e37-ab0a-94238050a61a"
  },
  {
    "code": "42",
    "label": "Student Loan and/or Tuition Fee Loan",
    "exclusive": false,
    "is_missing": false,
    "category_idenfifier": "240a5ea0-a7d8-4037-8a1d-4be401788079"
  },
  {
    "code": "43",
    "label": "Child Disability Payment",
    "exclusive": false,
    "is_missing": false,
    "category_idenfifier": "1d371069-5e67-4887-907e-890dae395d4d"
  },
  {
    "code": "44",
    "label": "Adult Disability Payment",
    "exclusive": false,
    "is_missing": false,
    "category_idenfifier": "f8f6526f-b0e5-4b3b-9a50-540d534ca0c4"
  },
  {
    "code": "45",
    "label": "Pension Age Disability Payment",
    "exclusive": false,

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    "is_missing": false,
    "category_idenfifier": "1a03fd6d-2c21-4633-a40e-99610e836a1b"
  },
  {
    "code": "46",
    "label": "Scottish Child Payment",
    "exclusive": false,
    "is_missing": false,
    "category_idenfifier": "dcd70049-d9b0-4e57-9c2a-3b3155da1114"
  },
  {
    "code": "47",
    "label": "Carer Support Payment",
    "exclusive": false,
    "is_missing": false,
    "category_idenfifier": "c48a6266-e323-42b4-94c5-5a527165d445"
  },
  {
    "code": "98",
    "label": "DON'T KNOW",
    "exclusive": false,
    "is_missing": false,
    "category_idenfifier": "e62f0cc0-e59f-49eb-be4b-1a059c2b220b"
  },
  {
    "code": "99",
    "label": "REFUSED",
    "exclusive": false,
    "is_missing": false,
    "category_idenfifier": "e1d03f0e-93af-476d-a657-ec49dcb01a84"
  },
  {
    "code": "-1",
    "label": "Don't know",
    "exclusive": false,
    "is_missing": true,
    "category_idenfifier": "df93b9f7-0f81-44b1-bb42-a68ae122e3c1"
  },
  {
    "code": "-2",
    "label": "Refused",
    "exclusive": false,
    "is_missing": true,
    "category_idenfifier": "158ab848-b837-406d-9588-130060f4e3df"
  },
  {
    "code": "-8",
    "label": "Inapplicable",
    "exclusive": false,
    "is_missing": true,
    "category_idenfifier": "b04e36eb-7569-44d7-bf3f-e4ff7cc54012"
  },
  {
    "code": "-9",
    "label": "Missing",
    "exclusive": false,
    "is_missing": true,
    "category_idenfifier": "96fae899-a4cc-4e46-bf80-dff40b244afb"
  }
}

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    ],
    "help_text": "",
    "interviewer_instruction": "",
    "sequence_index": 13
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  {
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    "label": "Income receipt sequence number",
    "text": "",
    "question_type": "number",
    "options": [
      {
        "code": "-1",
        "label": "Don't know",
        "exclusive": false,
        "is_missing": true,
        "category_idenfifier": "e16db4e7-f2dc-4a78-a167-3de3a7554381"
      },
      {
        "code": "-2",
        "label": "Refused",
        "exclusive": false,
        "is_missing": true,
        "category_idenfifier": "849fe9ee-0a05-4769-be10-d4e2982a4660"
      },
      {
        "code": "-8",
        "label": "Inapplicable",
        "exclusive": false,
        "is_missing": true,
        "category_idenfifier": "58fcc7c5-5ff1-4be8-8a71-13a99450cac8"
      },
      {
        "code": "-9",
        "label": "Missing",
        "exclusive": false,
        "is_missing": true,
        "category_idenfifier": "74776546-edf5-41fb-a3ca-27ec0436e9f3"
      }
    ],
    "help_text": "",
    "interviewer_instruction": "",
    "sequence_index": 14
  },
  {
    "name": "FRVAL",
    "label": "Last amount received",
    "text": "How much was the last payment of **`[FICODE]`** you received (to nearest \u00a3)?",
    "question_type": "number",
    "options": [
      {
        "code": "-1",
        "label": "Don't know",
        "exclusive": false,
        "is_missing": true,
        "category_idenfifier": "e16db4e7-f2dc-4a78-a167-3de3a7554381"
      },
      {
        "code": "-2",

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        "label": "Refused",
        "exclusive": false,
        "is_missing": true,
        "category_idenfifier": "849fe9ee-0a05-4769-be10-d4e2982a4660"
    },
    {
        "code": "-8",
        "label": "Inapplicable",
        "exclusive": false,
        "is_missing": true,
        "category_idenfifier": "58fcc7c5-5ff1-4be8-8a71-13a99450cac8"
    },
    {
        "code": "-9",
        "label": "Missing",
        "exclusive": false,
        "is_missing": true,
        "category_idenfifier": "74776546-edf5-41fb-a3ca-27ec0436e9f3"
    }
],
"help_text": "",
"interviewer_instruction": "IF 'DON'T KNOW/CAN'T REMEMBER' PROBE FOR APPROXIMATE AMOUNT\r\nIF RESPONDENT CANNOT SEPARATE OUT THE AMOUNTS FOR A SPECIFIC STATE BENEFIT, RECORD TOTAL AMOUNT FOR ALL COMBINED STATE BENEFITS THEN INDICATE AT SUBSEQUENT STATE BENEFITS THAT THE SPECIFIC AMOUNT HAS BEEN INCLUDED AS PART OF A PAYMENT ALREADY RECORDED BY ENTERING A VALUE OF \u00a30.",
"sequence_index": 15
},
{
    "name": "FRWC",
    "label": "Period covered by last amount received",
    "text": "What period did that cover?",
    "question_type": "single_choice",
    "options": [
        {
            "code": "1",
            "label": "One week",
            "exclusive": false,
            "is_missing": false,
            "category_idenfifier": "876f6757-e77d-4e04-a2fc-d46443ad71cf"
        },
        {
            "code": "2",
            "label": "Two weeks",
            "exclusive": false,
            "is_missing": false,
            "category_idenfifier": "0e3f2ad3-7aa6-46a8-81b9-9625be2dd7fa"
        },
        {
            "code": "3",
            "label": "Three weeks",
            "exclusive": false,
            "is_missing": false,
            "category_idenfifier": "4e1269a6-bb5c-443a-a8b8-b46618c2bbea"
        },
        {
            "code": "4",
            "label": "Four weeks",
            "exclusive": false,
            "is_missing": false,

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    "category_idenfifier": "b050fdfd-90d6-4d5d-b11b-6ad6744909bf"
  },
  {
    "code": "5",
    "label": "Calendar month",
    "exclusive": false,
    "is_missing": false,
    "category_idenfifier": "6158d631-bed7-4328-bac6-368fa1f43cd8"
  },
  {
    "code": "7",
    "label": "Two calendar months",
    "exclusive": false,
    "is_missing": false,
    "category_idenfifier": "bd826194-ffd1-4d33-ad1f-09e8d8182462"
  },
  {
    "code": "8",
    "label": "Eight times a year",
    "exclusive": false,
    "is_missing": false,
    "category_idenfifier": "4282f0e2-29e7-4548-be56-29c8e1ab4845"
  },
  {
    "code": "9",
    "label": "Nine times a year",
    "exclusive": false,
    "is_missing": false,
    "category_idenfifier": "8a76f6c5-fb54-49db-b1e3-b6c8d801c96f"
  },
  {
    "code": "10",
    "label": "Ten times a year",
    "exclusive": false,
    "is_missing": false,
    "category_idenfifier": "64d6b130-6b63-4460-85fd-666f69b8a3a2"
  },
  {
    "code": "13",
    "label": "Three months/13 weeks",
    "exclusive": false,
    "is_missing": false,
    "category_idenfifier": "c7f81ccc-730f-4744-b43d-5e0f3f520096"
  },
  {
    "code": "26",
    "label": "Six months/26 weeks",
    "exclusive": false,
    "is_missing": false,
    "category_idenfifier": "27dba035-c82a-4ef1-90b8-18f307b6e8c8"
  },
  {
    "code": "33",
    "label": "Academic year",
    "exclusive": false,
    "is_missing": false,
    "category_idenfifier": "b2d94cfb-8d2e-4121-b73c-c39f2cba8bb8"
  },
  {

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    "code": "52",
    "label": "One year/12 months/52 weeks",
    "exclusive": false,
    "is_missing": false,
    "category_idenfifier": "4c70d1d6-b2e7-4872-8dce-46b0a28e9f90"
  },
  {
    "code": "90",
    "label": "Less than one week",
    "exclusive": false,
    "is_missing": false,
    "category_idenfifier": "b843d025-6201-4eb4-87d1-ef5ebf1f5639"
  },
  {
    "code": "95",
    "label": "One off/lump sum",
    "exclusive": false,
    "is_missing": false,
    "category_idenfifier": "db89b918-e3a0-4b9d-aa89-b1d3ee54958b"
  },
  {
    "code": "96",
    "label": "None of these",
    "exclusive": true,
    "is_missing": false,
    "category_idenfifier": "afd9c70d-477d-45ca-906d-300f5fb33d26"
  },
  {
    "code": "-1",
    "label": "Don't know",
    "exclusive": false,
    "is_missing": true,
    "category_idenfifier": "df93b9f7-0f81-44b1-bb42-a68ae122e3c1"
  },
  {
    "code": "-2",
    "label": "Refused",
    "exclusive": false,
    "is_missing": true,
    "category_idenfifier": "158ab848-b837-406d-9588-130060f4e3df"
  },
  {
    "code": "-8",
    "label": "Inapplicable",
    "exclusive": false,
    "is_missing": true,
    "category_idenfifier": "b04e36eb-7569-44d7-bf3f-e4ff7cc54012"
  },
  {
    "code": "-9",
    "label": "Missing",
    "exclusive": false,
    "is_missing": true,
    "category_idenfifier": "96fae899-a4cc-4e46-bf80-dff40b244afb"
  }
],
"help_text": "",
"interviewer_instruction": "",
"sequence_index": 16

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},
{
  "name": "FRJT",
  "label": "Last payment: Sole or joint recipient",
  "text": "Do you receive that solely in your name or jointly with someone else?",
  "question_type": "single_choice",
  "options": [
    {
      "code": "1",
      "label": "Sole",
      "exclusive": false,
      "is_missing": false,
      "category_idenfifier": "f3c26d65-de38-49ba-a917-2a513006eadb"
    },
    {
      "code": "2",
      "label": "Joint",
      "exclusive": false,
      "is_missing": false,
      "category_idenfifier": "570599d5-b8d1-42e5-85b8-c4ce064618ed"
    },
    {
      "code": "-1",
      "label": "Don't know",
      "exclusive": false,
      "is_missing": true,
      "category_idenfifier": "df93b9f7-0f81-44b1-bb42-a68ae122e3c1"
    },
    {
      "code": "-2",
      "label": "Refused",
      "exclusive": false,
      "is_missing": true,
      "category_idenfifier": "158ab848-b837-406d-9588-130060f4e3df"
    },
    {
      "code": "-8",
      "label": "Inapplicable",
      "exclusive": false,
      "is_missing": true,
      "category_idenfifier": "b04e36eb-7569-44d7-bf3f-e4ff7cc54012"
    },
    {
      "code": "-9",
      "label": "Missing",
      "exclusive": false,
      "is_missing": true,
      "category_idenfifier": "96fae899-a4cc-4e46-bf80-dff40b244afb"
    }
  ],
  "help_text": "",
  "interviewer_instruction": "",
  "sequence_index": 17
},
{
  "name": "FRJTPN",
  "label": "PNO of joint recipient",
  "text": "With whom do you receive `[FICODE]`?",
  "question_type": "single_choice",

```

```

"options": [
  {
    "code": "0",
    "label": "Not resident here",
    "exclusive": false,
    "is_missing": false,
    "category_idenfifier": "8a164c11-d3eb-4f6b-882b-4c4d5b8766d2"
  },
  {
    "code": "1",
    "label": "[hhgrid.hhgrid.name where hhgrid.hhgrid.pno = 1]",
    "exclusive": false,
    "is_missing": false,
    "category_idenfifier": "2aec2bb5-fef7-442f-9717-48199b57776d"
  },
  {
    "code": "2",
    "label": "[hhgrid.hhgrid.name where hhgrid.hhgrid.pno = 2]",
    "exclusive": false,
    "is_missing": false,
    "category_idenfifier": "de4e5f36-922d-4a81-8aeb-af18d9840434"
  },
  {
    "code": "3",
    "label": "[hhgrid.hhgrid.name where hhgrid.hhgrid.pno = 3]",
    "exclusive": false,
    "is_missing": false,
    "category_idenfifier": "cb03a4ac-5621-4368-8c0f-4b01ca7d0f24"
  },
  {
    "code": "4",
    "label": "[hhgrid.hhgrid.name where hhgrid.hhgrid.pno = 4]",
    "exclusive": false,
    "is_missing": false,
    "category_idenfifier": "58679d39-a80e-420a-b4ba-f0d0bdd053b8"
  },
  {
    "code": "5",
    "label": "[hhgrid.hhgrid.name where hhgrid.hhgrid.pno = 5]",
    "exclusive": false,
    "is_missing": false,
    "category_idenfifier": "bebd2951-49c4-4989-bdac-b4debd09616f"
  },
  {
    "code": "6",
    "label": "[hhgrid.hhgrid.name where hhgrid.hhgrid.pno = 6]",
    "exclusive": false,
    "is_missing": false,
    "category_idenfifier": "c0153341-2c85-4bf1-b9a8-14b96d6a1f1f"
  },
  {
    "code": "7",
    "label": "[hhgrid.hhgrid.name where hhgrid.hhgrid.pno = 7]",
    "exclusive": false,
    "is_missing": false,
    "category_idenfifier": "951ce031-e133-4c6d-b716-5f1a274b7779"
  },
  {
    "code": "8",

```

```

    "label": "[hhgrid.hhgrid.name where hhgrid.hhgrid.pno = 8]",
    "exclusive": false,
    "is_missing": false,
    "category_idenfifier": "bcbfa5f5-ae38-422f-8079-0bcf64802c5b"
  },
  {
    "code": "9",
    "label": "[hhgrid.hhgrid.name where hhgrid.hhgrid.pno = 9]",
    "exclusive": false,
    "is_missing": false,
    "category_idenfifier": "cc2a219e-c391-4484-9b22-a6712be4da24"
  },
  {
    "code": "10",
    "label": "[hhgrid.hhgrid.name where hhgrid.hhgrid.pno = 10]",
    "exclusive": false,
    "is_missing": false,
    "category_idenfifier": "8bc7cd32-0544-4353-87c4-e187f072ce8b"
  },
  {
    "code": "11",
    "label": "[hhgrid.hhgrid.name where hhgrid.hhgrid.pno = 11]",
    "exclusive": false,
    "is_missing": false,
    "category_idenfifier": "9a870d55-ed3f-4d7a-9a64-5f8b59add35d"
  },
  {
    "code": "12",
    "label": "[hhgrid.hhgrid.name where hhgrid.hhgrid.pno = 12]",
    "exclusive": false,
    "is_missing": false,
    "category_idenfifier": "7696b30b-ada5-41a4-b32d-6a9c35291c8f"
  },
  {
    "code": "13",
    "label": "[hhgrid.hhgrid.name where hhgrid.hhgrid.pno = 13]",
    "exclusive": false,
    "is_missing": false,
    "category_idenfifier": "e75bb323-8e30-4323-b27c-89be54e3cab0"
  },
  {
    "code": "14",
    "label": "[hhgrid.hhgrid.name where hhgrid.hhgrid.pno = 14]",
    "exclusive": false,
    "is_missing": false,
    "category_idenfifier": "3c5a96c2-a109-4954-ae50-a271f51e032b"
  },
  {
    "code": "15",
    "label": "[hhgrid.hhgrid.name where hhgrid.hhgrid.pno = 15]",
    "exclusive": false,
    "is_missing": false,
    "category_idenfifier": "2f6b6ea5-537c-4fee-8220-428d2f6259c6"
  },
  {
    "code": "16",
    "label": "[hhgrid.hhgrid.name where hhgrid.hhgrid.pno = 16]",
    "exclusive": false,
    "is_missing": false,

```

```

    "category_identifier": "337e4800-ffa9-42f3-8228-f0b188bbc59b"
  },
  {
    "code": "-1",
    "label": "Don't know",
    "exclusive": false,
    "is_missing": true,
    "category_identifier": "df93b9f7-0f81-44b1-bb42-a68ae122e3c1"
  },
  {
    "code": "-2",
    "label": "Refused",
    "exclusive": false,
    "is_missing": true,
    "category_identifier": "15ebf48f-b0a7-4217-b73f-cd1326d9c0da"
  },
  {
    "code": "-8",
    "label": "Inapplicable",
    "exclusive": false,
    "is_missing": true,
    "category_identifier": "0a12b02e-b9a1-4dbf-9446-fe114fba84ad"
  },
  {
    "code": "-9",
    "label": "Missing",
    "exclusive": false,
    "is_missing": true,
    "category_identifier": "96fae899-a4cc-4e46-bf80-dff40b244afb"
  }
],
"help_text": "",
"interviewer_instruction": "IF PERSON NOT IN HOUSEHOLD SELECT \"NON-HOUSEHOLD MEMBER\"",
"sequence_index": 18
},
{
  "name": "FITAX",
  "label": "Pre/post tax income",
  "text": "Was this amount received before tax was owed or after tax was paid?",
  "question_type": "single_choice",
  "options": [
    {
      "code": "1",
      "label": "Before tax",
      "exclusive": false,
      "is_missing": false,
      "category_identifier": "8ffc2515-fa43-4f79-ae02-64c1ab71ff5b"
    },
    {
      "code": "2",
      "label": "After tax",
      "exclusive": false,
      "is_missing": false,
      "category_identifier": "11986c0e-a4b6-4a1f-91cd-b2a9e4f0c964"
    },
    {
      "code": "-1",
      "label": "Don't know",
      "exclusive": false,

```

```

      "is_missing": true,
      "category_idenfifier": "df93b9f7-0f81-44b1-bb42-a68ae122e3c1"
    },
    {
      "code": "-2",
      "label": "Refused",
      "exclusive": false,
      "is_missing": true,
      "category_idenfifier": "158ab848-b837-406d-9588-130060f4e3df"
    },
    {
      "code": "-8",
      "label": "Inapplicable",
      "exclusive": false,
      "is_missing": true,
      "category_idenfifier": "b04e36eb-7569-44d7-bf3f-e4ff7cc54012"
    },
    {
      "code": "-9",
      "label": "Missing",
      "exclusive": false,
      "is_missing": true,
      "category_idenfifier": "96fae899-a4cc-4e46-bf80-dff40b244afb"
    }
  ],
  "help_text": "",
  "interviewer_instruction": "",
  "sequence_index": 19
},
{
  "name": "FRADDL",
  "label": "Received additional payment from source",
  "text": "And are you currently receiving any other payments of **`[FICODE]`**?",
  "question_type": "single_choice",
  "options": [
    {
      "code": "1",
      "label": "Yes",
      "exclusive": false,
      "is_missing": false,
      "category_idenfifier": "13eebd1b-2ba6-4f29-a559-e95d9dec7fbd"
    },
    {
      "code": "2",
      "label": "No",
      "exclusive": false,
      "is_missing": false,
      "category_idenfifier": "1a5d0476-4269-467b-bc14-8c0dfed78e76"
    },
    {
      "code": "-1",
      "label": "Don't know",
      "exclusive": false,
      "is_missing": true,
      "category_idenfifier": "94a95a70-47e8-44d3-b6d6-761a8563dec9"
    },
    {
      "code": "-2",
      "label": "Refused",

```

```
    "exclusive": false,
    "is_missing": true,
    "category_identifier": "fb95b220-d9f4-4407-92e1-3277a4d04058"
  },
  {
    "code": "-8",
    "label": "Inapplicable",
    "exclusive": false,
    "is_missing": true,
    "category_identifier": "2f0c6df3-107e-40a9-84cd-a7e37c5e0830"
  },
  {
    "code": "-9",
    "label": "Missing",
    "exclusive": false,
    "is_missing": true,
    "category_identifier": "1d43b6f5-edaa-44e7-93e6-1a36aade019"
  }
],
"help_text": "",
"interviewer_instruction": "",
"sequence_index": 20
}
]
```


get_question

Scope: Colectica-only

MCP prompt: /showQuestion

Description

Single question lookup, case-insensitive on name.

Python signature

```
get_question(
    module_id_or_name: str,
    question_name: str
) -> dict
```

Example prompt shortcut

```
/showQuestion
```

"Using the get_question tool, show me the full detail for question 'FICODE' in module 'Benefits W18'."

Example request

```
get_question(module_id_or_name="Benefits W18", question_name="FICODE")
```

Example response (real output)

```
{
  "module_id": "c8171847-7442-41ef-b534-04682184a14f",
  "module_name": "Benefits W18",
  "name": "FICODE",
  "label": "Income receipt code",
  "text": "",
  "question_type": "single_choice",
  "options": [
    {
      "code": "1",
      "label": "NI Retirement/State Retirement (Old Age) Pension",
      "exclusive": false,
      "is_missing": false,
      "category_identifier": "4ade0afa-0280-49f7-9055-4b3032566c2f"
    },
    {
      "code": "2",
      "label": "A Pension from a previous employer",
      "exclusive": false,
      "is_missing": false,
      "category_identifier": "7cbf0468-250e-4e9a-b723-0a00197f7f7b"
    },
    {
      "code": "3",
      "label": "A Pension from a spouse's previous employer",
      "exclusive": false,
      "is_missing": false,
      "category_identifier": "41452729-a535-4d3f-86f1-8be3485c9845"
    },
    {
      "code": "4",
      "label": "A Private Pension/Annuity",
      "exclusive": false,
      "is_missing": false,
      "category_identifier": "88ff7317-37ac-40b6-af2a-fc81d6984070"
    }
  ]
}
```

```

},
{
  "code": "5",
  "label": "A Widow's or War Widow's Pension",
  "exclusive": false,
  "is_missing": false,
  "category_identifier": "99ef6cf1-786b-4d6f-9ea8-a686517b0301"
},
{
  "code": "6",
  "label": "A Widowed Mother's Allowance / Widowed Parent's Allowance / Bereavement Allowance",
  "exclusive": false,
  "is_missing": false,
  "category_identifier": "93a20cc1-b047-4209-8700-6e610b38ec61"
},
{
  "code": "7",
  "label": "Pension Credit (includes Guarantee Credit & Saving Credit)",
  "exclusive": false,
  "is_missing": false,
  "category_identifier": "ca837629-7b22-4e31-8812-274a1c58bcb3"
},
{
  "code": "8",
  "label": "Severe Disablement Allowance",
  "exclusive": false,
  "is_missing": false,
  "category_identifier": "92d4cb92-e608-4b43-9f1a-d9f7dedbb132"
},
{
  "code": "9",
  "label": "Industrial Injury Disablement Allowance",
  "exclusive": false,
  "is_missing": false,
  "category_identifier": "339c46aa-7566-4357-90ef-e2ecb4ef99e7"
},
{
  "code": "10",
  "label": "`Disability Living Allowance [if REGION =1|2|4]`      `Disability Living Allowance / Scottish Adult  
Disability Living Allowance [if REGION=3]`",
  "exclusive": false,
  "is_missing": false,
  "category_identifier": "8b3cab5d-3afd-45a8-92a3-81f22f86642c"
},
{
  "code": "11",
  "label": "Attendance Allowance",
  "exclusive": false,
  "is_missing": false,
  "category_identifier": "0325f523-5f8c-4e48-8ae0-aa4cd0a954b0"
},
{
  "code": "12",
  "label": "Carer's Allowance (formerly Invalid Care Allowance)",
  "exclusive": false,
  "is_missing": false,
  "category_identifier": "4ab029e4-c56c-4d83-9497-69e54825b766"
},
{

```

```

    "code": "13",
    "label": "War Disablement Pension",
    "exclusive": false,
    "is_missing": false,
    "category_identifier": "a11b2480-de9b-46fe-afde-ace8b88830f9"
  },
  {
    "code": "14",
    "label": "Incapacity Benefit",
    "exclusive": false,
    "is_missing": false,
    "category_identifier": "1a4a4dd4-779d-41e3-877f-cfc452489981"
  },
  {
    "code": "15",
    "label": "Income Support",
    "exclusive": false,
    "is_missing": false,
    "category_identifier": "e6c63f8d-5629-465a-9c98-f9b425a89133"
  },
  {
    "code": "16",
    "label": "Job Seeker's Allowance",
    "exclusive": false,
    "is_missing": false,
    "category_identifier": "d92cb98c-7ded-4112-9a75-971bca199254"
  },
  {
    "code": "18",
    "label": "Child Benefit (including Lone-Parent Child Benefit payments)",
    "exclusive": false,
    "is_missing": false,
    "category_identifier": "2f5c9d9e-5c3b-43c7-b9ea-2e427d10ff75"
  },
  {
    "code": "19",
    "label": "Child Tax Credit",
    "exclusive": false,
    "is_missing": false,
    "category_identifier": "3c5c6363-9f85-46f9-ae74-5cf02c3b3f11"
  },
  {
    "code": "20",
    "label": "Working Tax Credit (includes Disabled Person's Tax Credit)",
    "exclusive": false,
    "is_missing": false,
    "category_identifier": "55026e61-6564-49a1-b69b-6b740d9a6037"
  },
  {
    "code": "21",
    "label": "Maternity Allowance",
    "exclusive": false,
    "is_missing": false,
    "category_identifier": "9d216045-5d57-4f1b-8596-c2ab4d2e7b16"
  },
  {
    "code": "22",
    "label": "Housing Benefit",
    "exclusive": false,

```

```

    "is_missing": false,
    "category_identifier": "e4879540-160d-4768-820d-2ce9588d968c"
  },
  {
    "code": "23",
    "label": "Council Tax Benefit",
    "exclusive": false,
    "is_missing": false,
    "category_identifier": "280c3b3b-f48d-488b-8f43-471e2c308fe1"
  },
  {
    "code": "24",
    "label": "Educational Grant (not Student Loan or Tuition Fee Loan)",
    "exclusive": false,
    "is_missing": false,
    "category_identifier": "b1eb92dc-f4ab-4972-a7c1-0f171acc0464"
  },
  {
    "code": "25",
    "label": "Trade Union / Friendly Society Payment",
    "exclusive": false,
    "is_missing": false,
    "category_identifier": "688b8b7d-ab2f-485f-b054-a50d2d7ee245"
  },
  {
    "code": "26",
    "label": "Maintenance or Alimony",
    "exclusive": false,
    "is_missing": false,
    "category_identifier": "a4497d1a-3563-497a-b17c-91a1b8d122b8"
  },
  {
    "code": "27",
    "label": "Payments from a family member not living here",
    "exclusive": false,
    "is_missing": false,
    "category_identifier": "769a99c0-7292-4470-9777-335e50bc39fe"
  },
  {
    "code": "28",
    "label": "Rent from Boarders or Lodgers (not family members) living here with you",
    "exclusive": false,
    "is_missing": false,
    "category_identifier": "ba8e4805-c83d-40d8-9315-797f7106770d"
  },
  {
    "code": "29",
    "label": "Rent from any other property",
    "exclusive": false,
    "is_missing": false,
    "category_identifier": "b45101a6-8c0c-47f3-8709-3b9341c5c293"
  },
  {
    "code": "30",
    "label": "Foster Allowance / Guardian Allowance",
    "exclusive": false,
    "is_missing": false,
    "category_identifier": "05cc54b8-41f2-4330-96d4-12fef59d7c95"
  },

```

```

{
  "code": "31",
  "label": "Rent Rebate",
  "exclusive": false,
  "is_missing": false,
  "category_identifier": "4ab82894-b59f-4bf7-a0cf-d4273ec052d1"
},
{
  "code": "32",
  "label": "Rate Rebate",
  "exclusive": false,
  "is_missing": false,
  "category_identifier": "f0383eaa-d86a-4162-bed1-0d5ddd40aa90"
},
{
  "code": "33",
  "label": "Employment and Support Allowance",
  "exclusive": false,
  "is_missing": false,
  "category_identifier": "36a2c51c-7625-4c44-ba5e-dfdd9b608aaf"
},
{
  "code": "35",
  "label": "Sickness and Accident Insurance",
  "exclusive": false,
  "is_missing": false,
  "category_identifier": "cd345453-4ad5-45ff-9077-95bae9bc0e61"
},
{
  "code": "37",
  "label": "Other Disability Related Benefit or Payment",
  "exclusive": false,
  "is_missing": false,
  "category_identifier": "fabf5893-cd26-40cc-b555-1a5bc7755d8e"
},
{
  "code": "38",
  "label": "Any other regular payment",
  "exclusive": false,
  "is_missing": false,
  "category_identifier": "7c14fe52-3280-4a39-9d87-49b4f98fcf60"
},
{
  "code": "39",
  "label": "Any other state benefit",
  "exclusive": false,
  "is_missing": false,
  "category_identifier": "fb1ee672-d6f6-490a-974d-a40aca19769e"
},
{
  "code": "40",
  "label": "Universal Credit",
  "exclusive": false,
  "is_missing": false,
  "category_identifier": "3cabc78d-a099-4df8-9208-f53f3a8271ef"
},
{
  "code": "41",
  "label": "Personal Independence Payments",

```

```

    "exclusive": false,
    "is_missing": false,
    "category_identifier": "2bccf9d7-d426-4e37-ab0a-94238050a61a"
  },
  {
    "code": "42",
    "label": "Student Loan and/or Tuition Fee Loan",
    "exclusive": false,
    "is_missing": false,
    "category_identifier": "240a5ea0-a7d8-4037-8a1d-4be401788079"
  },
  {
    "code": "43",
    "label": "Child Disability Payment",
    "exclusive": false,
    "is_missing": false,
    "category_identifier": "1d371069-5e67-4887-907e-890dae395d4d"
  },
  {
    "code": "44",
    "label": "Adult Disability Payment",
    "exclusive": false,
    "is_missing": false,
    "category_identifier": "f8f6526f-b0e5-4b3b-9a50-540d534ca0c4"
  },
  {
    "code": "45",
    "label": "Pension Age Disability Payment",
    "exclusive": false,
    "is_missing": false,
    "category_identifier": "1a03fd6d-2c21-4633-a40e-99610e836a1b"
  },
  {
    "code": "46",
    "label": "Scottish Child Payment",
    "exclusive": false,
    "is_missing": false,
    "category_identifier": "dcd70049-d9b0-4e57-9c2a-3b3155da1114"
  },
  {
    "code": "47",
    "label": "Carer Support Payment",
    "exclusive": false,
    "is_missing": false,
    "category_identifier": "c48a6266-e323-42b4-94c5-5a527165d445"
  },
  {
    "code": "98",
    "label": "DON'T KNOW",
    "exclusive": false,
    "is_missing": false,
    "category_identifier": "e62f0cc0-e59f-49eb-be4b-1a059c2b220b"
  },
  {
    "code": "99",
    "label": "REFUSED",
    "exclusive": false,
    "is_missing": false,
    "category_identifier": "e1d03f0e-93af-476d-a657-ec49dcb01a84"
  }

```

```

    },
    {
      "code": "-1",
      "label": "Don't know",
      "exclusive": false,
      "is_missing": true,
      "category_identifier": "df93b9f7-0f81-44b1-bb42-a68ae122e3c1"
    },
    {
      "code": "-2",
      "label": "Refused",
      "exclusive": false,
      "is_missing": true,
      "category_identifier": "158ab848-b837-406d-9588-130060f4e3df"
    },
    {
      "code": "-8",
      "label": "Inapplicable",
      "exclusive": false,
      "is_missing": true,
      "category_identifier": "b04e36eb-7569-44d7-bf3f-e4ff7cc54012"
    },
    {
      "code": "-9",
      "label": "Missing",
      "exclusive": false,
      "is_missing": true,
      "category_identifier": "96fae899-a4cc-4e46-bf80-dff40b244afb"
    }
  ],
  "help_text": "",
  "interviewer_instruction": "",
  "sequence_index": 13
}

```

get_routing_edges

Scope: Colectica-only

MCP prompt: /listRoutingEdges

Description

All RoutingEdge rows for a Colectica module (source, target, condition, type).

Python signature

```
get_routing_edges(  
    module_id_or_name: str  
) -> dict
```

Example prompt shortcut

```
/listRoutingEdges
```

"Using the get_routing_edges tool, list every routing edge in module 'Benefits W18' (source, target, condition, edge type)."

Example request

```
get_routing_edges(module_id_or_name="Benefits W18")
```

Example response (real output)

```
{  
  "module_id": "c8171847-7442-41ef-b534-04682184a14f",  
  "module_name": "Benefits W18",  
  "count": 35,  
  "edges": [  
    {  
      "source_question": "BENBASE",  
      "target_question": "BENCTC",  
      "condition_text": "if [BenBase = 3 & BenBase <> 4]",  
      "edge_type": "conditional",  
      "sequence_index": 1  
    },  
    {  
      "source_question": "BENBASE",  
      "target_question": "BENWCA",  
      "condition_text": "if [BenBase = 4]",  
      "edge_type": "conditional",  
      "sequence_index": 2  
    },  
    {  
      "source_question": "BENWCA",  
      "target_question": "BENWCAOTC",  
      "condition_text": "if [BenWCA = 1]",  
      "edge_type": "conditional",  
      "sequence_index": 3  
    },  
    {  
      "source_question": "BENDIS",  
      "target_question": "BENESA",  
      "condition_text": "if [BenDis = 2]",  
      "edge_type": "conditional",  
      "sequence_index": 4  
    },  
    {  
      "source_question": "FRVAL",
```



```

    "target_question": "FRWC",
    "condition_text": "if [FRVAL > 0]",
    "edge_type": "conditional",
    "sequence_index": 5
  },
  {
    "source_question": "FRJT",
    "target_question": "FRJTPN",
    "condition_text": "if [FRJT = 2]",
    "edge_type": "conditional",
    "sequence_index": 6
  },
  {
    "source_question": "FICCODE, FRVAL",
    "target_question": "FITAX",
    "condition_text": "if [(FICCODE IN 2,3,4) AND (FRVAL >= 0)]",
    "edge_type": "conditional",
    "sequence_index": 7
  },
  {
    "source_question": "FICCODE",
    "target_question": "FRADDL",
    "condition_text": "if [FICCODE IN 2,3,4,24,25,26,27,28,29,30,35,38,39]",
    "edge_type": "conditional",
    "sequence_index": 8
  },
  {
    "source_question": "MISSOURCELoop",
    "target_question": "MISSOURCE",
    "condition_text": "",
    "edge_type": "loop",
    "sequence_index": 9
  },
  {
    "source_question": "NFHLoop",
    "target_question": "NFH",
    "condition_text": "",
    "edge_type": "loop",
    "sequence_index": 10
  },
  {
    "source_question": "FRVALLoop",
    "target_question": "FRVAL",
    "condition_text": "",
    "edge_type": "loop",
    "sequence_index": 11
  },
  {
    "source_question": "FRVALLoop",
    "target_question": "FRWC",
    "condition_text": "",
    "edge_type": "loop",
    "sequence_index": 12
  },
  {
    "source_question": "FRVALLoop",
    "target_question": "FRJT",
    "condition_text": "",
    "edge_type": "loop",

```

```

    "sequence_index": 13
  },
  {
    "source_question": "FRVALLoop",
    "target_question": "FRJTPN",
    "condition_text": "",
    "edge_type": "loop",
    "sequence_index": 14
  },
  {
    "source_question": "FRVALLoop",
    "target_question": "FITAX",
    "condition_text": "",
    "edge_type": "loop",
    "sequence_index": 15
  },
  {
    "source_question": "FRVALLoop",
    "target_question": "FRADDL",
    "condition_text": "",
    "edge_type": "loop",
    "sequence_index": 16
  },
  {
    "source_question": "INTBENEFIT",
    "target_question": "BENBASE",
    "condition_text": "",
    "edge_type": "sequential",
    "sequence_index": 17
  },
  {
    "source_question": "BENBASE",
    "target_question": "BENCTC",
    "condition_text": "",
    "edge_type": "sequential",
    "sequence_index": 18
  },
  {
    "source_question": "BENCTC",
    "target_question": "BENWCA",
    "condition_text": "",
    "edge_type": "sequential",
    "sequence_index": 19
  },
  {
    "source_question": "BENWCA",
    "target_question": "BENWCA0TC",
    "condition_text": "",
    "edge_type": "sequential",
    "sequence_index": 20
  },
  {
    "source_question": "BENWCA0TC",
    "target_question": "BENPEN",
    "condition_text": "",
    "edge_type": "sequential",
    "sequence_index": 21
  },
  {

```

```

    "source_question": "BENPEN",
    "target_question": "BENDIS",
    "condition_text": "",
    "edge_type": "sequential",
    "sequence_index": 22
  },
  {
    "source_question": "BENDIS",
    "target_question": "BENESA",
    "condition_text": "",
    "edge_type": "sequential",
    "sequence_index": 23
  },
  {
    "source_question": "BENESA",
    "target_question": "OTHBEN",
    "condition_text": "",
    "edge_type": "sequential",
    "sequence_index": 24
  },
  {
    "source_question": "OTHBEN",
    "target_question": "BENSTA",
    "condition_text": "",
    "edge_type": "sequential",
    "sequence_index": 25
  },
  {
    "source_question": "BENSTA",
    "target_question": "MISSOURCE",
    "condition_text": "",
    "edge_type": "sequential",
    "sequence_index": 26
  },
  {
    "source_question": "MISSOURCE",
    "target_question": "NFH",
    "condition_text": "",
    "edge_type": "sequential",
    "sequence_index": 27
  },
  {
    "source_question": "NFH",
    "target_question": "FICODE",
    "condition_text": "",
    "edge_type": "sequential",
    "sequence_index": 28
  },
  {
    "source_question": "FICODE",
    "target_question": "FISEQ",
    "condition_text": "",
    "edge_type": "sequential",
    "sequence_index": 29
  },
  {
    "source_question": "FISEQ",
    "target_question": "FRVAL",
    "condition_text": "",

```

```

    "edge_type": "sequential",
    "sequence_index": 30
  },
  {
    "source_question": "FRVAL",
    "target_question": "FRWC",
    "condition_text": "",
    "edge_type": "sequential",
    "sequence_index": 31
  },
  {
    "source_question": "FRWC",
    "target_question": "FRJT",
    "condition_text": "",
    "edge_type": "sequential",
    "sequence_index": 32
  },
  {
    "source_question": "FRJT",
    "target_question": "FRJTPN",
    "condition_text": "",
    "edge_type": "sequential",
    "sequence_index": 33
  },
  {
    "source_question": "FRJTPN",
    "target_question": "FITAX",
    "condition_text": "",
    "edge_type": "sequential",
    "sequence_index": 34
  },
  {
    "source_question": "FITAX",
    "target_question": "FRADDL",
    "condition_text": "",
    "edge_type": "sequential",
    "sequence_index": 35
  }
]
}

```

trace_variable

Scope: Colectica-only

MCP prompt: /traceVariable

Description

Find every RoutingEdge whose condition references this variable, across one Colectica module or all Colectica modules if module_id_or_name is omitted.

Uses the same external-variable rule as agentflow_payload_builder._extract_external_variables: a referenced variable is "external" if it is not the name of any NormalizedQuestion in that module and does not end in "Loop" (loop containers aren't external variables).

Python signature

```
trace_variable(
    variable_name: str,
    module_id_or_name: str | None = None
) -> dict
```

Example prompt shortcut

/traceVariable

"Using the trace_variable tool, find every routing edge referencing variable 'FICODE' across module 'Benefits W18', and tell me which references are external variables."

Example request

```
trace_variable(variable_name="FICODE", module_id_or_name="Benefits W18") # scoped to one module
```

Example response (real output)

```
{
  "variable_name": "FICODE",
  "match_count": 3,
  "matches": [
    {
      "module_id": "c8171847-7442-41ef-b534-04682184a14f",
      "module_name": "Benefits W18",
      "source_question": "FICODE, FRVAL",
      "target_question": "FITAX",
      "condition_text": "if [(FICODE IN 2,3,4) AND (FRVAL >= 0)]",
      "edge_type": "conditional",
      "sequence_index": 7,
      "is_external_variable": false
    },
    {
      "module_id": "c8171847-7442-41ef-b534-04682184a14f",
      "module_name": "Benefits W18",
      "source_question": "FICODE",
      "target_question": "FRADDL",
      "condition_text": "if [FICODE IN 2,3,4,24,25,26,27,28,29,30,35,38,39]",
      "edge_type": "conditional",
      "sequence_index": 8,
      "is_external_variable": false
    },
    {
      "module_id": "c8171847-7442-41ef-b534-04682184a14f",
      "module_name": "Benefits W18",
```

```

    "source_question": "FICODE",
    "target_question": "FISEQ",
    "condition_text": "",
    "edge_type": "sequential",
    "sequence_index": 29,
    "is_external_variable": false
  }
]
}
```

Second example: neighborhood mode

```

trace_variable(variable_name="GRIDVARIABLES.MODETYPE",
module_id_or_name="3a2de4bf-432b-42f2-b74c-2ceffd868e8a") # across the full W18 module -- a genuine
external variable
```

Example response (real output)

```

{
  "variable_name": "GRIDVARIABLES.MODETYPE",
  "match_count": 122,
  "matches": [
    {
      "module_id": "3a2de4bf-432b-42f2-b74c-2ceffd868e8a",
      "module_name": "W18",
      "source_question": "GRIDVARIABLES.ModeType",
      "target_question": "CHKWEBDOBD",
      "condition_text": "if [GRIDVARIABLES.ModeType = 3]",
      "edge_type": "conditional",
      "sequence_index": 1,
      "is_external_variable": true
    },
    {
      "module_id": "3a2de4bf-432b-42f2-b74c-2ceffd868e8a",
      "module_name": "W18",
      "source_question": "GRIDVARIABLES.ModeType",
      "target_question": "CHKWEBDOBM",
      "condition_text": "if [GRIDVARIABLES.ModeType = 3]",
      "edge_type": "conditional",
      "sequence_index": 2,
      "is_external_variable": true
    },
    {
      "module_id": "3a2de4bf-432b-42f2-b74c-2ceffd868e8a",
      "module_name": "W18",
      "source_question": "GRIDVARIABLES.ModeType",
      "target_question": "CHKWEBDOBY",
      "condition_text": "if [GRIDVARIABLES.ModeType = 3]",
      "edge_type": "conditional",
      "sequence_index": 3,
      "is_external_variable": true
    },
    {
      "module_id": "3a2de4bf-432b-42f2-b74c-2ceffd868e8a",
      "module_name": "W18",
      "source_question": "GRIDVARIABLES.MODETYPE",
      "target_question": "HGLIENG",
      "condition_text": "if [GRIDVARIABLES.MODETYPE = 1 | 2]",
      "edge_type": "conditional",
      "sequence_index": 6,
      "is_external_variable": true
    }
  ]
}
```

```
    },
    {
      "module_id": "3a2de4bf-432b-42f2-b74c-2ceffd868e8a",
      "module_name": "W18",
      "source_question": "GRIDVARIABLES.ModeType, PERGRID",
      "target_question": "PTITLE",
      "condition_text": "if [(GRIDVARIABLES.ModeType = 1 | 2) & (PERGRID = 0)]",
      "edge_type": "conditional",
      "sequence_index": 8,
      "is_external_variable": true
    }
  ],
  "_matches_truncated_for_manual": "Showing 5 of 122 real entries here for readability. Calling this tool live returns all 122."
}
```

CROSS-FORMAT TOOLS (COLECTICA + FORSTA+)

get_module_graph

Scope: Colectica + Forsta+**MCP prompt:** /showModuleGraph

Description

Enriched routing graph for a module -- Colectica OR Forsta+, both are supported. Returns the same incoming_routes/outgoing_routes/prerequisite_questions/start_semantics per node, and graph_start_summary overall, that the "View Graph" GUI page shows. Never builds the graph itself (that's a write path) -- raises ToolError if none has been built yet for this module.

Large modules cannot be returned in full: pass question_name to scope the response to that question's neighborhood (depth hops out, default 1, max 3) -- this is the primary way to explore a large module. Omit question_name only on modules with 50 or fewer questions ("mode": "full" in the response); on larger modules, omitting it returns "mode": "summary" (counts + graph_start_summary only, no per-node detail) rather than risk a huge response -- pass question_name in that case.

"mermaid" is Mermaid flowchart syntax (``mermaid fenced code renders as an actual diagram in most MCP clients, including Claude) for whatever is actually being returned -- the full graph in "full" mode, the scoped subgraph in "neighborhood" mode. Always None in "summary" mode, for the same size reason "graph" is None there.

Python signature

```
get_module_graph(
    module_id_or_name: str,
    question_name: str | None = None,
    depth: int = 1
) -> dict
```

Example prompt shortcut

```
/showModuleGraph
```

"Using the get_module_graph tool, show me the routing graph for module 'Benefits W18'. If it's too large to return in full, summarize graph_start_summary and the node/edge counts, then ask me which question to zoom into. If a "mermaid" field is returned, render it as a ``mermaid fenced code block."

Example request

```
get_module_graph(module_id_or_name="Benefits W18") # full mode (20 questions, under the 50-question threshold)
```

Example response (real output)

```
{
  "module_id": "c8171847-7442-41ef-b534-04682184a14f",
  "module_name": "Benefits W18",
  "source_format": "colectica_json",
  "scoped_to_question": null,
  "total_node_count": 23,
  "total_edge_count": 36,
```



```

"mode": "full",
"graph_start_summary": {
  "first_module_question_id": "INTBENEFIT",
  "true_start_question_ids": [
    "INTBENEFIT"
  ],
  "external_entry_prerequisites": [],
  "has_true_unconditional_start": true,
  "has_external_entry_prerequisites": false
},
"graph": {
  "nodes_json": [
    {
      "id": "INTBENEFIT",
      "data": {
        "label": "INTBENEFIT",
        "title": "Benefit module intro",
        "option_count": 5,
        "question_type": "single_choice"
      },
      "type": "question",
      "incoming_routes": [],
      "outgoing_routes": [
        {
          "id": "INTBENEFIT__BENBASE__17",
          "source": "INTBENEFIT",
          "source_parts": [
            "INTBENEFIT"
          ],
          "target": "BENBASE",
          "type": "sequential",
          "label": "sequential",
          "condition_text": "sequential"
        }
      ],
      "prerequisite_questions": [],
      "start_semantics": {
        "is_first_module_question": true,
        "is_true_unconditional_start": true,
        "is_entry_with_external_prerequisites": false,
        "has_incoming_routes": false,
        "has_external_prerequisites": false,
        "has_internal_prerequisites": false,
        "has_conditional_incoming_routes": false,
        "start_kind": "true_unconditional_start",
        "display_label": "True unconditional start",
        "explanation": "This is the first question inside the module and no incoming or external prerequisite routes were detected."
      },
      "dependency_summary": {
        "incoming_route_count": 0,
        "outgoing_route_count": 1,
        "prerequisite_question_count": 0,
        "has_external_prerequisites": false,
        "has_conditional_incoming_routes": false
      }
    },
    {
      "id": "BENBASE",

```

```

"data": {
  "label": "BENBASE",
  "title": "Income: Receives core benefits",
  "option_count": 9,
  "question_type": "multi_choice"
},
"type": "question",
"incoming_routes": [
  {
    "id": "INTBENEFIT__BENBASE__17",
    "source": "INTBENEFIT",
    "source_parts": [
      "INTBENEFIT"
    ],
    "target": "BENBASE",
    "type": "sequential",
    "label": "sequential",
    "condition_text": "sequential"
  }
],
"outgoing_routes": [
  {
    "id": "BENBASE__BENCTC__1",
    "source": "BENBASE",
    "source_parts": [
      "BENBASE"
    ],
    "target": "BENCTC",
    "type": "conditional",
    "label": "if [BenBase = 3 & BenBase <> 4]",
    "condition_text": "if [BenBase = 3 & BenBase <> 4]"
  },
  {
    "id": "BENBASE__BENWCA__2",
    "source": "BENBASE",
    "source_parts": [
      "BENBASE"
    ],
    "target": "BENWCA",
    "type": "conditional",
    "label": "if [BenBase = 4]",
    "condition_text": "if [BenBase = 4]"
  },
  {
    "id": "BENBASE__BENCTC__18",
    "source": "BENBASE",
    "source_parts": [
      "BENBASE"
    ],
    "target": "BENCTC",
    "type": "sequential",
    "label": "sequential",
    "condition_text": "sequential"
  }
],
"prerequisite_questions": [
  {
    "id": "INTBENEFIT",
    "exists_in_graph": true,

```

```

        "is_external": false,
        "type": "question",
        "label": "INTBENEFIT"
    }
],
"start_semantics": {
    "is_first_module_question": false,
    "is_true_unconditional_start": false,
    "is_entry_with_external_prerequisites": false,
    "has_incoming_routes": true,
    "has_external_prerequisites": false,
    "has_internal_prerequisites": true,
    "has_conditional_incoming_routes": false,
    "start_kind": "not_start",
    "display_label": "Not a module start",
    "explanation": "This is not the first question detected inside this module."
},
"dependency_summary": {
    "incoming_route_count": 1,
    "outgoing_route_count": 3,
    "prerequisite_question_count": 1,
    "has_external_prerequisites": false,
    "has_conditional_incoming_routes": false
}
},
{
    "id": "BENCTC",
    "data": {
        "label": "BENCTC",
        "title": "Income:  Receives Child Tax Credit",
        "option_count": 6,
        "question_type": "single_choice"
    },
    "type": "question",
    "incoming_routes": [
        {
            "id": "BENBASE__BENCTC__1",
            "source": "BENBASE",
            "source_parts": [
                "BENBASE"
            ],
            "target": "BENCTC",
            "type": "conditional",
            "label": "if [BenBase = 3 & BenBase <> 4]",
            "condition_text": "if [BenBase = 3 & BenBase <> 4]"
        },
        {
            "id": "BENBASE__BENCTC__18",
            "source": "BENBASE",
            "source_parts": [
                "BENBASE"
            ],
            "target": "BENCTC",
            "type": "sequential",
            "label": "sequential",
            "condition_text": "sequential"
        }
    ],
    "outgoing_routes": [

```

```

    {
      "id": "BENCTC__BENWCA__19",
      "source": "BENCTC",
      "source_parts": [
        "BENCTC"
      ],
      "target": "BENWCA",
      "type": "sequential",
      "label": "sequential",
      "condition_text": "sequential"
    }
  ],
  "prerequisite_questions": [
    {
      "id": "BENBASE",
      "exists_in_graph": true,
      "is_external": false,
      "type": "question",
      "label": "BENBASE"
    }
  ],
  "start_semantics": {
    "is_first_module_question": false,
    "is_true_unconditional_start": false,
    "is_entry_with_external_prerequisites": false,
    "has_incoming_routes": true,
    "has_external_prerequisites": false,
    "has_internal_prerequisites": true,
    "has_conditional_incoming_routes": true,
    "start_kind": "not_start",
    "display_label": "Not a module start",
    "explanation": "This is not the first question detected inside this module."
  },
  "dependency_summary": {
    "incoming_route_count": 2,
    "outgoing_route_count": 1,
    "prerequisite_question_count": 1,
    "has_external_prerequisites": false,
    "has_conditional_incoming_routes": true
  }
},
{
  "id": "BENWCA",
  "data": {
    "label": "BENWCA",
    "title": "Work Capability Assessment",
    "option_count": 7,
    "question_type": "single_choice"
  },
  "type": "question",
  "incoming_routes": [
    {
      "id": "BENBASE__BENWCA__2",
      "source": "BENBASE",
      "source_parts": [
        "BENBASE"
      ],
      "target": "BENWCA",
      "type": "conditional",

```

```

        "label": "if [BenBase = 4]",
        "condition_text": "if [BenBase = 4]"
    },
    {
        "id": "BENCTC__BENWCA__19",
        "source": "BENCTC",
        "source_parts": [
            "BENCTC"
        ],
        "target": "BENWCA",
        "type": "sequential",
        "label": "sequential",
        "condition_text": "sequential"
    }
],
"outgoing_routes": [
    {
        "id": "BENWCA__BENWCAOTC__3",
        "source": "BENWCA",
        "source_parts": [
            "BENWCA"
        ],
        "target": "BENWCAOTC",
        "type": "conditional",
        "label": "if [BenWCA = 1]",
        "condition_text": "if [BenWCA = 1]"
    },
    {
        "id": "BENWCA__BENWCAOTC__20",
        "source": "BENWCA",
        "source_parts": [
            "BENWCA"
        ],
        "target": "BENWCAOTC",
        "type": "sequential",
        "label": "sequential",
        "condition_text": "sequential"
    }
],
"prerequisite_questions": [
    {
        "id": "BENBASE",
        "exists_in_graph": true,
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        "exists_in_graph": true,
        "is_external": false,
        "type": "question",
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    "is_entry_with_external_prerequisites": false,

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    "outgoing_route_count": 2,
    "prerequisite_question_count": 2,
    "has_external_prerequisites": false,
    "has_conditional_incoming_routes": true
  }
},
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  "id": "BENWCA0TC",
  "data": {
    "label": "BENWCA0TC",
    "title": "Work Capability Assessment outcome",
    "option_count": 7,
    "question_type": "single_choice"
  },
  "type": "question",
  "incoming_routes": [
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      "source": "BENWCA",
      "source_parts": [
        "BENWCA"
      ],
      "target": "BENWCA0TC",
      "type": "conditional",
      "label": "if [BenWCA = 1]",
      "condition_text": "if [BenWCA = 1]"
    },
    {
      "id": "BENWCA__BENWCA0TC__20",
      "source": "BENWCA",
      "source_parts": [
        "BENWCA"
      ],
      "target": "BENWCA0TC",
      "type": "sequential",
      "label": "sequential",
      "condition_text": "sequential"
    }
  ],
  "outgoing_routes": [
    {
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      "source": "BENWCA0TC",
      "source_parts": [
        "BENWCA0TC"
      ],
      "target": "BENPEN",
      "type": "sequential",
      "label": "sequential",

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        "condition_text": "sequential"
    }
],
"prerequisite_questions": [
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        "type": "question",
        "label": "BENWCA"
    }
],
"start_semantics": {
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    "is_true_unconditional_start": false,
    "is_entry_with_external_prerequisites": false,
    "has_incoming_routes": true,
    "has_external_prerequisites": false,
    "has_internal_prerequisites": true,
    "has_conditional_incoming_routes": true,
    "start_kind": "not_start",
    "display_label": "Not a module start",
    "explanation": "This is not the first question detected inside this module."
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    "outgoing_route_count": 1,
    "prerequisite_question_count": 1,
    "has_external_prerequisites": false,
    "has_conditional_incoming_routes": true
}
},
{
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        "label": "BENPEN",
        "title": "Income: Pensions",
        "option_count": 13,
        "question_type": "multi_choice"
    },
    "type": "question",
    "incoming_routes": [
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            "source": "BENWCA0TC",
            "source_parts": [
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            ],
            "target": "BENPEN",
            "type": "sequential",
            "label": "sequential",
            "condition_text": "sequential"
        }
    ],
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            "id": "BENPEN__BENDIS__22",
            "source": "BENPEN",
            "source_parts": [

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        "BENPEN"
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      "type": "sequential",
      "label": "sequential",
      "condition_text": "sequential"
    }
  ],
  "prerequisite_questions": [
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      "exists_in_graph": true,
      "is_external": false,
      "type": "question",
      "label": "BENWCAOTC"
    }
  ],
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    "is_true_unconditional_start": false,
    "is_entry_with_external_prerequisites": false,
    "has_incoming_routes": true,
    "has_external_prerequisites": false,
    "has_internal_prerequisites": true,
    "has_conditional_incoming_routes": false,
    "start_kind": "not_start",
    "display_label": "Not a module start",
    "explanation": "This is not the first question detected inside this module."
  },
  "dependency_summary": {
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    "outgoing_route_count": 1,
    "prerequisite_question_count": 1,
    "has_external_prerequisites": false,
    "has_conditional_incoming_routes": false
  }
},
{
  "id": "BENDIS",
  "data": {
    "label": "BENDIS",
    "title": "Income: Disability benefits",
    "option_count": 15,
    "question_type": "multi_choice"
  },
  "type": "question",
  "incoming_routes": [
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      "id": "BENPEN__BENDIS__22",
      "source": "BENPEN",
      "source_parts": [
        "BENPEN"
      ],
      "target": "BENDIS",
      "type": "sequential",
      "label": "sequential",
      "condition_text": "sequential"
    }
  ]
},

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    "source": "BENDIS",
    "source_parts": [
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    "target": "BENESA",
    "type": "conditional",
    "label": "if [BenDis = 2]",
    "condition_text": "if [BenDis = 2]"
  },
  {
    "id": "BENDIS__BENESA__23",
    "source": "BENDIS",
    "source_parts": [
      "BENDIS"
    ],
    "target": "BENESA",
    "type": "sequential",
    "label": "sequential",
    "condition_text": "sequential"
  }
],
"prerequisite_questions": [
  {
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    "exists_in_graph": true,
    "is_external": false,
    "type": "question",
    "label": "BENPEN"
  }
],
"start_semantics": {
  "is_first_module_question": false,
  "is_true_unconditional_start": false,
  "is_entry_with_external_prerequisites": false,
  "has_incoming_routes": true,
  "has_external_prerequisites": false,
  "has_internal_prerequisites": true,
  "has_conditional_incoming_routes": false,
  "start_kind": "not_start",
  "display_label": "Not a module start",
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  "prerequisite_question_count": 1,
  "has_external_prerequisites": false,
  "has_conditional_incoming_routes": false
}
},
{
  "id": "BENESA",
  "data": {
    "label": "BENESA",
    "title": "Employment and Support Allowance: Work-Related Activity Group or Support Group",
    "option_count": 6,

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    "question_type": "single_choice"
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  "type": "question",
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      "source": "BENDIS",
      "source_parts": [
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      "target": "BENESA",
      "type": "conditional",
      "label": "if [BenDis = 2]",
      "condition_text": "if [BenDis = 2]"
    },
    {
      "id": "BENDIS__BENESA__23",
      "source": "BENDIS",
      "source_parts": [
        "BENDIS"
      ],
      "target": "BENESA",
      "type": "sequential",
      "label": "sequential",
      "condition_text": "sequential"
    }
  ],
  "outgoing_routes": [
    {
      "id": "BENESA__OTHBEN__24",
      "source": "BENESA",
      "source_parts": [
        "BENESA"
      ],
      "target": "OTHBEN",
      "type": "sequential",
      "label": "sequential",
      "condition_text": "sequential"
    }
  ],
  "prerequisite_questions": [
    {
      "id": "BENDIS",
      "exists_in_graph": true,
      "is_external": false,
      "type": "question",
      "label": "BENDIS"
    }
  ],
  "start_semantics": {
    "is_first_module_question": false,
    "is_true_unconditional_start": false,
    "is_entry_with_external_prerequisites": false,
    "has_incoming_routes": true,
    "has_external_prerequisites": false,
    "has_internal_prerequisites": true,
    "has_conditional_incoming_routes": true,
    "start_kind": "not_start",
    "display_label": "Not a module start",

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    "explanation": "This is not the first question detected inside this module."
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    "prerequisite_question_count": 1,
    "has_external_prerequisites": false,
    "has_conditional_incoming_routes": true
  }
},
{
  "id": "OTHBEN",
  "data": {
    "label": "OTHBEN",
    "title": "Other benefits or credits",
    "option_count": 14,
    "question_type": "multi_choice"
  },
  "type": "question",
  "incoming_routes": [
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      "id": "BENESA__OTHBEN__24",
      "source": "BENESA",
      "source_parts": [
        "BENESA"
      ],
      "target": "OTHBEN",
      "type": "sequential",
      "label": "sequential",
      "condition_text": "sequential"
    }
  ],
  "outgoing_routes": [
    {
      "id": "OTHBEN__BENSTA__25",
      "source": "OTHBEN",
      "source_parts": [
        "OTHBEN"
      ],
      "target": "BENSTA",
      "type": "sequential",
      "label": "sequential",
      "condition_text": "sequential"
    }
  ],
  "prerequisite_questions": [
    {
      "id": "BENESA",
      "exists_in_graph": true,
      "is_external": false,
      "type": "question",
      "label": "BENESA"
    }
  ],
  "start_semantics": {
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    "is_true_unconditional_start": false,
    "is_entry_with_external_prerequisites": false,
    "has_incoming_routes": true,

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      "has_external_prerequisites": false,
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      "has_conditional_incoming_routes": false,
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      "explanation": "This is not the first question detected inside this module."
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      "outgoing_route_count": 1,
      "prerequisite_question_count": 1,
      "has_external_prerequisites": false,
      "has_conditional_incoming_routes": false
    }
  },
  {
    "id": "BENSTA",
    "data": {
      "label": "BENSTA",
      "title": "Income: Other sources",
      "option_count": 13,
      "question_type": "multi_choice"
    },
    "type": "question",
    "incoming_routes": [
      {
        "id": "OTHBEN__BENSTA__25",
        "source": "OTHBEN",
        "source_parts": [
          "OTHBEN"
        ],
        "target": "BENSTA",
        "type": "sequential",
        "label": "sequential",
        "condition_text": "sequential"
      }
    ],
    "outgoing_routes": [
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        "id": "BENSTA__MISSOURCE__26",
        "source": "BENSTA",
        "source_parts": [
          "BENSTA"
        ],
        "target": "MISSOURCE",
        "type": "sequential",
        "label": "sequential",
        "condition_text": "sequential"
      }
    ],
    "prerequisite_questions": [
      {
        "id": "OTHBEN",
        "exists_in_graph": true,
        "is_external": false,
        "type": "question",
        "label": "OTHBEN"
      }
    ]
  },

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  "has_external_prerequisites": false,
  "has_internal_prerequisites": true,
  "has_conditional_incoming_routes": false,
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  "explanation": "This is not the first question detected inside this module."
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  "prerequisite_question_count": 1,
  "has_external_prerequisites": false,
  "has_conditional_incoming_routes": false
}
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{
  "id": "MISSOURCE",
  "data": {
    "label": "MISSOURCE",
    "title": "Income source enumerated at previous interview but not current interview",
    "option_count": 6,
    "question_type": "single_choice"
  },
  "type": "question",
  "incoming_routes": [
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      "source": "MISSOURCELoop",
      "source_parts": [
        "MISSOURCELoop"
      ],
      "target": "MISSOURCE",
      "type": "loop",
      "label": "loop",
      "condition_text": "loop"
    },
    {
      "id": "BENSTA__MISSOURCE__26",
      "source": "BENSTA",
      "source_parts": [
        "BENSTA"
      ],
      "target": "MISSOURCE",
      "type": "sequential",
      "label": "sequential",
      "condition_text": "sequential"
    }
  ],
  "outgoing_routes": [
    {
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      "source": "MISSOURCE",
      "source_parts": [
        "MISSOURCE"
      ]
    }
  ]
}

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    "label": "sequential",
    "condition_text": "sequential"
  }
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"prerequisite_questions": [
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    "exists_in_graph": true,
    "is_external": true,
    "type": "loop",
    "label": "MISSOURCELoop"
  },
  {
    "id": "BENSTA",
    "exists_in_graph": true,
    "is_external": false,
    "type": "question",
    "label": "BENSTA"
  }
],
"start_semantics": {
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  "is_true_unconditional_start": false,
  "is_entry_with_external_prerequisites": false,
  "has_incoming_routes": true,
  "has_external_prerequisites": true,
  "has_internal_prerequisites": true,
  "has_conditional_incoming_routes": false,
  "start_kind": "not_start",
  "display_label": "Not a module start",
  "explanation": "This is not the first question detected inside this module."
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  "outgoing_route_count": 1,
  "prerequisite_question_count": 2,
  "has_external_prerequisites": true,
  "has_conditional_incoming_routes": false
}
},
{
  "id": "NFH",
  "data": {
    "label": "NFH",
    "title": "Further income sources",
    "option_count": 6,
    "question_type": "single_choice"
  },
  "type": "question",
  "incoming_routes": [
    {
      "id": "NFHLoop__NFH__10",
      "source": "NFHLoop",
      "source_parts": [
        "NFHLoop"
      ]
    }
  ],

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        "target": "NFH",
        "type": "loop",
        "label": "loop",
        "condition_text": "loop"
    },
    {
        "id": "MISSOURCE__NFH__27",
        "source": "MISSOURCE",
        "source_parts": [
            "MISSOURCE"
        ],
        "target": "NFH",
        "type": "sequential",
        "label": "sequential",
        "condition_text": "sequential"
    }
],
"outgoing_routes": [
    {
        "id": "NFH__FICODE__28",
        "source": "NFH",
        "source_parts": [
            "NFH"
        ],
        "target": "FICODE",
        "type": "sequential",
        "label": "sequential",
        "condition_text": "sequential"
    }
],
"prerequisite_questions": [
    {
        "id": "NFHLoop",
        "exists_in_graph": true,
        "is_external": true,
        "type": "loop",
        "label": "NFHLoop"
    },
    {
        "id": "MISSOURCE",
        "exists_in_graph": true,
        "is_external": false,
        "type": "question",
        "label": "MISSOURCE"
    }
],
"start_semantics": {
    "is_first_module_question": false,
    "is_true_unconditional_start": false,
    "is_entry_with_external_prerequisites": false,
    "has_incoming_routes": true,
    "has_external_prerequisites": true,
    "has_internal_prerequisites": true,
    "has_conditional_incoming_routes": false,
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    "display_label": "Not a module start",
    "explanation": "This is not the first question detected inside this module."
},
"dependency_summary": {

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        "outgoing_route_count": 1,
        "prerequisite_question_count": 2,
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        "has_conditional_incoming_routes": false
    }
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    "data": {
        "label": "FICODE",
        "title": "Income receipt code",
        "option_count": 50,
        "question_type": "single_choice"
    },
    "type": "question",
    "incoming_routes": [
        {
            "id": "NFH_FICODE__28",
            "source": "NFH",
            "source_parts": [
                "NFH"
            ],
            "target": "FICODE",
            "type": "sequential",
            "label": "sequential",
            "condition_text": "sequential"
        }
    ],
    "outgoing_routes": [
        {
            "id": "FICODE_FITAX__7",
            "source": "FICODE",
            "source_parts": [
                "FICODE"
            ],
            "target": "FITAX",
            "type": "conditional",
            "label": "if [(FICODE IN 2,3,4) AND (FRVAL >= 0)]",
            "condition_text": "if [(FICODE IN 2,3,4) AND (FRVAL >= 0)]"
        },
        {
            "id": "FICODE_FRADDL__8",
            "source": "FICODE",
            "source_parts": [
                "FICODE"
            ],
            "target": "FRADDL",
            "type": "conditional",
            "label": "if [FICODE IN 2,3,4,24,25,26,27,28,29,30,35,38,39]",
            "condition_text": "if [FICODE IN 2,3,4,24,25,26,27,28,29,30,35,38,39]"
        },
        {
            "id": "FICODE_FISEQ__29",
            "source": "FICODE",
            "source_parts": [
                "FICODE"
            ],
            "target": "FISEQ",

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        "type": "sequential",
        "label": "sequential",
        "condition_text": "sequential"
    }
],
"prerequisite_questions": [
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        "id": "NFH",
        "exists_in_graph": true,
        "is_external": false,
        "type": "question",
        "label": "NFH"
    }
],
"start_semantics": {
    "is_first_module_question": false,
    "is_true_unconditional_start": false,
    "is_entry_with_external_prerequisites": false,
    "has_incoming_routes": true,
    "has_external_prerequisites": false,
    "has_internal_prerequisites": true,
    "has_conditional_incoming_routes": false,
    "start_kind": "not_start",
    "display_label": "Not a module start",
    "explanation": "This is not the first question detected inside this module."
},
"dependency_summary": {
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    "outgoing_route_count": 3,
    "prerequisite_question_count": 1,
    "has_external_prerequisites": false,
    "has_conditional_incoming_routes": false
}
},
{
    "id": "FISEQ",
    "data": {
        "label": "FISEQ",
        "title": "Income receipt sequence number",
        "option_count": 4,
        "question_type": "number"
    },
    "type": "question",
    "incoming_routes": [
        {
            "id": "FICODE__FISEQ__29",
            "source": "FICODE",
            "source_parts": [
                "FICODE"
            ],
            "target": "FISEQ",
            "type": "sequential",
            "label": "sequential",
            "condition_text": "sequential"
        }
    ],
    "outgoing_routes": [
        {
            "id": "FISEQ__FRVAL__30",

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        "source": "FISEQ",
        "source_parts": [
            "FISEQ"
        ],
        "target": "FRVAL",
        "type": "sequential",
        "label": "sequential",
        "condition_text": "sequential"
    }
],
"prerequisite_questions": [
    {
        "id": "FICODE",
        "exists_in_graph": true,
        "is_external": false,
        "type": "question",
        "label": "FICODE"
    }
],
"start_semantics": {
    "is_first_module_question": false,
    "is_true_unconditional_start": false,
    "is_entry_with_external_prerequisites": false,
    "has_incoming_routes": true,
    "has_external_prerequisites": false,
    "has_internal_prerequisites": true,
    "has_conditional_incoming_routes": false,
    "start_kind": "not_start",
    "display_label": "Not a module start",
    "explanation": "This is not the first question detected inside this module."
},
"dependency_summary": {
    "incoming_route_count": 1,
    "outgoing_route_count": 1,
    "prerequisite_question_count": 1,
    "has_external_prerequisites": false,
    "has_conditional_incoming_routes": false
}
},
{
    "id": "FRVAL",
    "data": {
        "label": "FRVAL",
        "title": "Last amount received",
        "option_count": 4,
        "question_type": "number"
    },
    "type": "question",
    "incoming_routes": [
        {
            "id": "FRVALLoop__FRVAL__11",
            "source": "FRVALLoop",
            "source_parts": [
                "FRVALLoop"
            ],
            "target": "FRVAL",
            "type": "loop",
            "label": "loop",
            "condition_text": "loop"
        }
    ]
}

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    },
    {
      "id": "FISEQ__FRVAL__30",
      "source": "FISEQ",
      "source_parts": [
        "FISEQ"
      ],
      "target": "FRVAL",
      "type": "sequential",
      "label": "sequential",
      "condition_text": "sequential"
    }
  ],
  "outgoing_routes": [
    {
      "id": "FRVAL__FRWC__5",
      "source": "FRVAL",
      "source_parts": [
        "FRVAL"
      ],
      "target": "FRWC",
      "type": "conditional",
      "label": "if [FRVAL > 0]",
      "condition_text": "if [FRVAL > 0]"
    },
    {
      "id": "FRVAL__FITAX__7",
      "source": "FRVAL",
      "source_parts": [
        "FRVAL"
      ],
      "target": "FITAX",
      "type": "conditional",
      "label": "if [(FICODE IN 2,3,4) AND (FRVAL >= 0)]",
      "condition_text": "if [(FICODE IN 2,3,4) AND (FRVAL >= 0)]"
    },
    {
      "id": "FRVAL__FRWC__31",
      "source": "FRVAL",
      "source_parts": [
        "FRVAL"
      ],
      "target": "FRWC",
      "type": "sequential",
      "label": "sequential",
      "condition_text": "sequential"
    }
  ],
  "prerequisite_questions": [
    {
      "id": "FRVALLoop",
      "exists_in_graph": true,
      "is_external": true,
      "type": "loop",
      "label": "FRVALLoop"
    },
    {
      "id": "FISEQ",
      "exists_in_graph": true,

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        "is_external": false,
        "type": "question",
        "label": "FISEQ"
    }
],
"start_semantics": {
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    "is_true_unconditional_start": false,
    "is_entry_with_external_prerequisites": false,
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    "has_external_prerequisites": true,
    "has_internal_prerequisites": true,
    "has_conditional_incoming_routes": false,
    "start_kind": "not_start",
    "display_label": "Not a module start",
    "explanation": "This is not the first question detected inside this module."
},
"dependency_summary": {
    "incoming_route_count": 2,
    "outgoing_route_count": 3,
    "prerequisite_question_count": 2,
    "has_external_prerequisites": true,
    "has_conditional_incoming_routes": false
}
},
{
    "id": "FRWC",
    "data": {
        "label": "FRWC",
        "title": "Period covered by last amount received",
        "option_count": 20,
        "question_type": "single_choice"
    },
    "type": "question",
    "incoming_routes": [
        {
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            "source": "FRVAL",
            "source_parts": [
                "FRVAL"
            ],
            "target": "FRWC",
            "type": "conditional",
            "label": "if [FRVAL > 0]",
            "condition_text": "if [FRVAL > 0]"
        },
        {
            "id": "FRVALLoop__FRWC__12",
            "source": "FRVALLoop",
            "source_parts": [
                "FRVALLoop"
            ],
            "target": "FRWC",
            "type": "loop",
            "label": "loop",
            "condition_text": "loop"
        },
        {
            "id": "FRVAL__FRWC__31",

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        "source": "FRVAL",
        "source_parts": [
            "FRVAL"
        ],
        "target": "FRWC",
        "type": "sequential",
        "label": "sequential",
        "condition_text": "sequential"
    }
],
"outgoing_routes": [
    {
        "id": "FRWC_FRJT_32",
        "source": "FRWC",
        "source_parts": [
            "FRWC"
        ],
        "target": "FRJT",
        "type": "sequential",
        "label": "sequential",
        "condition_text": "sequential"
    }
],
"prerequisite_questions": [
    {
        "id": "FRVAL",
        "exists_in_graph": true,
        "is_external": false,
        "type": "question",
        "label": "FRVAL"
    },
    {
        "id": "FRVALLoop",
        "exists_in_graph": true,
        "is_external": true,
        "type": "loop",
        "label": "FRVALLoop"
    }
],
"start_semantics": {
    "is_first_module_question": false,
    "is_true_unconditional_start": false,
    "is_entry_with_external_prerequisites": false,
    "has_incoming_routes": true,
    "has_external_prerequisites": true,
    "has_internal_prerequisites": true,
    "has_conditional_incoming_routes": true,
    "start_kind": "not_start",
    "display_label": "Not a module start",
    "explanation": "This is not the first question detected inside this module."
},
"dependency_summary": {
    "incoming_route_count": 3,
    "outgoing_route_count": 1,
    "prerequisite_question_count": 2,
    "has_external_prerequisites": true,
    "has_conditional_incoming_routes": true
}
},

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{
  "id": "FRJT",
  "data": {
    "label": "FRJT",
    "title": "Last payment: Sole or joint recipient",
    "option_count": 6,
    "question_type": "single_choice"
  },
  "type": "question",
  "incoming_routes": [
    {
      "id": "FRVALLoop__FRJT__13",
      "source": "FRVALLoop",
      "source_parts": [
        "FRVALLoop"
      ],
      "target": "FRJT",
      "type": "loop",
      "label": "loop",
      "condition_text": "loop"
    },
    {
      "id": "FRWC__FRJT__32",
      "source": "FRWC",
      "source_parts": [
        "FRWC"
      ],
      "target": "FRJT",
      "type": "sequential",
      "label": "sequential",
      "condition_text": "sequential"
    }
  ],
  "outgoing_routes": [
    {
      "id": "FRJT__FRJTPN__6",
      "source": "FRJT",
      "source_parts": [
        "FRJT"
      ],
      "target": "FRJTPN",
      "type": "conditional",
      "label": "if [FRJT = 2]",
      "condition_text": "if [FRJT = 2]"
    },
    {
      "id": "FRJT__FRJTPN__33",
      "source": "FRJT",
      "source_parts": [
        "FRJT"
      ],
      "target": "FRJTPN",
      "type": "sequential",
      "label": "sequential",
      "condition_text": "sequential"
    }
  ],
  "prerequisite_questions": [
    {

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      "id": "FRVALLoop",
      "exists_in_graph": true,
      "is_external": true,
      "type": "loop",
      "label": "FRVALLoop"
    },
    {
      "id": "FRWC",
      "exists_in_graph": true,
      "is_external": false,
      "type": "question",
      "label": "FRWC"
    }
  ],
  "start_semantics": {
    "is_first_module_question": false,
    "is_true_unconditional_start": false,
    "is_entry_with_external_prerequisites": false,
    "has_incoming_routes": true,
    "has_external_prerequisites": true,
    "has_internal_prerequisites": true,
    "has_conditional_incoming_routes": false,
    "start_kind": "not_start",
    "display_label": "Not a module start",
    "explanation": "This is not the first question detected inside this module."
  },
  "dependency_summary": {
    "incoming_route_count": 2,
    "outgoing_route_count": 2,
    "prerequisite_question_count": 2,
    "has_external_prerequisites": true,
    "has_conditional_incoming_routes": false
  }
},
{
  "id": "FRJTPN",
  "data": {
    "label": "FRJTPN",
    "title": "PNO of joint recipient",
    "option_count": 21,
    "question_type": "single_choice"
  },
  "type": "question",
  "incoming_routes": [
    {
      "id": "FRJT__FRJTPN__6",
      "source": "FRJT",
      "source_parts": [
        "FRJT"
      ],
      "target": "FRJTPN",
      "type": "conditional",
      "label": "if [FRJT = 2]",
      "condition_text": "if [FRJT = 2]"
    },
    {
      "id": "FRVALLoop__FRJTPN__14",
      "source": "FRVALLoop",
      "source_parts": [

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        "FRVALLoop"
      ],
      "target": "FRJTPN",
      "type": "loop",
      "label": "loop",
      "condition_text": "loop"
    },
    {
      "id": "FRJT__FRJTPN__33",
      "source": "FRJT",
      "source_parts": [
        "FRJT"
      ],
      "target": "FRJTPN",
      "type": "sequential",
      "label": "sequential",
      "condition_text": "sequential"
    }
  ],
  "outgoing_routes": [
    {
      "id": "FRJTPN__FITAX__34",
      "source": "FRJTPN",
      "source_parts": [
        "FRJTPN"
      ],
      "target": "FITAX",
      "type": "sequential",
      "label": "sequential",
      "condition_text": "sequential"
    }
  ],
  "prerequisite_questions": [
    {
      "id": "FRJT",
      "exists_in_graph": true,
      "is_external": false,
      "type": "question",
      "label": "FRJT"
    },
    {
      "id": "FRVALLoop",
      "exists_in_graph": true,
      "is_external": true,
      "type": "loop",
      "label": "FRVALLoop"
    }
  ],
  "start_semantics": {
    "is_first_module_question": false,
    "is_true_unconditional_start": false,
    "is_entry_with_external_prerequisites": false,
    "has_incoming_routes": true,
    "has_external_prerequisites": true,
    "has_internal_prerequisites": true,
    "has_conditional_incoming_routes": true,
    "start_kind": "not_start",
    "display_label": "Not a module start",
    "explanation": "This is not the first question detected inside this module."
  }
}

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    },
    "dependency_summary": {
      "incoming_route_count": 3,
      "outgoing_route_count": 1,
      "prerequisite_question_count": 2,
      "has_external_prerequisites": true,
      "has_conditional_incoming_routes": true
    }
  },
  {
    "id": "FITAX",
    "data": {
      "label": "FITAX",
      "title": "Pre/post tax income",
      "option_count": 6,
      "question_type": "single_choice"
    },
    "type": "question",
    "incoming_routes": [
      {
        "id": "FICODE__FITAX__7",
        "source": "FICODE",
        "source_parts": [
          "FICODE"
        ],
        "target": "FITAX",
        "type": "conditional",
        "label": "if [(FICODE IN 2,3,4) AND (FRVAL >= 0)]",
        "condition_text": "if [(FICODE IN 2,3,4) AND (FRVAL >= 0)]"
      },
      {
        "id": "FRVAL__FITAX__7",
        "source": "FRVAL",
        "source_parts": [
          "FRVAL"
        ],
        "target": "FITAX",
        "type": "conditional",
        "label": "if [(FICODE IN 2,3,4) AND (FRVAL >= 0)]",
        "condition_text": "if [(FICODE IN 2,3,4) AND (FRVAL >= 0)]"
      },
      {
        "id": "FRVALLoop__FITAX__15",
        "source": "FRVALLoop",
        "source_parts": [
          "FRVALLoop"
        ],
        "target": "FITAX",
        "type": "loop",
        "label": "loop",
        "condition_text": "loop"
      },
      {
        "id": "FRJTPN__FITAX__34",
        "source": "FRJTPN",
        "source_parts": [
          "FRJTPN"
        ],
        "target": "FITAX",

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        "type": "sequential",
        "label": "sequential",
        "condition_text": "sequential"
    }
],
"outgoing_routes": [
    {
        "id": "FITAX__FRADDL__35",
        "source": "FITAX",
        "source_parts": [
            "FITAX"
        ],
        "target": "FRADDL",
        "type": "sequential",
        "label": "sequential",
        "condition_text": "sequential"
    }
],
"prerequisite_questions": [
    {
        "id": "FICODE",
        "exists_in_graph": true,
        "is_external": false,
        "type": "question",
        "label": "FICODE"
    },
    {
        "id": "FRVAL",
        "exists_in_graph": true,
        "is_external": false,
        "type": "question",
        "label": "FRVAL"
    },
    {
        "id": "FRVALLoop",
        "exists_in_graph": true,
        "is_external": true,
        "type": "loop",
        "label": "FRVALLoop"
    },
    {
        "id": "FRJTPN",
        "exists_in_graph": true,
        "is_external": false,
        "type": "question",
        "label": "FRJTPN"
    }
],
"start_semantics": {
    "is_first_module_question": false,
    "is_true_unconditional_start": false,
    "is_entry_with_external_prerequisites": false,
    "has_incoming_routes": true,
    "has_external_prerequisites": true,
    "has_internal_prerequisites": true,
    "has_conditional_incoming_routes": true,
    "start_kind": "not_start",
    "display_label": "Not a module start",
    "explanation": "This is not the first question detected inside this module."
}

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    },
    "dependency_summary": {
      "incoming_route_count": 4,
      "outgoing_route_count": 1,
      "prerequisite_question_count": 4,
      "has_external_prerequisites": true,
      "has_conditional_incoming_routes": true
    }
  },
  {
    "id": "FRADDL",
    "data": {
      "label": "FRADDL",
      "title": "Received additional payment from source",
      "option_count": 6,
      "question_type": "single_choice"
    },
    "type": "question",
    "incoming_routes": [
      {
        "id": "FICODE__FRADDL__8",
        "source": "FICODE",
        "source_parts": [
          "FICODE"
        ],
        "target": "FRADDL",
        "type": "conditional",
        "label": "if [FICODE IN 2,3,4,24,25,26,27,28,29,30,35,38,39]",
        "condition_text": "if [FICODE IN 2,3,4,24,25,26,27,28,29,30,35,38,39]"
      },
      {
        "id": "FRVALLoop__FRADDL__16",
        "source": "FRVALLoop",
        "source_parts": [
          "FRVALLoop"
        ],
        "target": "FRADDL",
        "type": "loop",
        "label": "loop",
        "condition_text": "loop"
      },
      {
        "id": "FITAX__FRADDL__35",
        "source": "FITAX",
        "source_parts": [
          "FITAX"
        ],
        "target": "FRADDL",
        "type": "sequential",
        "label": "sequential",
        "condition_text": "sequential"
      }
    ],
    "outgoing_routes": [],
    "prerequisite_questions": [
      {
        "id": "FICODE",
        "exists_in_graph": true,
        "is_external": false,

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        "type": "question",
        "label": "FICODE"
    },
    {
        "id": "FRVALLoop",
        "exists_in_graph": true,
        "is_external": true,
        "type": "loop",
        "label": "FRVALLoop"
    },
    {
        "id": "FITAX",
        "exists_in_graph": true,
        "is_external": false,
        "type": "question",
        "label": "FITAX"
    }
],
"start_semantics": {
    "is_first_module_question": false,
    "is_true_unconditional_start": false,
    "is_entry_with_external_prerequisites": false,
    "has_incoming_routes": true,
    "has_external_prerequisites": true,
    "has_internal_prerequisites": true,
    "has_conditional_incoming_routes": true,
    "start_kind": "not_start",
    "display_label": "Not a module start",
    "explanation": "This is not the first question detected inside this module."
},
"dependency_summary": {
    "incoming_route_count": 3,
    "outgoing_route_count": 0,
    "prerequisite_question_count": 3,
    "has_external_prerequisites": true,
    "has_conditional_incoming_routes": true
}
},
{
    "id": "MISSOURCELoop",
    "data": {
        "label": "MISSOURCELoop",
        "title": "MISSOURCELoop",
        "option_count": 0,
        "question_type": "loop"
    },
    "type": "loop"
},
{
    "id": "NFHLoop",
    "data": {
        "label": "NFHLoop",
        "title": "NFHLoop",
        "option_count": 0,
        "question_type": "loop"
    },
    "type": "loop"
},
{

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      "id": "FRVALLoop",
      "data": {
        "label": "FRVALLoop",
        "title": "FRVALLoop",
        "option_count": 0,
        "question_type": "loop"
      },
      "type": "loop"
    }
  ],
  "edges_json": [
    {
      "id": "BENBASE__BENCTC__1",
      "type": "conditional",
      "label": "if [BenBase = 3 & BenBase <> 4]",
      "source": "BENBASE",
      "target": "BENCTC",
      "condition": "if [BenBase = 3 & BenBase <> 4]",
      "sequence_index": 1
    },
    {
      "id": "BENBASE__BENWCA__2",
      "type": "conditional",
      "label": "if [BenBase = 4]",
      "source": "BENBASE",
      "target": "BENWCA",
      "condition": "if [BenBase = 4]",
      "sequence_index": 2
    },
    {
      "id": "BENWCA__BENWCAOTC__3",
      "type": "conditional",
      "label": "if [BenWCA = 1]",
      "source": "BENWCA",
      "target": "BENWCAOTC",
      "condition": "if [BenWCA = 1]",
      "sequence_index": 3
    },
    {
      "id": "BENDIS__BENESA__4",
      "type": "conditional",
      "label": "if [BenDis = 2]",
      "source": "BENDIS",
      "target": "BENESA",
      "condition": "if [BenDis = 2]",
      "sequence_index": 4
    },
    {
      "id": "FRVAL__FRWC__5",
      "type": "conditional",
      "label": "if [FRVAL > 0]",
      "source": "FRVAL",
      "target": "FRWC",
      "condition": "if [FRVAL > 0]",
      "sequence_index": 5
    },
    {
      "id": "FRJT__FRJTPN__6",
      "type": "conditional",

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    "label": "if [FRJT = 2]",
    "source": "FRJT",
    "target": "FRJTPN",
    "condition": "if [FRJT = 2]",
    "sequence_index": 6
  },
  {
    "id": "FICODE__FITAX__7",
    "type": "conditional",
    "label": "if [(FICODE IN 2,3,4) AND (FRVAL >= 0)]",
    "source": "FICODE",
    "target": "FITAX",
    "condition": "if [(FICODE IN 2,3,4) AND (FRVAL >= 0)]",
    "sequence_index": 7
  },
  {
    "id": "FRVAL__FITAX__7",
    "type": "conditional",
    "label": "if [(FICODE IN 2,3,4) AND (FRVAL >= 0)]",
    "source": "FRVAL",
    "target": "FITAX",
    "condition": "if [(FICODE IN 2,3,4) AND (FRVAL >= 0)]",
    "sequence_index": 7
  },
  {
    "id": "FICODE__FRADDL__8",
    "type": "conditional",
    "label": "if [FICODE IN 2,3,4,24,25,26,27,28,29,30,35,38,39]",
    "source": "FICODE",
    "target": "FRADDL",
    "condition": "if [FICODE IN 2,3,4,24,25,26,27,28,29,30,35,38,39]",
    "sequence_index": 8
  },
  {
    "id": "MISSOURCELoop__MISSOURCE__9",
    "type": "loop",
    "label": "loop",
    "source": "MISSOURCELoop",
    "target": "MISSOURCE",
    "condition": "",
    "sequence_index": 9
  },
  {
    "id": "NFHLoop__NFH__10",
    "type": "loop",
    "label": "loop",
    "source": "NFHLoop",
    "target": "NFH",
    "condition": "",
    "sequence_index": 10
  },
  {
    "id": "FRVALLoop__FRVAL__11",
    "type": "loop",
    "label": "loop",
    "source": "FRVALLoop",
    "target": "FRVAL",
    "condition": "",
    "sequence_index": 11
  }

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},
{
  "id": "FRVALLoop__FRWC__12",
  "type": "loop",
  "label": "loop",
  "source": "FRVALLoop",
  "target": "FRWC",
  "condition": "",
  "sequence_index": 12
},
{
  "id": "FRVALLoop__FRJT__13",
  "type": "loop",
  "label": "loop",
  "source": "FRVALLoop",
  "target": "FRJT",
  "condition": "",
  "sequence_index": 13
},
{
  "id": "FRVALLoop__FRJTPN__14",
  "type": "loop",
  "label": "loop",
  "source": "FRVALLoop",
  "target": "FRJTPN",
  "condition": "",
  "sequence_index": 14
},
{
  "id": "FRVALLoop__FITAX__15",
  "type": "loop",
  "label": "loop",
  "source": "FRVALLoop",
  "target": "FITAX",
  "condition": "",
  "sequence_index": 15
},
{
  "id": "FRVALLoop__FRADDL__16",
  "type": "loop",
  "label": "loop",
  "source": "FRVALLoop",
  "target": "FRADDL",
  "condition": "",
  "sequence_index": 16
},
{
  "id": "INTBENEFIT__BENBASE__17",
  "type": "sequential",
  "label": "sequential",
  "source": "INTBENEFIT",
  "target": "BENBASE",
  "condition": "",
  "sequence_index": 17
},
{
  "id": "BENBASE__BENCTC__18",
  "type": "sequential",
  "label": "sequential",

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    "source": "BENBASE",
    "target": "BENCTC",
    "condition": "",
    "sequence_index": 18
  },
  {
    "id": "BENCTC__BENWCA__19",
    "type": "sequential",
    "label": "sequential",
    "source": "BENCTC",
    "target": "BENWCA",
    "condition": "",
    "sequence_index": 19
  },
  {
    "id": "BENWCA__BENWCAOTC__20",
    "type": "sequential",
    "label": "sequential",
    "source": "BENWCA",
    "target": "BENWCAOTC",
    "condition": "",
    "sequence_index": 20
  },
  {
    "id": "BENWCAOTC__BENPEN__21",
    "type": "sequential",
    "label": "sequential",
    "source": "BENWCAOTC",
    "target": "BENPEN",
    "condition": "",
    "sequence_index": 21
  },
  {
    "id": "BENPEN__BENDIS__22",
    "type": "sequential",
    "label": "sequential",
    "source": "BENPEN",
    "target": "BENDIS",
    "condition": "",
    "sequence_index": 22
  },
  {
    "id": "BENDIS__BENESA__23",
    "type": "sequential",
    "label": "sequential",
    "source": "BENDIS",
    "target": "BENESA",
    "condition": "",
    "sequence_index": 23
  },
  {
    "id": "BENESA__OTHBEN__24",
    "type": "sequential",
    "label": "sequential",
    "source": "BENESA",
    "target": "OTHBEN",
    "condition": "",
    "sequence_index": 24
  },
  },

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{
  "id": "OTHBEN__BENSTA__25",
  "type": "sequential",
  "label": "sequential",
  "source": "OTHBEN",
  "target": "BENSTA",
  "condition": "",
  "sequence_index": 25
},
{
  "id": "BENSTA__MISSOURCE__26",
  "type": "sequential",
  "label": "sequential",
  "source": "BENSTA",
  "target": "MISSOURCE",
  "condition": "",
  "sequence_index": 26
},
{
  "id": "MISSOURCE__NFH__27",
  "type": "sequential",
  "label": "sequential",
  "source": "MISSOURCE",
  "target": "NFH",
  "condition": "",
  "sequence_index": 27
},
{
  "id": "NFH__FICODE__28",
  "type": "sequential",
  "label": "sequential",
  "source": "NFH",
  "target": "FICODE",
  "condition": "",
  "sequence_index": 28
},
{
  "id": "FICODE__FISEQ__29",
  "type": "sequential",
  "label": "sequential",
  "source": "FICODE",
  "target": "FISEQ",
  "condition": "",
  "sequence_index": 29
},
{
  "id": "FISEQ__FRVAL__30",
  "type": "sequential",
  "label": "sequential",
  "source": "FISEQ",
  "target": "FRVAL",
  "condition": "",
  "sequence_index": 30
},
{
  "id": "FRVAL__FRWC__31",
  "type": "sequential",
  "label": "sequential",
  "source": "FRVAL",

```

```

    "target": "FRWC",
    "condition": "",
    "sequence_index": 31
  },
  {
    "id": "FRWC__FRJT_32",
    "type": "sequential",
    "label": "sequential",
    "source": "FRWC",
    "target": "FRJT",
    "condition": "",
    "sequence_index": 32
  },
  {
    "id": "FRJT__FRJTPN_33",
    "type": "sequential",
    "label": "sequential",
    "source": "FRJT",
    "target": "FRJTPN",
    "condition": "",
    "sequence_index": 33
  },
  {
    "id": "FRJTPN__FITAX_34",
    "type": "sequential",
    "label": "sequential",
    "source": "FRJTPN",
    "target": "FITAX",
    "condition": "",
    "sequence_index": 34
  },
  {
    "id": "FITAX__FRADDL_35",
    "type": "sequential",
    "label": "sequential",
    "source": "FITAX",
    "target": "FRADDL",
    "condition": "",
    "sequence_index": 35
  }
],
"graph_start_summary": {
  "first_module_question_id": "INTBENEFIT",
  "true_start_question_ids": [
    "INTBENEFIT"
  ],
  "external_entry_prerequisites": [],
  "has_true_unconditional_start": true,
  "has_external_entry_prerequisites": false
}
},
"mermaid": "flowchart TD\n    INTBENEFIT[\\\"INTBENEFIT\\\"]\n    BENBASE[\\\"BENBASE\\\"]\n    BENCTC[\\\"BENCTC\\\"]\n    BENWCA[\\\"BENWCA\\\"]\n    BENWCAOTC[\\\"BENWCAOTC\\\"]\n    BENPEN[\\\"BENPEN\\\"]\n    BENDIS[\\\"BENDIS\\\"]\n    BENESA[\\\"BENESA\\\"]\n    OTHBEN[\\\"OTHBEN\\\"]\n    BENSTA[\\\"BENSTA\\\"]\n    MISSOURCE[\\\"MISSOURCE\\\"]\n    NFH[\\\"NFH\\\"]\n    FICODE[\\\"FICODE\\\"]\n    FISEQ[\\\"FISEQ\\\"]\n    FRVAL[\\\"FRVAL\\\"]\n    FRWC[\\\"FRWC\\\"]\n    FRJT[\\\"FRJT\\\"]\n    FRJTPN[\\\"FRJTPN\\\"]\n    FITAX[\\\"FITAX\\\"]\n    FRADDL[\\\"FRADDL\\\"]\n    MISSOURCELoop{\\\"MISSOURCELoop\\\"}\n    NFHLoop{\\\"NFHLoop\\\"}\n    FRVALLoop{\\\"FRVALLoop\\\"}\n    BENBASE -->|\\\"if [BenBase = 3 & BenBase <> 4]\\\"|\n    BENCTC\n    BENBASE -->|\\\"if [BenBase = 4]\\\"|\n    BENWCA\n    BENWCA -->|\\\"if [BenWCA = 1]\\\"|\n    BENWCAOTC\n    BENDIS -->|\\\"if [BenDis = 2]\\\"|\n    BENESA\n    FRVAL -->|\\\"if [FRVAL > 0]\\\"|\n    FRWC\n    FRJT
```

```
-->|\\"if [FRJT = 2]\\"| FRJT\\n      FICODE -->|\\"if [(FICODE IN 2,3,4) AND (FRVAL >= 0)]\\"| FITAX\\n      FRVAL
-->|\\"if [(FICODE IN 2,3,4) AND (FRVAL >= 0)]\\"| FITAX\\n      FICODE -->|\\"if [FICODE IN
2,3,4,24,25,26,27,28,29,30,35,38,39]\\"| FRADDL\\n      MISSOURCELoop -->|\\"loop\\"| MISSOURCE\\n      NFHLoop
-->|\\"loop\\"| NFH\\n      FRVALLoop -->|\\"loop\\"| FRVAL\\n      FRVALLoop -->|\\"loop\\"| FRWC\\n      FRVALLoop -->|\\"loop\\"|
FRJT\\n      FRVALLoop -->|\\"loop\\"| FRJT\\n      FRVALLoop -->|\\"loop\\"| FITAX\\n      FRVALLoop -->|\\"loop\\"| FRADDL\\n
INTBENEFIT -->|\\"sequential\\"| BENBASE\\n      BENBASE -->|\\"sequential\\"| BENCTC\\n      BENCTC -->|\\"sequential\\"|
BENWCA\\n      BENWCA -->|\\"sequential\\"| BENWCAOTC\\n      BENWCAOTC -->|\\"sequential\\"| BENPEN\\n      BENPEN
-->|\\"sequential\\"| BENDIS\\n      BENDIS -->|\\"sequential\\"| BENESA\\n      BENESA -->|\\"sequential\\"| OTHBEN\\n
OTHBEN -->|\\"sequential\\"| BENSTA\\n      BENSTA -->|\\"sequential\\"| MISSOURCE\\n      MISSOURCE -->|\\"sequential\\"|
NFH\\n      NFH -->|\\"sequential\\"| FICODE\\n      FICODE -->|\\"sequential\\"| FISEQ\\n      FISEQ -->|\\"sequential\\"|
FRVAL\\n      FRVAL -->|\\"sequential\\"| FRWC\\n      FRWC -->|\\"sequential\\"| FRJT\\n      FRJT -->|\\"sequential\\"| FRJT\\n
FRJT -->|\\"sequential\\"| FITAX\\n      FITAX -->|\\"sequential\\"| FRADDL"
}
```

Second example: neighborhood mode

```
get_module_graph(module_id_or_name="Benefits W18", question_name="FICODE", depth=1) # neighborhood
mode
```

Example response (real output)

```
{
  "module_id": "c8171847-7442-41ef-b534-04682184a14f",
  "module_name": "Benefits W18",
  "source_format": "colectica_json",
  "scoped_to_question": "FICODE",
  "total_node_count": 23,
  "total_edge_count": 36,
  "mode": "neighborhood",
  "graph_start_summary": {
    "first_module_question_id": "NFH",
    "true_start_question_ids": [
      "NFH"
    ],
  },
  "external_entry_prerequisites": [],
  "has_true_unconditional_start": true,
  "has_external_entry_prerequisites": false
},
"graph": {
  "nodes_json": [
    {
      "id": "NFH",
      "data": {
        "label": "NFH",
        "title": "Further income sources",
        "option_count": 6,
        "question_type": "single_choice"
      },
    },
    {
      "type": "question",
      "incoming_routes": [],
      "outgoing_routes": [
        {
          "id": "NFH_FICODE__28",
          "source": "NFH",
          "source_parts": [
            "NFH"
          ],
          "target": "FICODE",
          "type": "sequential",
          "label": "sequential",
          "condition_text": "sequential"
        }
      ]
    }
  ]
}
```

```

    }
  ],
  "prerequisite_questions": [],
  "start_semantics": {
    "is_first_module_question": true,
    "is_true_unconditional_start": true,
    "is_entry_with_external_prerequisites": false,
    "has_incoming_routes": false,
    "has_external_prerequisites": false,
    "has_internal_prerequisites": false,
    "has_conditional_incoming_routes": false,
    "start_kind": "true_unconditional_start",
    "display_label": "True unconditional start",
    "explanation": "This is the first question inside the module and no incoming or external prerequisite
routes were detected."
  },
  "dependency_summary": {
    "incoming_route_count": 0,
    "outgoing_route_count": 1,
    "prerequisite_question_count": 0,
    "has_external_prerequisites": false,
    "has_conditional_incoming_routes": false
  }
},
{
  "id": "FICODE",
  "data": {
    "label": "FICODE",
    "title": "Income receipt code",
    "option_count": 50,
    "question_type": "single_choice"
  },
  "type": "question",
  "incoming_routes": [
    {
      "id": "NFH_FICODE_28",
      "source": "NFH",
      "source_parts": [
        "NFH"
      ],
      "target": "FICODE",
      "type": "sequential",
      "label": "sequential",
      "condition_text": "sequential"
    }
  ],
  "outgoing_routes": [
    {
      "id": "FICODE_FITAX_7",
      "source": "FICODE",
      "source_parts": [
        "FICODE"
      ],
      "target": "FITAX",
      "type": "conditional",
      "label": "if [(FICODE IN 2,3,4) AND (FRVAL >= 0)]",
      "condition_text": "if [(FICODE IN 2,3,4) AND (FRVAL >= 0)]"
    }
  ],
  {

```

```

      "id": "FICODE__FRADDL__8",
      "source": "FICODE",
      "source_parts": [
        "FICODE"
      ],
      "target": "FRADDL",
      "type": "conditional",
      "label": "if [FICODE IN 2,3,4,24,25,26,27,28,29,30,35,38,39]",
      "condition_text": "if [FICODE IN 2,3,4,24,25,26,27,28,29,30,35,38,39]"
    },
    {
      "id": "FICODE__FISEQ__29",
      "source": "FICODE",
      "source_parts": [
        "FICODE"
      ],
      "target": "FISEQ",
      "type": "sequential",
      "label": "sequential",
      "condition_text": "sequential"
    }
  ],
  "prerequisite_questions": [
    {
      "id": "NFH",
      "exists_in_graph": true,
      "is_external": false,
      "type": "question",
      "label": "NFH"
    }
  ],
  "start_semantics": {
    "is_first_module_question": false,
    "is_true_unconditional_start": false,
    "is_entry_with_external_prerequisites": false,
    "has_incoming_routes": true,
    "has_external_prerequisites": false,
    "has_internal_prerequisites": true,
    "has_conditional_incoming_routes": false,
    "start_kind": "not_start",
    "display_label": "Not a module start",
    "explanation": "This is not the first question detected inside this module."
  },
  "dependency_summary": {
    "incoming_route_count": 1,
    "outgoing_route_count": 3,
    "prerequisite_question_count": 1,
    "has_external_prerequisites": false,
    "has_conditional_incoming_routes": false
  }
},
{
  "id": "FISEQ",
  "data": {
    "label": "FISEQ",
    "title": "Income receipt sequence number",
    "option_count": 4,
    "question_type": "number"
  }
},

```

```

    "type": "question",
    "incoming_routes": [
      {
        "id": "FICODE__FISEQ__29",
        "source": "FICODE",
        "source_parts": [
          "FICODE"
        ],
        "target": "FISEQ",
        "type": "sequential",
        "label": "sequential",
        "condition_text": "sequential"
      }
    ],
    "outgoing_routes": [],
    "prerequisite_questions": [
      {
        "id": "FICODE",
        "exists_in_graph": true,
        "is_external": false,
        "type": "question",
        "label": "FICODE"
      }
    ],
    "start_semantics": {
      "is_first_module_question": false,
      "is_true_unconditional_start": false,
      "is_entry_with_external_prerequisites": false,
      "has_incoming_routes": true,
      "has_external_prerequisites": false,
      "has_internal_prerequisites": true,
      "has_conditional_incoming_routes": false,
      "start_kind": "not_start",
      "display_label": "Not a module start",
      "explanation": "This is not the first question detected inside this module."
    },
    "dependency_summary": {
      "incoming_route_count": 1,
      "outgoing_route_count": 0,
      "prerequisite_question_count": 1,
      "has_external_prerequisites": false,
      "has_conditional_incoming_routes": false
    }
  },
  {
    "id": "FITAX",
    "data": {
      "label": "FITAX",
      "title": "Pre/post tax income",
      "option_count": 6,
      "question_type": "single_choice"
    },
    "type": "question",
    "incoming_routes": [
      {
        "id": "FICODE__FITAX__7",
        "source": "FICODE",
        "source_parts": [
          "FICODE"
        ]
      }
    ]
  }

```

```

    ],
    "target": "FITAX",
    "type": "conditional",
    "label": "if [(FICODE IN 2,3,4) AND (FRVAL >= 0)]",
    "condition_text": "if [(FICODE IN 2,3,4) AND (FRVAL >= 0)]"
  }
],
"outgoing_routes": [],
"prerequisite_questions": [
  {
    "id": "FICODE",
    "exists_in_graph": true,
    "is_external": false,
    "type": "question",
    "label": "FICODE"
  }
],
"start_semantics": {
  "is_first_module_question": false,
  "is_true_unconditional_start": false,
  "is_entry_with_external_prerequisites": false,
  "has_incoming_routes": true,
  "has_external_prerequisites": false,
  "has_internal_prerequisites": true,
  "has_conditional_incoming_routes": true,
  "start_kind": "not_start",
  "display_label": "Not a module start",
  "explanation": "This is not the first question detected inside this module."
},
"dependency_summary": {
  "incoming_route_count": 1,
  "outgoing_route_count": 0,
  "prerequisite_question_count": 1,
  "has_external_prerequisites": false,
  "has_conditional_incoming_routes": true
}
},
{
  "id": "FRADDL",
  "data": {
    "label": "FRADDL",
    "title": "Received additional payment from source",
    "option_count": 6,
    "question_type": "single_choice"
  },
  "type": "question",
  "incoming_routes": [
    {
      "id": "FICODE__FRADDL__8",
      "source": "FICODE",
      "source_parts": [
        "FICODE"
      ],
      "target": "FRADDL",
      "type": "conditional",
      "label": "if [FICODE IN 2,3,4,24,25,26,27,28,29,30,35,38,39]",
      "condition_text": "if [FICODE IN 2,3,4,24,25,26,27,28,29,30,35,38,39]"
    }
  ]
},

```

```

    "outgoing_routes": [],
    "prerequisite_questions": [
      {
        "id": "FICODE",
        "exists_in_graph": true,
        "is_external": false,
        "type": "question",
        "label": "FICODE"
      }
    ],
    "start_semantics": {
      "is_first_module_question": false,
      "is_true_unconditional_start": false,
      "is_entry_with_external_prerequisites": false,
      "has_incoming_routes": true,
      "has_external_prerequisites": false,
      "has_internal_prerequisites": true,
      "has_conditional_incoming_routes": true,
      "start_kind": "not_start",
      "display_label": "Not a module start",
      "explanation": "This is not the first question detected inside this module."
    },
    "dependency_summary": {
      "incoming_route_count": 1,
      "outgoing_route_count": 0,
      "prerequisite_question_count": 1,
      "has_external_prerequisites": false,
      "has_conditional_incoming_routes": true
    }
  },
  "edges_json": [
    {
      "id": "FICODE__FITAX__7",
      "type": "conditional",
      "label": "if [(FICODE IN 2,3,4) AND (FRVAL >= 0)]",
      "source": "FICODE",
      "target": "FITAX",
      "condition": "if [(FICODE IN 2,3,4) AND (FRVAL >= 0)]",
      "sequence_index": 7
    },
    {
      "id": "FICODE__FRADDL__8",
      "type": "conditional",
      "label": "if [FICODE IN 2,3,4,24,25,26,27,28,29,30,35,38,39]",
      "source": "FICODE",
      "target": "FRADDL",
      "condition": "if [FICODE IN 2,3,4,24,25,26,27,28,29,30,35,38,39]",
      "sequence_index": 8
    },
    {
      "id": "NFH__FICODE__28",
      "type": "sequential",
      "label": "sequential",
      "source": "NFH",
      "target": "FICODE",
      "condition": "",
      "sequence_index": 28
    }
  ],

```



```

    {
      "id": "FICODE__FISEQ__29",
      "type": "sequential",
      "label": "sequential",
      "source": "FICODE",
      "target": "FISEQ",
      "condition": "",
      "sequence_index": 29
    }
  ],
  "graph_start_summary": {
    "first_module_question_id": "NFH",
    "true_start_question_ids": [
      "NFH"
    ],
    "external_entry_prerequisites": [],
    "has_true_unconditional_start": true,
    "has_external_entry_prerequisites": false
  }
},
"mermaid": "flowchart TD\n    NFH[\"NFH\"]\n    FICODE[\"FICODE\"]\n    FISEQ[\"FISEQ\"]\n    FITAX[\"FITAX\"]\n    FRADDL[\"FRADDL\"]\n    FICODE -->|\"if [(FICODE IN 2,3,4) AND (FRVAL >= 0)]\"| FITAX\n    FITAX -->|\"if [FICODE IN 2,3,4,24,25,26,27,28,29,30,35,38,39]\"| FRADDL\n    FRADDL -->|\"sequential\"| FICODE\n    FICODE -->|\"sequential\"| FISEQ"
}

```

evaluate_edge_condition

Scope: Colectica + Forsta+

MCP prompt: /evaluateCondition

Description

Evaluate a hypothetical condition expression against hypothetical answers -- Colectica OR Forsta+, both supported (the grammar picked automatically from the module's source_format). condition_text does not need to belong to any real RoutingEdge in this module; this lets a client ask "what if" questions interactively (e.g. would this fire if AIDHH=1), not just replay edges that already exist.

Pure/side-effect-free. Returns status ("true"/"false"/"unknown"/"unsupported"), missing_inputs, used_inputs, normalized_expression, and warnings -- unchanged from the underlying evaluator.

Python signature

```
evaluate_edge_condition(
    module_id_or_name: str,
    condition_text: str,
    inputs: dict[str, Any]
) -> dict
```

Example prompt shortcut

/evaluateCondition

"Using the evaluate_edge_condition tool, evaluate the condition 'if [BenBase = 4]' for module 'Benefits W18' against these hypothetical answers: {'BenBase': '4'}. Tell me whether it fires (true/false/unknown/unsupported) and which inputs were used or missing."

Example request

```
evaluate_edge_condition(module_id_or_name="Benefits W18", condition_text="if [BenBase = 4]",
inputs={"BenBase": "4"})
```

Example response (real output)

```
{
  "module_id": "c8171847-7442-41ef-b534-04682184a14f",
  "module_name": "Benefits W18",
  "source_format": "colectica_json",
  "condition_text": "if [BenBase = 4]",
  "status": "true",
  "missing_inputs": [],
  "used_inputs": {
    "BenBase": "4"
  },
  "normalized_expression": "BenBase = 4",
  "warnings": []
}
```

COLECTICA VS FORSTA+ ROUTING DIFF

get_routing_diff_report

Scope: Colectica + Forsta+ pair

MCP prompt: /showRoutingDiffReport

Description

Structural routing discrepancies between a matched Colectica/Forsta+ module pair, exactly matching the routing-diff report GUI page (/questionnaires/routing-diff/<colectica_id>/<forsta_id>/). Does NOT run matching/comparison itself (build_question_matches/compare_routing_for_modules are expensive for large modules) -- if matching has never been run for this exact pair, has_run is False and discrepancies is empty rather than raising.

Each discrepancy is a *lead for manual review*, not a confirmed bug -- see each row's "meaning" field: this is a structural comparison only, it never evaluates what a condition means.

discrepancy_type filters to one DiscrepancyType value. limit caps how many discrepancies are returned (default 200) -- discrepancy_count always reports the true total regardless of limit, since large modules can have hundreds of rows with full explainer text each.

Python signature

```
get_routing_diff_report(
    colectica_module_id_or_name: str,
    forsta_module_id_or_name: str,
    discrepancy_type: str | None = None,
    limit: int = 200
) -> dict
```

Example prompt shortcut

/showRoutingDiffReport

"Using the get_routing_diff_report tool, show me the routing-diff report between Colectica module '3a2de4bf-432b-42f2-b74c-2ceffd868e8a' and Forsta+ module 'Forsta Full Export'. If has_run is false, tell me matching/comparison hasn't been run for this pair yet instead of guessing. Otherwise summarize the match_summary and group the discrepancies by type and severity."

Example request

```
get_routing_diff_report(colectica_module_id_or_name="3a2de4bf-432b-42f2-b74c-2ceffd868e8a",
forsta_module_id_or_name="Forsta Full Export", limit=3) # real pair: 1803 matched questions, 431 real
discrepancies -- limit caps the list, not the true count
```

Example response (real output)

```
{
  "colectica_module_id": "3a2de4bf-432b-42f2-b74c-2ceffd868e8a",
  "colectica_module_name": "W18",
  "forsta_module_id": "6e5a11e6-b6c8-44b3-8ea9-0fd487ac6f3f",
  "forsta_module_name": "Forsta Full Export",
  "has_run": true,
  "match_summary": {
    "total": 1803,
    "matched": 676,
```

```

    "unmatched": 1127
  },
  "discrepancy_count": 431,
  "discrepancies": [
    {
      "id": "95b2be00-e5d5-4d6e-8924-e980a2917dfa",
      "discrepancy_type": "missing_in_forsta",
      "severity": "warning",
      "colectica_question": "RELTOPWCNBDONOR",
      "forsta_question": "RELTOPWCNBDONOR",
      "summary": "Question RELTOPWCNBDONOR matched in both systems, but the routing edge to RELTOPWCNBLPAR (fires when RELTOPWCNBDONOR = 1|2) has no equivalent in Forsta+.",
      "meaning": "This is a structural comparison only: it checks whether an edge with the same source and target exists on both sides, not what the condition means or whether an alternate path exists. This flag means either (a) Forsta+ genuinely never routes to RELTOPWCNBLPAR from here -- a real gap worth fixing, or (b) Forsta+ reaches RELTOPWCNBLPAR some other way (a different source question, different condition logic, a default fallback) that this structural check can't recognize -- a false positive. Severity \"Warning\" means this needs a human to check, not that it's a confirmed bug.",
      "edge": {
        "origin_system": "Colectica",
        "source_question": "RELTOPWCNBDONOR",
        "target_question": "RELTOPWCNBLPAR",
        "condition_text": "RELTOPWCNBDONOR = 1|2"
      }
    },
    {
      "id": "9d6b1b37-9b3d-42d1-9d60-583f9929c309",
      "discrepancy_type": "missing_in_forsta",
      "severity": "warning",
      "colectica_question": "LCHNOFH",
      "forsta_question": "LCHNOFH",
      "summary": "Question LCHNOFH matched in both systems, but the routing edge to LCHCHKFH (fires when LCHNOFH = 1..16) has no equivalent in Forsta+.",
      "meaning": "This is a structural comparison only: it checks whether an edge with the same source and target exists on both sides, not what the condition means or whether an alternate path exists. This flag means either (a) Forsta+ genuinely never routes to LCHCHKFH from here -- a real gap worth fixing, or (b) Forsta+ reaches LCHCHKFH some other way (a different source question, different condition logic, a default fallback) that this structural check can't recognize -- a false positive. Severity \"Warning\" means this needs a human to check, not that it's a confirmed bug.",
      "edge": {
        "origin_system": "Colectica",
        "source_question": "LCHNOFH",
        "target_question": "LCHCHKFH",
        "condition_text": "LCHNOFH = 1..16"
      }
    },
    {
      "id": "1e00689d-4167-4046-882c-f91602a54b35",
      "discrepancy_type": "missing_in_forsta",
      "severity": "warning",
      "colectica_question": "LCHNO",
      "forsta_question": "LCHNO",
      "summary": "Question LCHNO matched in both systems, but the routing edge to LCHCHK (fires when LChNo <> 99|DK|REF|MIS) has no equivalent in Forsta+.",
      "meaning": "This is a structural comparison only: it checks whether an edge with the same source and target exists on both sides, not what the condition means or whether an alternate path exists. This flag means either (a) Forsta+ genuinely never routes to LCHCHK from here -- a real gap worth fixing, or (b) Forsta+ reaches LCHCHK some other way (a different source question, different condition logic, a default fallback) that this structural check can't recognize -- a false positive. Severity \"Warning\" means this needs a human to check, not that it's a confirmed bug."
    }
  ]
}

```

```
confirmed bug.",
  "edge": {
    "origin_system": "Colectica",
    "source_question": "LCHNO",
    "target_question": "LCHCHK",
    "condition_text": "LChNo <> 99|DK|REF|MIS"
  }
}
]
```

get_routing_discrepancy_detail

Scope: Colectica + Forsta+ pair

MCP prompt: /showColecticaForstaDiscrepancy

Description

One routing discrepancy in full detail, exactly matching the routing-diff discrepancy detail GUI page

(/questionnaires/routing-diff/<colectica_id>/<forsta_id>/discrepancy/<discrepancy_id>/)

-- including BOTH systems' routing graph neighborhoods around the focus

question (colectica_graph and forsta_graph), not just Colectica's side.

Either graph is None if that module has no graph built yet, rather than erroring the whole call.

Python signature

```
get_routing_discrepancy_detail(
    colectica_module_id_or_name: str,
    forsta_module_id_or_name: str,
    discrepancy_id: str
) -> dict
```

Example prompt shortcut

/showColecticaForstaDiscrepancy

"Using the get_routing_discrepancy_detail tool, show me full detail for discrepancy '95b2be00-e5d5-4d6e-8924-e980a2917dfa' between Colectica module '3a2de4bf-432b-42f2-b74c-2ceffd868e8a' and Forsta+ module 'Forsta Full Export' -- the summary, the meaning (why this is a lead not a confirmed bug), and describe the routing around the focus question on both the Colectica graph and the Forsta+ graph."

Example request

```
get_routing_discrepancy_detail(colectica_module_id_or_name="3a2de4bf-432b-42f2-b74c-2ceffd868e8a",
forsta_module_id_or_name="Forsta Full Export", discrepancy_id="95b2be00-e5d5-4d6e-8924-e980a2917dfa")
```

Example response (real output)

```
{
  "colectica_module_id": "3a2de4bf-432b-42f2-b74c-2ceffd868e8a",
  "forsta_module_id": "6e5a11e6-b6c8-44b3-8ea9-0fd487ac6f3f",
  "discrepancy_id": "95b2be00-e5d5-4d6e-8924-e980a2917dfa",
  "discrepancy_type": "missing_in_forsta",
  "severity": "warning",
  "summary": "Question RELTOPWCNBDONOR matched in both systems, but the routing edge to RELTOPWCNBLPAR (fires when RELTOPWCNBDONOR = 1|2) has no equivalent in Forsta+.",
  "meaning": "This is a structural comparison only: it checks whether an edge with the same source and target exists on both sides, not what the condition means or whether an alternate path exists. This flag means either (a) Forsta+ genuinely never routes to RELTOPWCNBLPAR from here -- a real gap worth fixing, or (b) Forsta+ reaches RELTOPWCNBLPAR some other way (a different source question, different condition logic, a default fallback) that this structural check can't recognize -- a false positive. Severity \"Warning\" means this needs a human to check, not that it's a confirmed bug.",
  "edge": {
    "origin_system": "Colectica",
    "source_question": "RELTOPWCNBDONOR",
    "target_question": "RELTOPWCNBLPAR",
    "condition_text": "RELTOPWCNBDONOR = 1|2"
  },
  "focus_question_name": "RELTOPWCNBDONOR",
  "colectica_focus_question_name": "RELTOPWCNBDONOR",
```

```

"forsta_focus_question_name": "RELTOPWCNBDONOR",
"highlight_edge": {
  "source": "RELTOPWCNBDONOR",
  "target": "RELTOPWCNBLPAR"
},
"colectica_graph": {
  "nodes_json": [
    {
      "id": "RELTOPWCNBBIRTH",
      "data": {
        "label": "RELTOPWCNBBIRTH",
        "title": "Relationship when child born",
        "option_count": 13,
        "question_type": "single_choice"
      },
      "type": "question"
    },
    {
      "id": "RELTOPWCNBDONOR",
      "data": {
        "label": "RELTOPWCNBDONOR",
        "title": "Partner sperm donor/ surrogacy",
        "option_count": 7,
        "question_type": "single_choice"
      },
      "type": "question"
    },
    {
      "id": "RELTOPWCNBLPAR",
      "data": {
        "label": "RELTOPWCNBLPAR",
        "title": "Partner child's adoptive/legal parent",
        "option_count": 6,
        "question_type": "single_choice"
      },
      "type": "question"
    },
    {
      "id": "PregnancyLoop",
      "data": {
        "label": "PregnancyLoop",
        "title": "PregnancyLoop",
        "option_count": 0,
        "question_type": "loop"
      },
      "type": "loop"
    },
    {
      "id": "LCHLVLoop",
      "data": {
        "label": "LCHLVLoop",
        "title": "LCHLVLoop",
        "option_count": 0,
        "question_type": "loop"
      },
      "type": "loop"
    }
  ],
  "edges_json": [

```

```

{
  "id": "RELTOPWCNBBIRTH__RELTOPWCNBDONOR__526",
  "type": "conditional",
  "label": "if [RELTOPWCNBBIRTH = 8]",
  "source": "RELTOPWCNBBIRTH",
  "target": "RELTOPWCNBDONOR",
  "condition": "if [RELTOPWCNBBIRTH = 8]",
  "sequence_index": 526
},
{
  "id": "RELTOPWCNBDONOR__RELTOPWCNBLPAR__527",
  "type": "conditional",
  "label": "if [RELTOPWCNBDONOR = 1|2]",
  "source": "RELTOPWCNBDONOR",
  "target": "RELTOPWCNBLPAR",
  "condition": "if [RELTOPWCNBDONOR = 1|2]",
  "sequence_index": 527
},
{
  "id": "PregnancyLoop__RELTOPWCNBDONOR__1678",
  "type": "loop",
  "label": "loop",
  "source": "PregnancyLoop",
  "target": "RELTOPWCNBDONOR",
  "condition": "",
  "sequence_index": 1678
},
{
  "id": "LCHLVLoop__RELTOPWCNBDONOR__1714",
  "type": "loop",
  "label": "loop",
  "source": "LCHLVLoop",
  "target": "RELTOPWCNBDONOR",
  "condition": "",
  "sequence_index": 1714
},
{
  "id": "RELTOPWCNBBIRTH__RELTOPWCNBDONOR__3058",
  "type": "sequential",
  "label": "sequential",
  "source": "RELTOPWCNBBIRTH",
  "target": "RELTOPWCNBDONOR",
  "condition": "",
  "sequence_index": 3058
},
{
  "id": "RELTOPWCNBDONOR__RELTOPWCNBLPAR__3059",
  "type": "sequential",
  "label": "sequential",
  "source": "RELTOPWCNBDONOR",
  "target": "RELTOPWCNBLPAR",
  "condition": "",
  "sequence_index": 3059
}
]
},
"forsta_graph": {
  "nodes_json": [
    {

```



```

    "id": "RELTOPWCNBIRTH",
    "data": {
      "label": "RELTOPWCNBIRTH",
      "title": "Relationship when child born",
      "option_count": 13,
      "question_type": "single_choice"
    },
    "type": "question"
  },
  {
    "id": "RELTOPWCNBDONOR",
    "data": {
      "label": "RELTOPWCNBDONOR",
      "title": "Partner sperm donor/ surrogacy",
      "option_count": 7,
      "question_type": "single_choice"
    },
    "type": "question"
  },
  {
    "id": "RELTOPWCNBLPAR",
    "data": {
      "label": "RELTOPWCNBLPAR",
      "title": "Partner child's adoptive/legal parent",
      "option_count": 2,
      "question_type": "single_choice"
    },
    "type": "question"
  },
  {
    "id": "Loop_27151",
    "data": {
      "label": "Loop_27151",
      "title": "Loop_27151",
      "option_count": 0,
      "question_type": "loop"
    },
    "type": "loop"
  },
  {
    "id": "Loop_27141",
    "data": {
      "label": "Loop_27141",
      "title": "Loop_27141",
      "option_count": 0,
      "question_type": "loop"
    },
    "type": "loop"
  }
],
"edges_json": [
  {
    "id": "Loop_27151__RELTOPWCNBDONOR__3422",
    "type": "loop",
    "label": "loop",
    "source": "Loop_27151",
    "target": "RELTOPWCNBDONOR",
    "condition": "",
    "sequence_index": 3422
  }
]

```

```
    },
    {
      "id": "Loop_27141__RELTOPWCNBDONOR__3523",
      "type": "loop",
      "label": "loop",
      "source": "Loop_27141",
      "target": "RELTOPWCNBDONOR",
      "condition": "",
      "sequence_index": 3523
    },
    {
      "id": "RELTOPWCNBBIRTH__RELTOPWCNBDONOR__5224",
      "type": "sequential",
      "label": "sequential",
      "source": "RELTOPWCNBBIRTH",
      "target": "RELTOPWCNBDONOR",
      "condition": "",
      "sequence_index": 5224
    },
    {
      "id": "RELTOPWCNBDONOR__RELTOPWCNBLPAR__5225",
      "type": "sequential",
      "label": "sequential",
      "source": "RELTOPWCNBDONOR",
      "target": "RELTOPWCNBLPAR",
      "condition": "",
      "sequence_index": 5225
    }
  ]
}
```

get_routing_simulation

Scope: Colectica-only

MCP prompt: /showRoutingSimulation

Description

Deterministic coverage-intent simulation for a Colectica module: for each of up to 10 auto-generated coverage intents, which conditional edges fired given seed answers, and whether each intent's target edge(s) were covered. Exactly replicates what the routing-simulation GUI JSON endpoint already computes -- not persisted, recomputed on every call, same cost profile as that existing endpoint. Bounded regardless of module size (coverage_intent_builder caps at 10 intents).

Python signature

```
get_routing_simulation(
    module_id_or_name: str
) -> dict
```

Example prompt shortcut

/showRoutingSimulation

"Using the get_routing_simulation tool, show me the routing coverage simulation for module 'Benefits W18'. Summarize status_counts and call out any intents that are not_covered or blocked_missing_inputs."

Example request

```
get_routing_simulation(module_id_or_name="Benefits W18")
```

Example response (real output)

```
{
  "module_id": "c8171847-7442-41ef-b534-04682184a14f",
  "module_name": "Benefits W18",
  "coverage_intent_count": 8,
  "status_counts": {
    "covered": 8
  },
  "results": [
    {
      "intent_id": "Trigger BENCTC",
      "intent_label": "Trigger BENCTC",
      "inputs": {
        "BENBASE": "3",
        "BenBase": "3"
      },
      "active_questions": [],
      "skipped_questions": [],
      "triggered_edges": [
        {
          "edge_index": 0,
          "edge_id": "6e9db933-602f-4d64-9551-36eaf14ee6cb",
          "source": "BENBASE",
          "target": "BENCTC",
          "condition_text": "if [BenBase = 3 & BenBase <> 4]",
          "evaluation_status": "true",
          "used_inputs": {
            "BenBase": "3"
          }
        },

```

```

    "missing_inputs": [],
    "normalized_expression": "BenBase = 3 & BenBase <> 4",
    "warnings": [],
    "is_target_edge": true
  }
],
"untriggered_edges": [
  {
    "edge_index": 1,
    "edge_id": "71e5f417-747e-484a-aac0-e21da20526d3",
    "source": "BENBASE",
    "target": "BENWCA",
    "condition_text": "if [BenBase = 4]",
    "evaluation_status": "false",
    "used_inputs": {
      "BenBase": "3"
    },
    "missing_inputs": [],
    "normalized_expression": "BenBase = 4",
    "warnings": [],
    "is_target_edge": false
  },
  {
    "edge_index": 2,
    "edge_id": "73222862-2f0c-47a0-96bd-c5d5727d6bcc",
    "source": "FICODE",
    "target": "FRADDL",
    "condition_text": "if [FICODE IN 2,3,4,24,25,26,27,28,29,30,35,38,39]",
    "evaluation_status": "unknown",
    "used_inputs": {},
    "missing_inputs": [
      "FICODE"
    ],
    "normalized_expression": "FICODE IN 2,3,4,24,25,26,27,28,29,30,35,38,39",
    "warnings": [],
    "is_target_edge": false
  },
  {
    "edge_index": 3,
    "edge_id": "82df3e3f-edf6-4e60-9cd5-a5400e956114",
    "source": "BENDIS",
    "target": "BENESA",
    "condition_text": "if [BenDis = 2]",
    "evaluation_status": "unknown",
    "used_inputs": {},
    "missing_inputs": [
      "BenDis"
    ],
    "normalized_expression": "BenDis = 2",
    "warnings": [],
    "is_target_edge": false
  },
  {
    "edge_index": 4,
    "edge_id": "93689ebc-bebc-4433-8769-799357b89f0c",
    "source": "FRVAL",
    "target": "FRWC",
    "condition_text": "if [FRVAL > 0]",
    "evaluation_status": "unknown",

```

```

    "used_inputs": {},
    "missing_inputs": [
      "FRVAL"
    ],
    "normalized_expression": "FRVAL > 0",
    "warnings": [],
    "is_target_edge": false
  },
  {
    "edge_index": 5,
    "edge_id": "af189980-e877-4fac-a3d3-41d0a8125069",
    "source": "BENWCA",
    "target": "BENWCAOTC",
    "condition_text": "if [BenWCA = 1]",
    "evaluation_status": "unknown",
    "used_inputs": {},
    "missing_inputs": [
      "BenWCA"
    ],
    "normalized_expression": "BenWCA = 1",
    "warnings": [],
    "is_target_edge": false
  },
  {
    "edge_index": 6,
    "edge_id": "b4ee61ad-7bb8-41d3-9684-5b5a75c6dc52",
    "source": "FRJT",
    "target": "FRJTPN",
    "condition_text": "if [FRJT = 2]",
    "evaluation_status": "unknown",
    "used_inputs": {},
    "missing_inputs": [
      "FRJT"
    ],
    "normalized_expression": "FRJT = 2",
    "warnings": [],
    "is_target_edge": false
  },
  {
    "edge_index": 7,
    "edge_id": "e1966439-7392-4dc3-b803-f09549a72264",
    "source": "FICODE, FRVAL",
    "target": "FITAX",
    "condition_text": "if [(FICODE IN 2,3,4) AND (FRVAL >= 0)]",
    "evaluation_status": "unknown",
    "used_inputs": {},
    "missing_inputs": [
      "FICODE",
      "FRVAL"
    ],
    "normalized_expression": "(FICODE IN 2,3,4) AND (FRVAL >= 0)",
    "warnings": [],
    "is_target_edge": false
  }
],
"target_edges": [
  {
    "edge_index": 0,
    "edge_id": "6e9db933-602f-4d64-9551-36eaf14ee6cb",

```

```

    "source": "BENBASE",
    "target": "BENCTC",
    "condition_text": "if [BenBase = 3 & BenBase <> 4]",
    "evaluation_status": "true",
    "used_inputs": {
      "BenBase": "3"
    },
    "missing_inputs": [],
    "normalized_expression": "BenBase = 3 & BenBase <> 4",
    "warnings": [],
    "is_target_edge": true
  }
],
"non_target_edges": [
  {
    "edge_index": 1,
    "edge_id": "71e5f417-747e-484a-aac0-e21da20526d3",
    "source": "BENBASE",
    "target": "BENWCA",
    "condition_text": "if [BenBase = 4]",
    "evaluation_status": "false",
    "used_inputs": {
      "BenBase": "3"
    },
    "missing_inputs": [],
    "normalized_expression": "BenBase = 4",
    "warnings": [],
    "is_target_edge": false
  },
  {
    "edge_index": 2,
    "edge_id": "73222862-2f0c-47a0-96bd-c5d5727d6bcc",
    "source": "FICODE",
    "target": "FRADDL",
    "condition_text": "if [FICODE IN 2,3,4,24,25,26,27,28,29,30,35,38,39]",
    "evaluation_status": "unknown",
    "used_inputs": {},
    "missing_inputs": [
      "FICODE"
    ],
    "normalized_expression": "FICODE IN 2,3,4,24,25,26,27,28,29,30,35,38,39",
    "warnings": [],
    "is_target_edge": false
  },
  {
    "edge_index": 3,
    "edge_id": "82df3e3f-edf6-4e60-9cd5-a5400e956114",
    "source": "BENDIS",
    "target": "BENESA",
    "condition_text": "if [BenDis = 2]",
    "evaluation_status": "unknown",
    "used_inputs": {},
    "missing_inputs": [
      "BenDis"
    ],
    "normalized_expression": "BenDis = 2",
    "warnings": [],
    "is_target_edge": false
  },

```

```

{
  "edge_index": 4,
  "edge_id": "93689ebc-bebc-4433-8769-799357b89f0c",
  "source": "FRVAL",
  "target": "FRWC",
  "condition_text": "if [FRVAL > 0]",
  "evaluation_status": "unknown",
  "used_inputs": {},
  "missing_inputs": [
    "FRVAL"
  ],
  "normalized_expression": "FRVAL > 0",
  "warnings": [],
  "is_target_edge": false
},
{
  "edge_index": 5,
  "edge_id": "af189980-e877-4fac-a3d3-41d0a8125069",
  "source": "BENWCA",
  "target": "BENWCAOTC",
  "condition_text": "if [BenWCA = 1]",
  "evaluation_status": "unknown",
  "used_inputs": {},
  "missing_inputs": [
    "BenWCA"
  ],
  "normalized_expression": "BenWCA = 1",
  "warnings": [],
  "is_target_edge": false
},
{
  "edge_index": 6,
  "edge_id": "b4ee61ad-7bb8-41d3-9684-5b5a75c6dc52",
  "source": "FRJT",
  "target": "FRJTPN",
  "condition_text": "if [FRJT = 2]",
  "evaluation_status": "unknown",
  "used_inputs": {},
  "missing_inputs": [
    "FRJT"
  ],
  "normalized_expression": "FRJT = 2",
  "warnings": [],
  "is_target_edge": false
},
{
  "edge_index": 7,
  "edge_id": "e1966439-7392-4dc3-b803-f09549a72264",
  "source": "FICODE, FRVAL",
  "target": "FITAX",
  "condition_text": "if [(FICODE IN 2,3,4) AND (FRVAL >= 0)]",
  "evaluation_status": "unknown",
  "used_inputs": {},
  "missing_inputs": [
    "FICODE",
    "FRVAL"
  ],
  "normalized_expression": "(FICODE IN 2,3,4) AND (FRVAL >= 0)",
  "warnings": [],

```

```

    "is_target_edge": false
  }
],
"coverage_status": "covered",
"coverage_summary": {
  "declared_target_edge_indexes": [
    0
  ],
  "matched_target_edge_indexes": [
    0
  ],
  "target_edge_count": 1,
  "triggered_target_edge_count": 1,
  "false_target_edge_count": 0,
  "missing_target_edge_count": 0,
  "unknown_target_edge_count": 0,
  "triggered_target_edge_indexes": [
    0
  ],
  "missing_target_edge_indexes": [],
  "false_target_edge_indexes": []
},
"missing_inputs": [
  "BENDIS",
  "BENWCA",
  "FICODE",
  "FRJT",
  "FRVAL"
],
"invalid_inputs": [],
"warnings": [
  "Edge index 2 (73222862-2f0c-47a0-96bd-c5d5727d6bcc): unsupported evaluator status 'unknown'.",
  "Edge index 3 (82df3e3f-edf6-4e60-9cd5-a5400e956114): unsupported evaluator status 'unknown'.",
  "Edge index 4 (93689ebc-bebc-4433-8769-799357b89f0c): unsupported evaluator status 'unknown'.",
  "Edge index 5 (af189980-e877-4fac-a3d3-41d0a8125069): unsupported evaluator status 'unknown'.",
  "Edge index 6 (b4ee61ad-7bb8-41d3-9684-5b5a75c6dc52): unsupported evaluator status 'unknown'.",
  "Edge index 7 (e1966439-7392-4dc3-b803-f09549a72264): unsupported evaluator status 'unknown'."
]
"Step 16.4 evaluates all conditional edges globally and separately reports target edge coverage.
Active/skipped question traversal is not implemented yet."
]
},
{
  "intent_id": "Trigger BENWCA",
  "intent_label": "Trigger BENWCA",
  "inputs": {
    "BENBASE": "4",
    "BenBase": "4"
  },
  "active_questions": [],
  "skipped_questions": [],
  "triggered_edges": [
    {
      "edge_index": 1,
      "edge_id": "71e5f417-747e-484a-aac0-e21da20526d3",
      "source": "BENBASE",
      "target": "BENWCA",
      "condition_text": "if [BenBase = 4]",
      "evaluation_status": "true",
      "used_inputs": {

```



```

        "BenBase": "4"
    },
    "missing_inputs": [],
    "normalized_expression": "BenBase = 4",
    "warnings": [],
    "is_target_edge": true
}
],
"untriggered_edges": [
    {
        "edge_index": 0,
        "edge_id": "6e9db933-602f-4d64-9551-36eaf14ee6cb",
        "source": "BENBASE",
        "target": "BENCTC",
        "condition_text": "if [BenBase = 3 & BenBase <> 4]",
        "evaluation_status": "false",
        "used_inputs": {
            "BenBase": "4"
        },
        "missing_inputs": [],
        "normalized_expression": "BenBase = 3 & BenBase <> 4",
        "warnings": [],
        "is_target_edge": false
    },
    {
        "edge_index": 2,
        "edge_id": "73222862-2f0c-47a0-96bd-c5d5727d6bcc",
        "source": "FICODE",
        "target": "FRADDL",
        "condition_text": "if [FICODE IN 2,3,4,24,25,26,27,28,29,30,35,38,39]",
        "evaluation_status": "unknown",
        "used_inputs": {},
        "missing_inputs": [
            "FICODE"
        ],
        "normalized_expression": "FICODE IN 2,3,4,24,25,26,27,28,29,30,35,38,39",
        "warnings": [],
        "is_target_edge": false
    },
    {
        "edge_index": 3,
        "edge_id": "82df3e3f-edf6-4e60-9cd5-a5400e956114",
        "source": "BENDIS",
        "target": "BENESA",
        "condition_text": "if [BenDis = 2]",
        "evaluation_status": "unknown",
        "used_inputs": {},
        "missing_inputs": [
            "BenDis"
        ],
        "normalized_expression": "BenDis = 2",
        "warnings": [],
        "is_target_edge": false
    },
    {
        "edge_index": 4,
        "edge_id": "93689ebc-bebc-4433-8769-799357b89f0c",
        "source": "FRVAL",
        "target": "FRWC",

```

```

    "condition_text": "if [FRVAL > 0]",
    "evaluation_status": "unknown",
    "used_inputs": {},
    "missing_inputs": [
      "FRVAL"
    ],
    "normalized_expression": "FRVAL > 0",
    "warnings": [],
    "is_target_edge": false
  },
  {
    "edge_index": 5,
    "edge_id": "af189980-e877-4fac-a3d3-41d0a8125069",
    "source": "BENWCA",
    "target": "BENWCAOTC",
    "condition_text": "if [BenWCA = 1]",
    "evaluation_status": "unknown",
    "used_inputs": {},
    "missing_inputs": [
      "BenWCA"
    ],
    "normalized_expression": "BenWCA = 1",
    "warnings": [],
    "is_target_edge": false
  },
  {
    "edge_index": 6,
    "edge_id": "b4ee61ad-7bb8-41d3-9684-5b5a75c6dc52",
    "source": "FRJT",
    "target": "FRJTPN",
    "condition_text": "if [FRJT = 2]",
    "evaluation_status": "unknown",
    "used_inputs": {},
    "missing_inputs": [
      "FRJT"
    ],
    "normalized_expression": "FRJT = 2",
    "warnings": [],
    "is_target_edge": false
  },
  {
    "edge_index": 7,
    "edge_id": "e1966439-7392-4dc3-b803-f09549a72264",
    "source": "FICODE, FRVAL",
    "target": "FITAX",
    "condition_text": "if [(FICODE IN 2,3,4) AND (FRVAL >= 0)]",
    "evaluation_status": "unknown",
    "used_inputs": {},
    "missing_inputs": [
      "FICODE",
      "FRVAL"
    ],
    "normalized_expression": "(FICODE IN 2,3,4) AND (FRVAL >= 0)",
    "warnings": [],
    "is_target_edge": false
  }
],
"target_edges": [
  {

```

```

    "edge_index": 1,
    "edge_id": "71e5f417-747e-484a-aac0-e21da20526d3",
    "source": "BENBASE",
    "target": "BENWCA",
    "condition_text": "if [BenBase = 4]",
    "evaluation_status": "true",
    "used_inputs": {
      "BenBase": "4"
    },
    "missing_inputs": [],
    "normalized_expression": "BenBase = 4",
    "warnings": [],
    "is_target_edge": true
  }
],
"non_target_edges": [
  {
    "edge_index": 0,
    "edge_id": "6e9db933-602f-4d64-9551-36eaf14ee6cb",
    "source": "BENBASE",
    "target": "BENCTC",
    "condition_text": "if [BenBase = 3 & BenBase <> 4]",
    "evaluation_status": "false",
    "used_inputs": {
      "BenBase": "4"
    },
    "missing_inputs": [],
    "normalized_expression": "BenBase = 3 & BenBase <> 4",
    "warnings": [],
    "is_target_edge": false
  },
  {
    "edge_index": 2,
    "edge_id": "73222862-2f0c-47a0-96bd-c5d5727d6bcc",
    "source": "FICODE",
    "target": "FRADDL",
    "condition_text": "if [FICODE IN 2,3,4,24,25,26,27,28,29,30,35,38,39]",
    "evaluation_status": "unknown",
    "used_inputs": {},
    "missing_inputs": [
      "FICODE"
    ],
    "normalized_expression": "FICODE IN 2,3,4,24,25,26,27,28,29,30,35,38,39",
    "warnings": [],
    "is_target_edge": false
  },
  {
    "edge_index": 3,
    "edge_id": "82df3e3f-edf6-4e60-9cd5-a5400e956114",
    "source": "BENDIS",
    "target": "BENESA",
    "condition_text": "if [BenDis = 2]",
    "evaluation_status": "unknown",
    "used_inputs": {},
    "missing_inputs": [
      "BenDis"
    ],
    "normalized_expression": "BenDis = 2",
    "warnings": [],

```

```

    "is_target_edge": false
  },
  {
    "edge_index": 4,
    "edge_id": "93689ebc-bebc-4433-8769-799357b89f0c",
    "source": "FRVAL",
    "target": "FRWC",
    "condition_text": "if [FRVAL > 0]",
    "evaluation_status": "unknown",
    "used_inputs": {},
    "missing_inputs": [
      "FRVAL"
    ],
    "normalized_expression": "FRVAL > 0",
    "warnings": [],
    "is_target_edge": false
  },
  {
    "edge_index": 5,
    "edge_id": "af189980-e877-4fac-a3d3-41d0a8125069",
    "source": "BENWCA",
    "target": "BENWCAOTC",
    "condition_text": "if [BenWCA = 1]",
    "evaluation_status": "unknown",
    "used_inputs": {},
    "missing_inputs": [
      "BenWCA"
    ],
    "normalized_expression": "BenWCA = 1",
    "warnings": [],
    "is_target_edge": false
  },
  {
    "edge_index": 6,
    "edge_id": "b4ee61ad-7bb8-41d3-9684-5b5a75c6dc52",
    "source": "FRJT",
    "target": "FRJTPN",
    "condition_text": "if [FRJT = 2]",
    "evaluation_status": "unknown",
    "used_inputs": {},
    "missing_inputs": [
      "FRJT"
    ],
    "normalized_expression": "FRJT = 2",
    "warnings": [],
    "is_target_edge": false
  },
  {
    "edge_index": 7,
    "edge_id": "e1966439-7392-4dc3-b803-f09549a72264",
    "source": "FICODE, FRVAL",
    "target": "FITAX",
    "condition_text": "if [(FICODE IN 2,3,4) AND (FRVAL >= 0)]",
    "evaluation_status": "unknown",
    "used_inputs": {},
    "missing_inputs": [
      "FICODE",
      "FRVAL"
    ],
  },

```

```

        "normalized_expression": "(FICODE IN 2,3,4) AND (FRVAL >= 0)",
        "warnings": [],
        "is_target_edge": false
    }
],
"coverage_status": "covered",
"coverage_summary": {
    "declared_target_edge_indexes": [
        1
    ],
    "matched_target_edge_indexes": [
        1
    ],
    "target_edge_count": 1,
    "triggered_target_edge_count": 1,
    "false_target_edge_count": 0,
    "missing_target_edge_count": 0,
    "unknown_target_edge_count": 0,
    "triggered_target_edge_indexes": [
        1
    ],
    "missing_target_edge_indexes": [],
    "false_target_edge_indexes": []
},
"missing_inputs": [
    "BENDIS",
    "BENWCA",
    "FICODE",
    "FRJT",
    "FRVAL"
],
"invalid_inputs": [],
"warnings": [
    "Edge index 2 (73222862-2f0c-47a0-96bd-c5d5727d6bcc): unsupported evaluator status 'unknown'.",
    "Edge index 3 (82df3e3f-edf6-4e60-9cd5-a5400e956114): unsupported evaluator status 'unknown'.",
    "Edge index 4 (93689ebc-bebc-4433-8769-799357b89f0c): unsupported evaluator status 'unknown'.",
    "Edge index 5 (af189980-e877-4fac-a3d3-41d0a8125069): unsupported evaluator status 'unknown'.",
    "Edge index 6 (b4ee61ad-7bb8-41d3-9684-5b5a75c6dc52): unsupported evaluator status 'unknown'.",
    "Edge index 7 (e1966439-7392-4dc3-b803-f09549a72264): unsupported evaluator status 'unknown'.",
    "Step 16.4 evaluates all conditional edges globally and separately reports target edge coverage. Active/skipped question traversal is not implemented yet."
]
}
],
"_results_truncated_for_manual": "Showing 2 of 8 real entries here for readability. Calling this tool live returns all 8."
}

```

COLECTICA VS COLECTICA ROUTING DIFF (VERSION DIFF)

get_version_diff_report

Scope: Colectica + Colectica pair

MCP prompt: /showVersionDiffReport

Description

Structural (plus condition-text) routing discrepancies between two Colectica-format modules -- e.g. two waves/versions of the same questionnaire -- exactly matching the version-diff report GUI page (/questionnaires/version-diff/<base_id>/<compare_id>/). Does NOT run matching/comparison itself -- if it's never been run for this exact pair, has_run is False and discrepancies is empty rather than raising.

Both modules must be Colectica JSON; a Forsta+ module raises ToolError.

This is a separate pipeline from get_routing_diff_report (Colectica vs Forsta+) -- different matcher priority (name-first) and comparator (adds a CONDITION_MISMATCH type, meaningful here specifically because both sides share the same Colectica grammar).

discrepancy_type filters to one VersionRoutingDiscrepancy.DiscrepancyType value (missing_in_compare / missing_in_base / condition_mismatch). limit caps how many discrepancies are returned (default 200) -- discrepancy_count always reports the true total regardless of limit.

Python signature

```
get_version_diff_report(
    base_module_id_or_name: str,
    compare_module_id_or_name: str,
    discrepancy_type: str | None = None,
    limit: int = 200
) -> dict
```

Example prompt shortcut

/showVersionDiffReport

"Using the get_version_diff_report tool, show me the version-diff report between base module '3a2de4bf-432b-42f2-b74c-2ceffd868e8a' and compare module 'W16' (e.g. two waves/versions of the same questionnaire). If has_run is false, tell me matching/comparison hasn't been run for this pair yet instead of guessing. Otherwise summarize the match_summary and group the discrepancies by type and severity, calling out condition_mismatch rows separately since those mean both modules route to the same place but the gating condition changed."

Example request

```
get_version_diff_report(base_module_id_or_name="3a2de4bf-432b-42f2-b74c-2ceffd868e8a",
compare_module_id_or_name="W16", limit=3) # real pair: 1803 matched questions, 256 real discrepancies
-- limit caps the list, not the true count
```

Example response (real output)

```
{
  "base_module_id": "3a2de4bf-432b-42f2-b74c-2ceffd868e8a",
  "base_module_name": "W18",
  "compare_module_id": "082097ea-506f-42bc-b93e-df24afe88190",
  "compare_module_name": "W16",
```

```

"has_run": true,
"match_summary": {
  "total": 1803,
  "matched": 1465,
  "unmatched": 338
},
"discrepancy_count": 256,
"discrepancies": [
  {
    "id": "cf0faf88-fafe-49a9-9093-01d8425502cf",
    "discrepancy_type": "missing_in_compare",
    "severity": "warning",
    "base_question": "PNAME",
    "compare_question": "PNAME",
    "summary": "Question PNAME matched in both modules, but the routing edge to RESPEMAIL (fires when (ff_remail = MIS & ((PERGRID > 0 & NAME & SNAME <> DK | REF | MIS)|(PNAME & PSNAME <> DK | REF | MIS))) | RESPEMAILCONF = 2) has no equivalent in the compare module.",
    "meaning": "This is a structural comparison only: it checks whether an edge with the same source and target exists in both modules, not what the condition means or whether an alternate path exists. This flag means either (a) the compare module genuinely never routes to RESPEMAIL from here -- a real routing change worth reviewing, or (b) the compare module reaches RESPEMAIL some other way (a different source question, different condition logic, a default fallthrough) that this structural check can't recognize -- a false positive. Severity \"Warning\" means this needs a human to check, not that it's a confirmed bug.",
    "edge": {
      "origin_module": "base",
      "source_question": "PERGRID, NAME, SNAME, PNAME, PSNAME, RESPEMAILCONF",
      "target_question": "RESPEMAIL",
      "condition_text": "(ff_remail = MIS & ((PERGRID > 0 & NAME & SNAME <> DK | REF | MIS)|(PNAME & PSNAME <> DK | REF | MIS))) | RESPEMAILCONF = 2"
    }
  },
  {
    "id": "49de0cee-3fcc-4f77-8253-9001b0c707cb",
    "discrepancy_type": "missing_in_compare",
    "severity": "warning",
    "base_question": "PSNAME",
    "compare_question": "PSNAME",
    "summary": "Question PSNAME matched in both modules, but the routing edge to RESPEMAIL (fires when (ff_remail = MIS & ((PERGRID > 0 & NAME & SNAME <> DK | REF | MIS)|(PNAME & PSNAME <> DK | REF | MIS))) | RESPEMAILCONF = 2) has no equivalent in the compare module.",
    "meaning": "This is a structural comparison only: it checks whether an edge with the same source and target exists in both modules, not what the condition means or whether an alternate path exists. This flag means either (a) the compare module genuinely never routes to RESPEMAIL from here -- a real routing change worth reviewing, or (b) the compare module reaches RESPEMAIL some other way (a different source question, different condition logic, a default fallthrough) that this structural check can't recognize -- a false positive. Severity \"Warning\" means this needs a human to check, not that it's a confirmed bug.",
    "edge": {
      "origin_module": "base",
      "source_question": "PERGRID, NAME, SNAME, PNAME, PSNAME, RESPEMAILCONF",
      "target_question": "RESPEMAIL",
      "condition_text": "(ff_remail = MIS & ((PERGRID > 0 & NAME & SNAME <> DK | REF | MIS)|(PNAME & PSNAME <> DK | REF | MIS))) | RESPEMAILCONF = 2"
    }
  },
  {
    "id": "c7147cdc-c565-4a72-8b98-fedab38b3bea",
    "discrepancy_type": "missing_in_compare",
    "severity": "warning",
    "base_question": "PERGRID",

```

```

    "compare_question": "PERGRID",
    "summary": "Question PERGRID matched in both modules, but the routing edge to RESPEMAIL (fires when (ff_remail = MIS & ((PERGRID > 0 & NAME & SNAME <> DK | REF | MIS)|(PNAME & PSNAME <> DK | REF | MIS))) | RESPEMAILCONF = 2) has no equivalent in the compare module.",
    "meaning": "This is a structural comparison only: it checks whether an edge with the same source and target exists in both modules, not what the condition means or whether an alternate path exists. This flag means either (a) the compare module genuinely never routes to RESPEMAIL from here -- a real routing change worth reviewing, or (b) the compare module reaches RESPEMAIL some other way (a different source question, different condition logic, a default fallthrough) that this structural check can't recognize -- a false positive. Severity \"Warning\" means this needs a human to check, not that it's a confirmed bug.",
    "edge": {
      "origin_module": "base",
      "source_question": "PERGRID, NAME, SNAME, PNAME, PSNAME, RESPEMAILCONF",
      "target_question": "RESPEMAIL",
      "condition_text": "(ff_remail = MIS & ((PERGRID > 0 & NAME & SNAME <> DK | REF | MIS)|(PNAME & PSNAME <> DK | REF | MIS))) | RESPEMAILCONF = 2"
    }
  }
]
}

```


get_version_diff_discrepancy_detail

Scope: Colectica + Colectica pair

MCP prompt: /showVersionDiffDiscrepancy

Description

One version-diff discrepancy in full detail, exactly matching the version-diff discrepancy detail GUI page

(/questionnaires/version-diff/<base_id>/<compare_id>/discrepancy/<discrepancy_id>/)

-- including BOTH modules' routing graph neighborhoods around the focus question (base_graph and compare_graph), not just the base side. Either graph is None if that module has no graph built yet, rather than erroring the whole call.

Python signature

```
get_version_diff_discrepancy_detail(
    base_module_id_or_name: str,
    compare_module_id_or_name: str,
    discrepancy_id: str
) -> dict
```

Example prompt shortcut

/showVersionDiffDiscrepancy

"Using the get_version_diff_discrepancy_detail tool, show me full detail for discrepancy '6e709dd7-d965-42c4-86a4-477132d3da3b' between base module '3a2de4bf-432b-42f2-b74c-2ceffd868e8a' and compare module 'W16' -- the summary, the meaning (why this is a lead not a confirmed bug, and if it's a condition_mismatch, that the comparison is textual not semantic), and describe the routing around the focus question on both the base graph and the compare graph."

Example request

```
get_version_diff_discrepancy_detail(base_module_id_or_name="3a2de4bf-432b-42f2-b74c-2ceffd868e8a",
compare_module_id_or_name="W16", discrepancy_id="6e709dd7-d965-42c4-86a4-477132d3da3b") # a real
CONDITION_MISMATCH: NAMEPERM -> NONAME fires under a narrower condition in W16
```

Example response (real output)

```
{
  "base_module_id": "3a2de4bf-432b-42f2-b74c-2ceffd868e8a",
  "compare_module_id": "082097ea-506f-42bc-b93e-df24afe88190",
  "discrepancy_id": "6e709dd7-d965-42c4-86a4-477132d3da3b",
  "discrepancy_type": "condition_mismatch",
  "severity": "warning",
  "summary": "Question NAMEPERM matched in both modules. Both modules route to NONAME, but the gating condition text differs: base fires when NAMEPERM =1 | NAME = DK|REF; compare fires when NAMEPERM =1.",
  "meaning": "This is a textual comparison of the two condition expressions after normalizing whitespace and casing -- it does not evaluate whether the conditions are logically equivalent. Two differently-worded conditions can still mean exactly the same thing (and would still be reported here as different), just as two identically-normalized conditions aren't guaranteed to behave identically in every runtime edge case. Both modules agree that NONAME is a valid route from here -- what changed is *when* it fires. Treat this as a lead: worth checking whether the gating logic change was intentional (e.g. a design refinement between waves) or a routing regression.",
  "edge": {
    "origin_module": "base",
    "source_question": "NAMEPERM, NAME",
    "target_question": "NONAME",
    "condition_text": "NAMEPERM =1 | NAME = DK|REF"
  },
}
```

```

"focus_question_name": "NAMEPERM",
"base_focus_question_name": "NAMEPERM",
"compare_focus_question_name": "NAMEPERM",
"highlight_edge": {
  "source": "NAMEPERM, NAME",
  "target": "NONAME"
},
"base_graph": {
  "nodes_json": [
    {
      "id": "MORES",
      "data": {
        "label": "MORES",
        "title": "More people resident",
        "option_count": 6,
        "question_type": "single_choice"
      },
      "type": "question"
    },
    {
      "id": "NAME",
      "data": {
        "label": "NAME",
        "title": "Name/identifier for household member",
        "option_count": 4,
        "question_type": "text"
      },
      "type": "question"
    },
    {
      "id": "NAMEPERM",
      "data": {
        "label": "NAMEPERM",
        "title": "Does not have permission to give name",
        "option_count": 3,
        "question_type": "single_choice"
      },
      "type": "question"
    },
    {
      "id": "SNAME",
      "data": {
        "label": "SNAME",
        "title": "Surname for household member",
        "option_count": 4,
        "question_type": "text"
      },
      "type": "question"
    },
    {
      "id": "TITLE",
      "data": {
        "label": "TITLE",
        "title": "Title for household member",
        "option_count": 11,
        "question_type": "single_choice"
      },
      "type": "question"
    }
  ],

```

```

{
  "id": "NONAME",
  "data": {
    "label": "NONAME",
    "title": "Identifier where no name is given",
    "option_count": 0,
    "question_type": "text"
  },
  "type": "question"
},
{
  "id": "EMAILINTRO",
  "data": {
    "label": "EMAILINTRO",
    "title": "Household grid email introduction",
    "option_count": 5,
    "question_type": "single_choice"
  },
  "type": "question"
},
{
  "id": "NameLoop",
  "data": {
    "label": "NameLoop",
    "title": "NameLoop",
    "option_count": 0,
    "question_type": "loop"
  },
  "type": "loop"
}
],
"edges_json": [
  {
    "id": "MORES__NAMEPERM__18",
    "type": "conditional",
    "label": "if [MORES = 1]",
    "source": "MORES",
    "target": "NAMEPERM",
    "condition": "if [MORES = 1]",
    "sequence_index": 18
  },
  {
    "id": "NAMEPERM__SNAME__19",
    "type": "conditional",
    "label": "if [NAMEPERM <> 1]",
    "source": "NAMEPERM",
    "target": "SNAME",
    "condition": "if [NAMEPERM <> 1]",
    "sequence_index": 19
  },
  {
    "id": "NAMEPERM__TITLE__20",
    "type": "conditional",
    "label": "if [NAMEPERM <> 1]",
    "source": "NAMEPERM",
    "target": "TITLE",
    "condition": "if [NAMEPERM <> 1]",
    "sequence_index": 20
  },

```

```

{
  "id": "NAMEPERM__NONAME__22",
  "type": "conditional",
  "label": "if [NAMEPERM =1 | NAME = DK|REF]",
  "source": "NAMEPERM",
  "target": "NONAME",
  "condition": "if [NAMEPERM =1 | NAME = DK|REF]",
  "sequence_index": 22
},
{
  "id": "NAMEPERM__EMAILINTRO__79",
  "type": "conditional",
  "label": "if [((GRIDVARIABLES.ModeType = 1) | (GRIDVARIABLES.ModeType = 2) | (GRIDVARIABLES.ModeType = 3 &
HHSIZE > 1)) & NAMEPERM <> 1 & NAME <> DK|REF]",
  "source": "NAMEPERM",
  "target": "EMAILINTRO",
  "condition": "if [((GRIDVARIABLES.ModeType = 1) | (GRIDVARIABLES.ModeType = 2) | (GRIDVARIABLES.ModeType =
3 & HHSIZE > 1)) & NAMEPERM <> 1 & NAME <> DK|REF]",
  "sequence_index": 79
},
{
  "id": "NameLoop__NAMEPERM__1446",
  "type": "loop",
  "label": "loop",
  "source": "NameLoop",
  "target": "NAMEPERM",
  "condition": "",
  "sequence_index": 1446
},
{
  "id": "NAME__NAMEPERM__2383",
  "type": "sequential",
  "label": "sequential",
  "source": "NAME",
  "target": "NAMEPERM",
  "condition": "",
  "sequence_index": 2383
},
{
  "id": "NAMEPERM__SNAME__2384",
  "type": "sequential",
  "label": "sequential",
  "source": "NAMEPERM",
  "target": "SNAME",
  "condition": "",
  "sequence_index": 2384
}
]
},
"compare_graph": {
  "nodes_json": [
    {
      "id": "MORES",
      "data": {
        "label": "MORES",
        "title": "More people resident",
        "option_count": 6,
        "question_type": "single_choice"
      }
    }
  ]
}

```

```

    "type": "question"
  },
  {
    "id": "NAME",
    "data": {
      "label": "NAME",
      "title": "Name/identifier for household member",
      "option_count": 4,
      "question_type": "text"
    },
    "type": "question"
  },
  {
    "id": "NAMEPERM",
    "data": {
      "label": "NAMEPERM",
      "title": "Does not have permission to give name",
      "option_count": 3,
      "question_type": "single_choice"
    },
    "type": "question"
  },
  {
    "id": "SNAME",
    "data": {
      "label": "SNAME",
      "title": "Surname for household member",
      "option_count": 4,
      "question_type": "text"
    },
    "type": "question"
  },
  {
    "id": "TITLE",
    "data": {
      "label": "TITLE",
      "title": "Title for household member",
      "option_count": 11,
      "question_type": "single_choice"
    },
    "type": "question"
  },
  {
    "id": "NONAME",
    "data": {
      "label": "NONAME",
      "title": "Identifier where no name is given",
      "option_count": 0,
      "question_type": "text"
    },
    "type": "question"
  },
  {
    "id": "REJOINER",
    "data": {
      "label": "REJOINER",
      "title": "Rejoiner indicator",
      "option_count": 11,
      "question_type": "single_choice"
    }
  }

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    },
    "type": "question"
  },
  {
    "id": "EMAILINTRO",
    "data": {
      "label": "EMAILINTRO",
      "title": "Household grid email introduction",
      "option_count": 5,
      "question_type": "single_choice"
    },
    "type": "question"
  },
  {
    "id": "RESPEMAILCONF",
    "data": {
      "label": "RESPEMAILCONF",
      "title": "Respondent email address check",
      "option_count": 6,
      "question_type": "single_choice"
    },
    "type": "question"
  },
  {
    "id": "NameLoop",
    "data": {
      "label": "NameLoop",
      "title": "NameLoop",
      "option_count": 0,
      "question_type": "loop"
    },
    "type": "loop"
  }
],
"edges_json": [
  {
    "id": "MORES__NAMEPERM__19",
    "type": "conditional",
    "label": "if [MORES = 1]",
    "source": "MORES",
    "target": "NAMEPERM",
    "condition": "if [MORES = 1]",
    "sequence_index": 19
  },
  {
    "id": "NAMEPERM__SNAME__20",
    "type": "conditional",
    "label": "if [NAMEPERM <> 1]",
    "source": "NAMEPERM",
    "target": "SNAME",
    "condition": "if [NAMEPERM <> 1]",
    "sequence_index": 20
  },
  {
    "id": "NAMEPERM__TITLE__21",
    "type": "conditional",
    "label": "if [NAMEPERM <> 1]",
    "source": "NAMEPERM",
    "target": "TITLE",

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    "condition": "if [NAMEPERM <> 1]",
    "sequence_index": 21
  },
  {
    "id": "NAMEPERM__NONAME__23",
    "type": "conditional",
    "label": "if [NAMEPERM =1]",
    "source": "NAMEPERM",
    "target": "NONAME",
    "condition": "if [NAMEPERM =1]",
    "sequence_index": 23
  },
  {
    "id": "NAMEPERM__REJOINER__28",
    "type": "conditional",
    "label": "if [(JOINER = 1 | 3) & NAMEPERM <> 1]",
    "source": "NAMEPERM",
    "target": "REJOINER",
    "condition": "if [(JOINER = 1 | 3) & NAMEPERM <> 1]",
    "sequence_index": 28
  },
  {
    "id": "NAMEPERM__EMAILINTRO__79",
    "type": "conditional",
    "label": "if [((GRIDVARIABLES.ModeType = 1) | (GRIDVARIABLES.ModeType = 2) | (GRIDVARIABLES.ModeType = 3 &
HHSIZE > 1)) & NAMEPERM <> 1]",
    "source": "NAMEPERM",
    "target": "EMAILINTRO",
    "condition": "if [((GRIDVARIABLES.ModeType = 1) | (GRIDVARIABLES.ModeType = 2) | (GRIDVARIABLES.ModeType =
3 & HHSIZE > 1)) & NAMEPERM <> 1]",
    "sequence_index": 79
  },
  {
    "id": "NAMEPERM__RESPEMAILCONF__82",
    "type": "conditional",
    "label": "if [ff_remail <> MIS & NAMEPERM <> 1]",
    "source": "NAMEPERM",
    "target": "RESPEMAILCONF",
    "condition": "if [ff_remail <> MIS & NAMEPERM <> 1]",
    "sequence_index": 82
  },
  {
    "id": "NameLoop__NAMEPERM__1673",
    "type": "loop",
    "label": "loop",
    "source": "NameLoop",
    "target": "NAMEPERM",
    "condition": "",
    "sequence_index": 1673
  },
  {
    "id": "NAME__NAMEPERM__2518",
    "type": "sequential",
    "label": "sequential",
    "source": "NAME",
    "target": "NAMEPERM",
    "condition": "",
    "sequence_index": 2518
  },

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{
  "id": "NAMEPERM__SNAME__2519",
  "type": "sequential",
  "label": "sequential",
  "source": "NAMEPERM",
  "target": "SNAME",
  "condition": "",
  "sequence_index": 2519
}
]
}
}
```